

University of Victoria



Final Design Report

Retractable Bridge Project

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Executive Summary

This report outlines the development of a retractable bridge concept designed to accommodate air, water, and land traffic within a highly constrained urban environment. The client's requirements specify that the bridge must not exceed 12 inches in height when retracted and must maintain a 12-inch-wide centered clearance for water traffic. The goal of this project is to design and prototype a mechanically actuated retractable bridge capable of spanning a 32-inch water channel while meeting strict dimensional and operational constraints.

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1.0 Introduction

Stemming from a client request of a retractable bridge design, the research, concept design, and selection of bridge designs was undertaken. A review of existing bridge designs was undertaken, providing valuable insight into practical design features which were considered when brainstorming concepts. To provide designs which best meet the needs of the client, a problem statement and need statement are formed, highlighting the context of the constraints provided. From this, concrete design objectives and constraints are formulated, providing a set of guidelines for the bridge design to adhere to.

This commenced the creation of the three concept designs, each providing a unique solution to the problem poised by the client. Utilizing a pairwise comparison table in conjunction with a weighted selection table, the concept designs were compared, allowing for the selection of the design which best fits the needs of the client. This design will then be further revised and prototyped to become the final bridge design presented.

2.0 Background

As referenced in the introduction, a need for a retractable bridge design was expressed by the client. The bridge connects the banks of the waterway to allow for traffic to cross on the road deck, while also located in an area in which air and water traffic must remain unobstructed by the bridge. A retractable bridge design must be employed to clear the waterway of obstruction.

The bridge is to be designed and prototyped at a reduced scale, spanning 32” across the waterway. The main structure bridge will be constructed of laser cut plywood as it is an economically and environmentally friendly method to prototype the early stage of the bridge design. Small parts may be constructed of alternative materials, such as shafts and latches, with methods such as

machining and 3D printing being feasible options. The bridge will be built on two plywood pylons to simulate the bridge's footprint on the banks of the waterway. Steel plates along the road deck will simulate the weight of the vehicles and road surface on the bridge.

An energy storage device must be contained in the bridge design to retract the movable portion of the bridge. The device may be mechanical or electrical (i.e. a motor and battery) but must be activated by a single human input. The input can be of any form, including (but not limited to) a lever, crank, button, or latch. To return the bridge back to its original position extending across the waterway, the bridge is to be manually moved by hand by the operator. The bridge is to withstand multiple cycles of extending and retracting to evaluate the longevity of the bridge design.

3.0 Review of Literature

To better approach the bridge design problem, an investigation into the designs of previous retracting bridges was conducted. As per the constraints of the design, the bridges considered that most closely met the design criteria all retracted completely to one side of the waterway. Other considerations such as the bridge folding perpendicular to the waterway, and rearward space constraints were considered when finding designs to analyze.

3.1 Hörn Bridge

The Hörn Bridge in Keil, Germany is a folding pedestrian bridge that uses a series of cables and pulleys to fold up into three articulating sections. Spanning 84ft with a width of 16ft, the bridge opens to allow ships to pass through the Kiel Fjord inlet.



Figure 1: Hörn Bridge Partially Folded [1]

To fold inwards, the rear-most upright's cable is retracted, which begins the folding motion of the three sections of the bridge. Once retracted enough, the furthest outwards section latches to the second section using the yellow mechanism seen in Figure 1. The second upright's cable is then retracted, drawing the entire bridge inwards. As the furthest inwards section rotates up, the cables act on the latch mechanism to rotate the furthest outwards section into the center section, then draw the center and outwards section up into the first section. Hydraulic motors are used to drive winches that retract the bridges cables [1], which would require the use of a fluid pump to mimic the design.

3.2 Rolling Bridge

The pedestrian Rolling Bridge, located in London, United Kingdom, spans 12 metres across the Paddington Basin. The bridge is composed of eight triangular prism shaped segments, connected by pinned joints and actuated by fourteen hydraulic cylinders, allowing the structure to collapse to a minimal compact form. The closest pair of pistons to the stationary segment of the bridge initiates the lift. As these pistons push against the middle of the handrails, the reduced distance between each segment allows the bridge to pivot and lift. The remaining pistons are then actuated, further reducing

the distance between the segments until the bridge forms an octagonal shape, measuring approximately half of its original length.

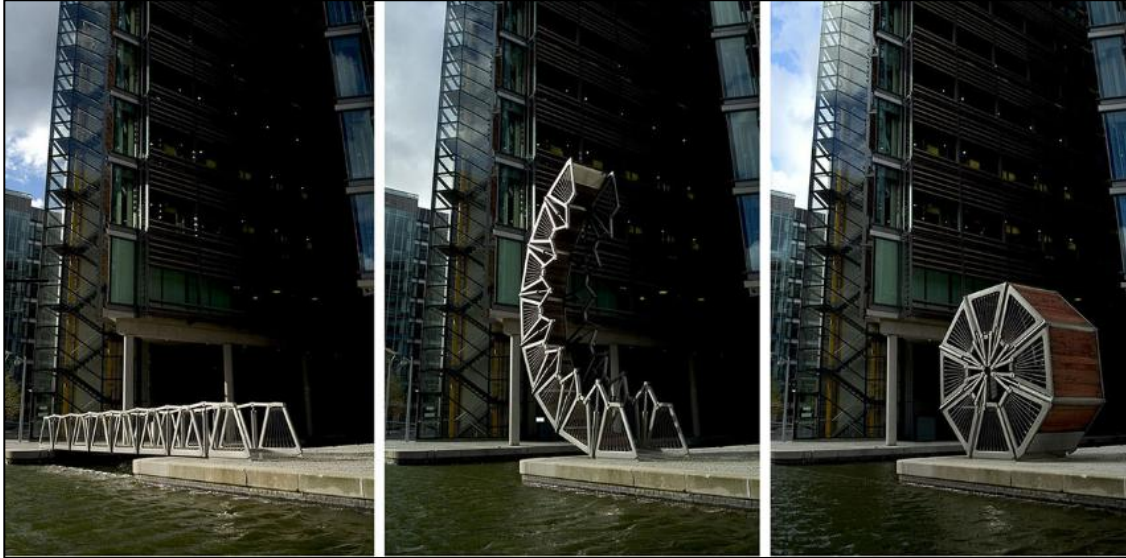


Figure 2: Three Stages of the Rolling Bridge [2]

3.3 Johan van Hasselt Canal Bridge

An existing concept for a folding bridge over the Johan van Hasselt Canal in the Netherlands, was developed by Atelier Kempe Thill, an international architectural firm. The bridge was designed with overall dimensions of 110 metres in length and 19 meters in width. The bridge features two movable sections, allowing boats to pass along the canal. Each section is actuated by hydraulic pistons. The first piston pushes the largest bridge segment beneath the car deck, enabling the bridge to lift. The second piston collapses the smaller section of the bridge, shortening the total overhead air space used.

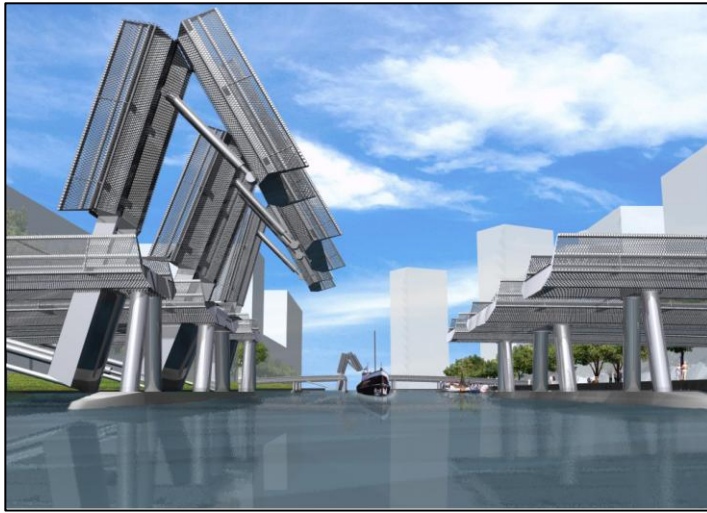


Figure 3: Johan van Hasselt Canal Bridge Folded [3]

4.0 Problem Statement

Urban environments with high volumes of water, road, and air traffic face challenges in maintaining commuter flow without disrupting shipping or airspace operations. Bridges such as bascule, vertical lift, and other more common designs are highly dependent on surrounding infrastructure making them ineffectual for spaces where vertical and horizontal clearances are limited. Thus, there is a need for a one-sided, retractable bridge system that can span a waterway while minimizing the bridges footprint to ensure water and air traffic are unobstructed.

This goal of this project is to design and construct a functional prototype of a mechanically retractable bridge capable of spanning a 32” water channel while fully retracting to provide a 12” clear passage for water traffic. The bridge must operate smoothly and reliably using stored mechanical energy or a single motor, remain within specified dimensional constraints, operate successfully at least 12 times, and deploys without damage. The design should balance load capacity, durability, ease of actuation, cost, safety, and aesthetics.

5.0 Design Objectives and Constraints

The following design objectives and constraints define the performance, geometric, and safety requirements that all bridge concepts in this project must follow. These were derived from the MECH 350 Major Project Outline [4] and from our own engineering judgement and manufacturability considerations. Collectively, they ensure that each proposed design remains within the design envelope. The constraints primarily address fixed physical and dimensional limits while the objectives focus on performance-based criteria. Together, Table 1 outlines the foundation for concept generation, evaluation, and selection.

Table 1: Design Objectives and Constraints

#	Design Objective/Constraint	Units	Target Value/Range
1	Bridge span	inches	32"
2	Max deployed height	inches	$\leq 12"$
3	Deck width	inches	$\leq 9"$ ($\geq 3"$ clear roadway)
4	Deck Flatness Tolerance	inches	$\leq 1/16"$
5	Load Capacity	lbs	$\geq$ steel plates (~ 4 lb)
6	Durability (cycle life)	# of cycles	≥ 12
7	Ease of Actuation	newtons	≤ 20 N pull force
8	Cost	CAD \$	$\leq \$250$
9	Safety	qualitative	limited sharp edges/pinch points
11	Retracted Bridge (Behind Pylon)	inches	$\leq 4"$
12	Retracted Center Clearance	inches	$\geq 12"$
13	Underside Clearance Limit	inches	0"
14	Aesthetic & Finish	qualitative	Smooth surface, clean/organized

Justification of Design Objectives:

- Load Capacity ($\geq$ steel plates, ~ 4 lb) – Specified in the project outline as the standard testing load. Meeting or exceeding this ensures sufficient structural rigidity and verifies mechanical strength under expected conditions.
- Durability (≥ 12 cycles) – Chosen to match the project requirement for repeated actuation without structural or mechanical failure. This target reflects the minimum cycles required as outlined by the Major Project Outline and reflects expected fatigue limits [4].
- Ease of Actuation (≤ 20 N force) – Based on ergonomic guidelines for hand-actuated mechanisms, this threshold ensures the bridge can be operated comfortably and consistently by a single user input.
- Cost ($\leq \$250$) – Specified by the Major Project Outline to maintain affordability and promote design efficiency. Staying within this limit ensures cost-effective material and manufacturing decisions.
- Safety (limited sharp edges/pinch points) – This is a key design requirement to minimize the risk during handling and operation of the bridge. Smooth edges and controlled movement make the bridge safe to handle.
- Aesthetic and Finish (Smooth surface, clean/organized) – This objective ensures the bridge looks professional and well-built. Clean finishes and organized components improve overall design quality.

6.0 Concept Alternatives

From the brainstorming stage, three concept designs were selected to be considered as the potential bridge design. They will be weighed against each other to select the final design.

6.1 Concept A – Sliding Rail Folding Bridge

Concept A, the Sliding Rail Folding Bridge, is a mechanically actuated retraction system designed to span the required 32” distance while folding down into a compact arrangement within the 12” maximum height constraint. The bridge achieves this functionality by combining a tower-based actuation mechanism with a sliding rail system and sequentially hinged deck panels. This arrangement ensures the deck can fold in a predictable, controlled manner while remaining structurally stable during deployment and retraction.

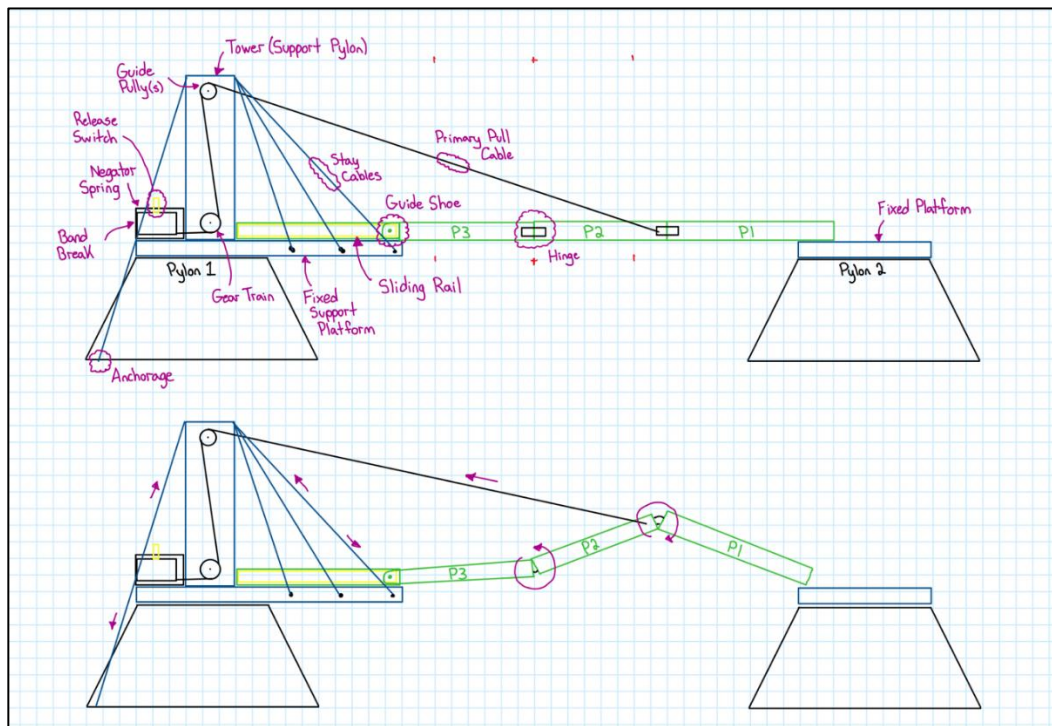


Figure 4: Concept A – Sliding Rail Folding Bridge in Deployed and Initial Retraction Position

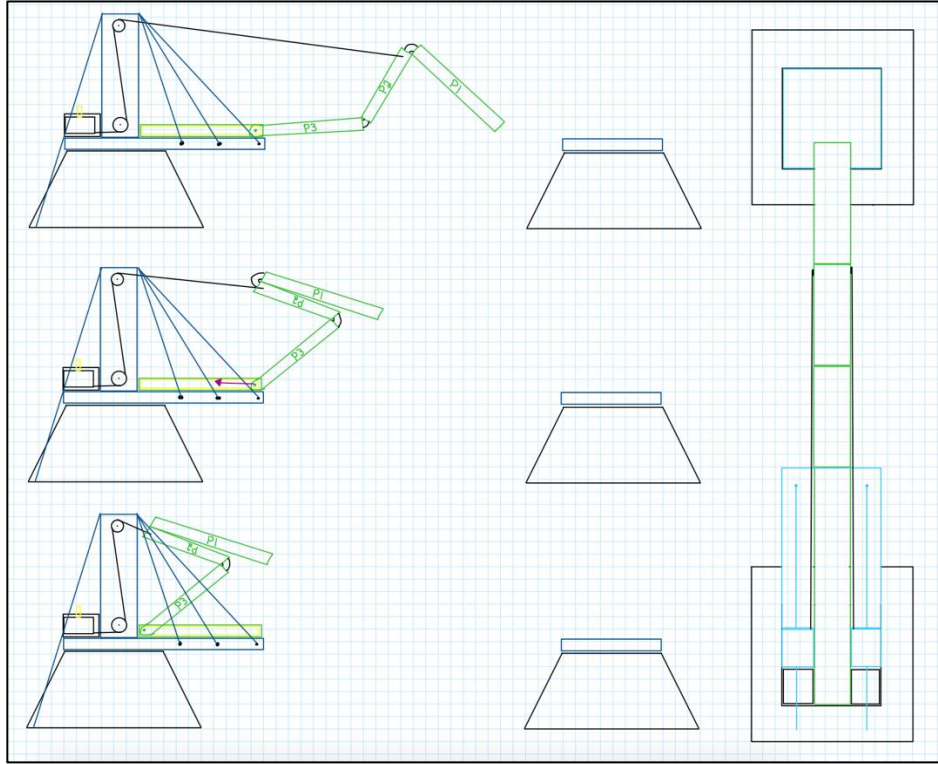


Figure 5: Concept A – Folding Sequence Progression and Top View Layout

The bridge deck is divided into three primary panels (P1-P3), as shown in Figure 4. Panel P3 is connected to the fixed road deck on Pylon 1 via a sliding rail and guide shoe system, while Panels P2 and P1 are attached in series through concealed rotating hinges. When fully deployed, P1 rests flush on a fixed support platform mounted to Pylon 2, ensuring proper weight support and alignment.

The bridge's motion is driven by a primary pull cable running through guide pulleys at the top of the tower. This cable is tensioned by a negator spring, which stores constant-force energy, and is restrained using a band brake system. Deployment is initiated by activating a mechanical release switch, which disengages the band brake and allows the spring to wind the cable back, pulling the deck into its folded configuration. In addition to the pulleys, the tower (support pylon) also

incorporates a gear train anchored to the pylon. This gearing allows the relatively short stroke of the negator spring to be translated into the longer cable travel required for bridge retraction.

The rear deck panel (P3) is supported by a sliding rail (yellow in Figure 4) that constrains its vertical position while allowing controlled horizontal translation. The guide show ensures that the panel does not lift or tilt unexpectedly during retraction. This linear control is important because it transforms the pulling force of the cable into a predictable sequence of hinge rotations, causing the panels to fold in order rather than collapsing chaotically. The rail is keyed so that the panel only moves horizontally once deck panel P3 has reached a desired height.

The panels are joined using a concealed hinge system (between P3-P2 and P2-P1). Unlike surface-mounted hinges, this design is embedded within the deck structure so that no hinge components obstruct the roadway surface. This ensures a smooth, flush deck when the bridge is deployed. The concealed hinges are directional in function, allowing rotation in only one direction. This guarantees that the panels fold upward and back rather than buckling downward. To further control the folding sequence, dampers are integrated at the hinge points. These dampers control the angular velocity of each panel, preventing sudden whipping as the cable pulls the system.

In addition to the primary pull cable, stay cables are used to provide secondary support. The primary stay cables are anchored from the tower to the fixed road deck section that houses the sliding rail. Further support cables can be added to the active deck panels (P1-P3). These cables would hang loosely when the bridge is retracted, but once the bridge is deployed, they would come under tension to support the weight of the moving panels. This way, the stay cables provide structural reinforcement without impeding the folding mechanism, ensuring that the deck panels can carry load safely in the deployed state.

Concept A, the sliding rail folding bridge, is defined by its simplicity in both structure and motion. Unlike more complex drawbridge systems that rely on multiple interconnected mechanisms, this design uses a minimal set of components (concealed hinges, sliding rail, a single pull cable) to

achieve smooth folding. The motion is initiated by the primary pull cable but naturally assisted by gravity, allowing the bridge to retract in a controlled and predictable manner without over-constraining movement. The guided rail ensures consistent alignment during folding and as a result, the bridge is expected to surpass the durability requirement of 12 cycles, with the only notable long-term concern being fatigue of the negator spring.

Ease of actuation is the key advantage of this concept. Because the mechanism is mechanically simple, minimal input force is required to initiate retraction. The integrated gear train and pulley system translate the spring's motion into a balanced pull across the structure, while the band brake ensures that the retraction occurs at a steady, controlled speed. However, as a purely mechanism system, it offers limited flexibility in adjusting retraction speed compared to a powered design.

The major weakness of Concept A is load capacity. When deployed, the bridge's weight and any applied loads are primarily carried by the main pull cable. While stay cable and rails add some structural stability, the design does not distribute loads as evenly as a truss or multi-cable system. This makes the bridge less capable of carrying weight and load performance will remain its most limiting factor.

In terms of cost, the system remains feasible. While the integration of spring, band brakes, and concealed hinges will introduce some manufacturing difficulty, these components are relatively small-scale and less expensive. It is expected that the final build will remain below the \$250 budget.

Safety is addressed through the concealed hinges and band brakes, while keep the road deck flush and prevent uncontrolled folding speeds. Some pinch points remain at hinge joints, but the controlled motion reduces risk. Therefore, Concept A is durable, simple, and safe, but has drawbacks in load capacity and actuation control.

6.2 Concept B – Trifold Motor Bridge

This prototype demonstrates a trifold bridge mechanism that leverages a coordinated system of gears, coupling bars/rods and a motor driven pulley to retract a bridge that spans the 32” gap. The bridge consists of three main deck sections linked through gears. The bridge operates by transmitting rotational motion from the motorized pulley system to lift the bridge. Figure 6 shows the primary components and layout of the bridge in its extended position.

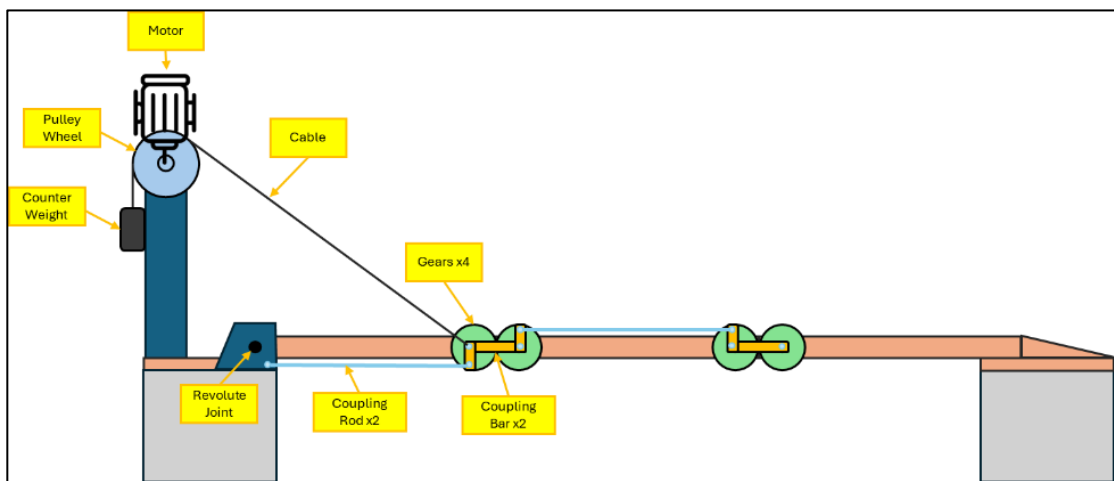


Figure 6: Concept B - Trifold bridge concept design

The overarching principle of this design relies on the synchronized motion of the stretch and contraction mechanism as shown in [5]. Once the motor is powered, the tension in the rope will result in the first segment of the bridge being raised. As it rotates, the combination of coupling rods and gears causes the second and third road decks to begin rotating inwards towards the collapsed position. This allows the bridge to operate using only one degree of actuation without having one of the bridge decks go below the top of the pylons. To help relieve torque from the motor during retraction, a counterweight is placed on the other side of the cable. The partially and fully contracted bridges can be seen in Figure 7 & Figure 8.

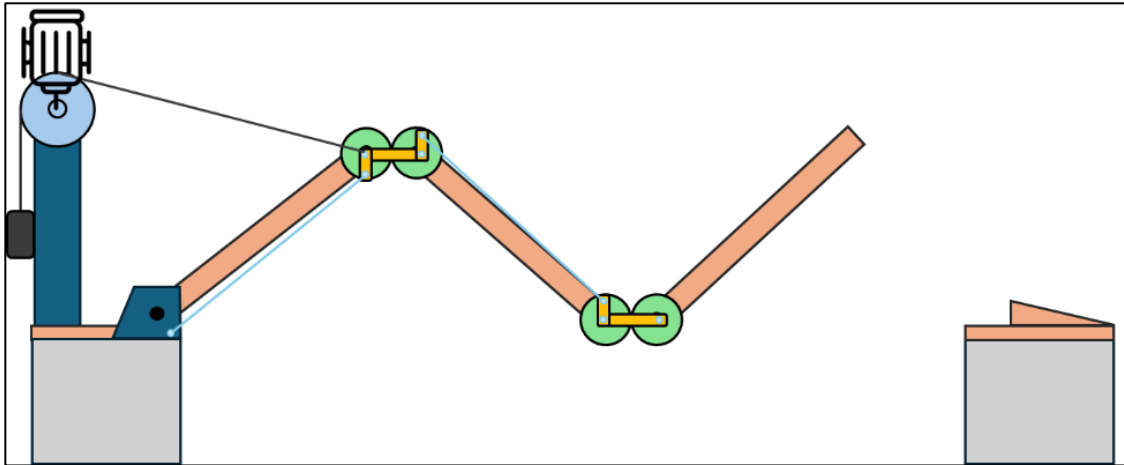


Figure 7: Concept B - Trifold bridge mid retraction

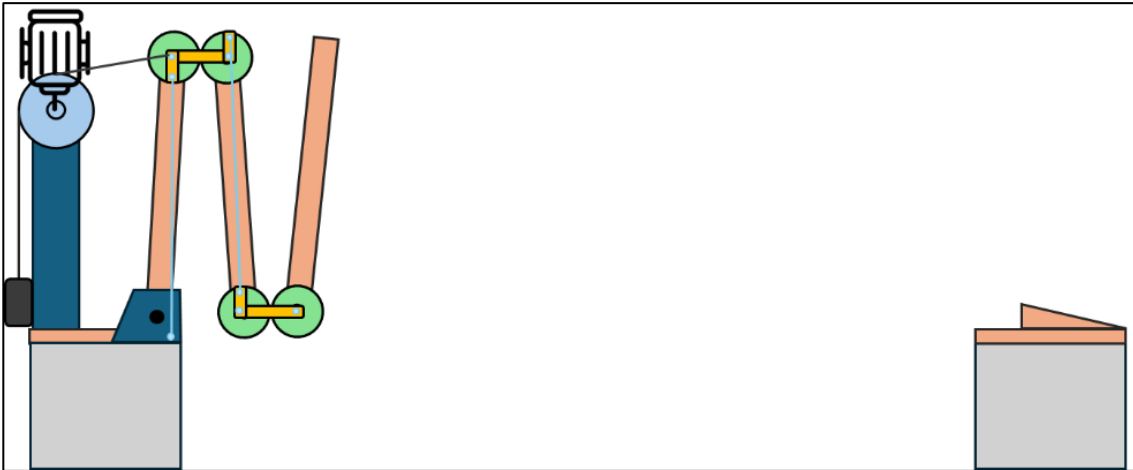


Figure 8: Concept B - Trifold bridge fully retracted

The trifold bridge presents a couple of key features that make it worth consideration. The counterweight mechanism enhances energy efficiency by offsetting part of the load on the motor allowing for ease of actuation. The use of mechanical coupling ensures that the motion remains smooth and synchronized even without sensors or electronic controllers resulting safer operation.

The design also presents several limitations that must be considered. The inclusion of multiple gears, rods, and joints introduces mechanical complexity that increases maintenance requirements and susceptibility to wear. The system's load capacity is limited by the strength of the gears and coupling mechanisms, which may not be able to hold under the desired load. Gears would need to be made from a strong material, likely increasing costs. Precise alignment of gears and rods is essential for reliable operation, and any misalignment could lead to binding or uneven folding.

Despite the limitations, the main mechanism serves as a useful thought experiment for how controlled contractions can be achieved without the need for multiple actuators. In addition, it also presents a solution for lessening the burden of actuation. It is possible that later in development, portions of this concept can be adapted and adopted to solve problems during prototyping.

6.3 Concept C – Four-Bar Retracting Bridge

Concept C, the “Four-Bar Retracting Bridge”, utilizes a rack and pinion and simple four-bar mechanism to achieve the desired design objectives of the client. The road deck is constructed at the maximum allowable height of 12 inches, enabling the design to harness gravity and store mechanical energy in the form of potential energy. The primary structural assembly consists of a large support structure on the active pylon, three road deck segments, and an extension of the opposite pylon that acts as a passive support during operation.

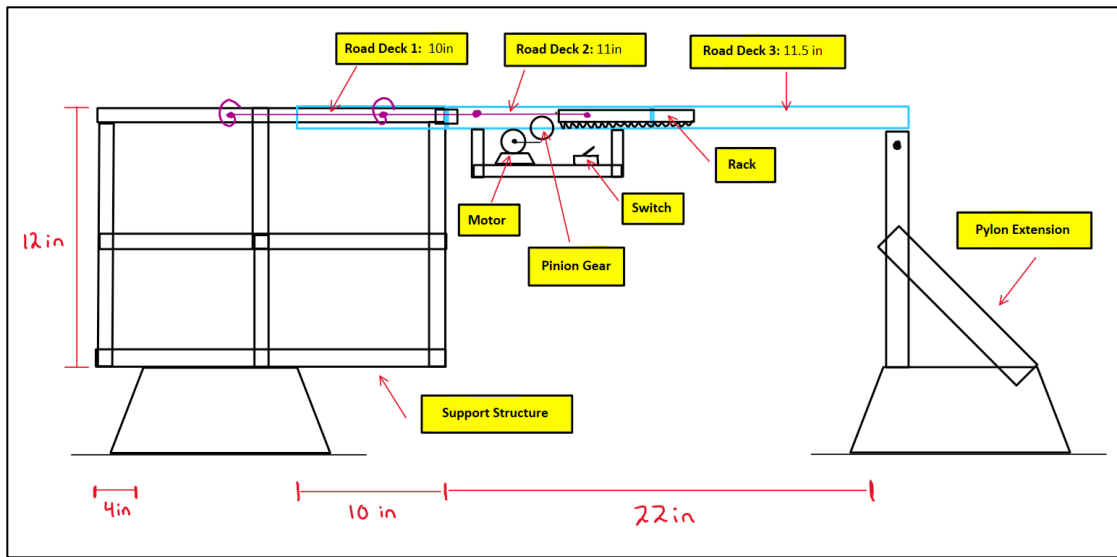


Figure 9: Concept C - Position 1

Once activated by a switch, the motor system mounted on road deck 2 drives a geartrain that engages the rack and pinion mechanism, producing the linear motion of road deck 3 and retracting it into road deck 2 in a telescoping motion. Once road deck 3 reaches the desired position, a latch mechanism between decks 1 and 2 is released, allowing decks 2 and 3 to descend and swing into the support structure under the influence of gravity by means of the four-bar linkages. Torsional springs or alternatively, rotational dampers are used to maintain controlled motion of the mechanism. Figure 10, Figure 11, and Figure 12 provide a visual representation of the bridge's motion.

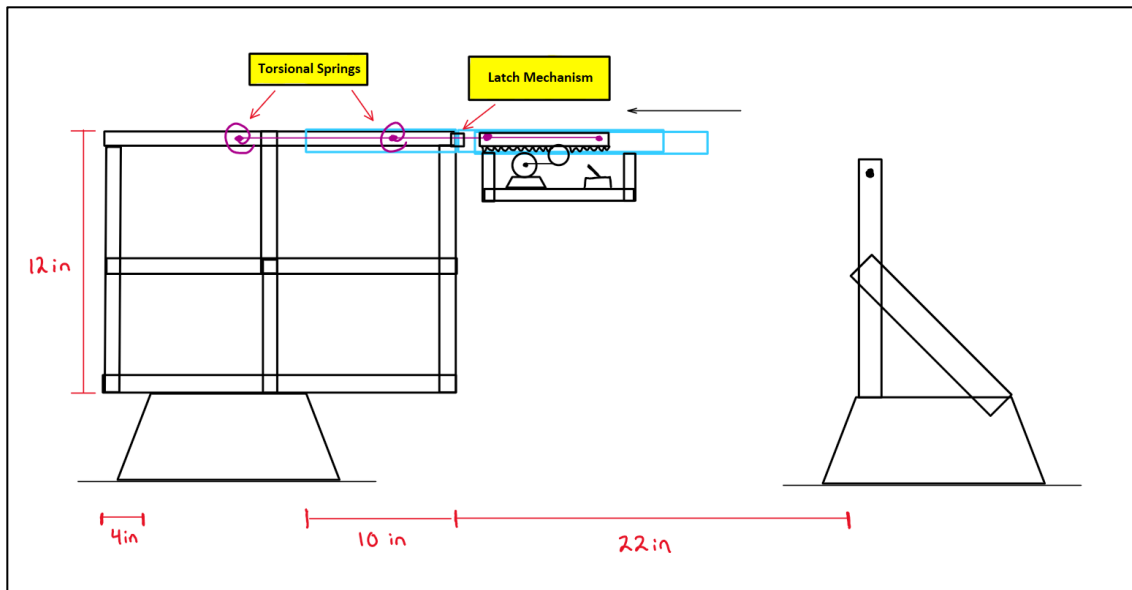


Figure 10: Concept C - Position 2

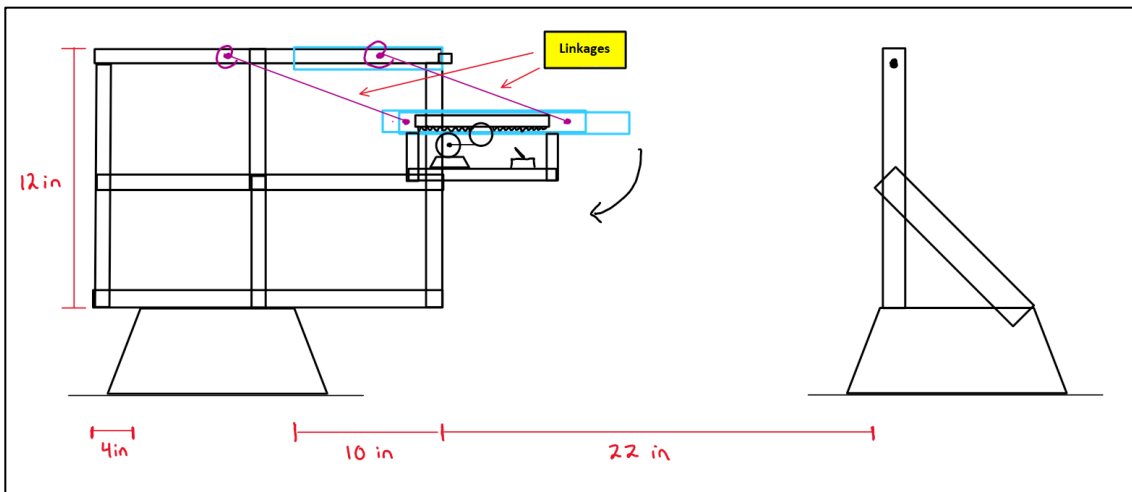


Figure 11: Concept C - Position 3

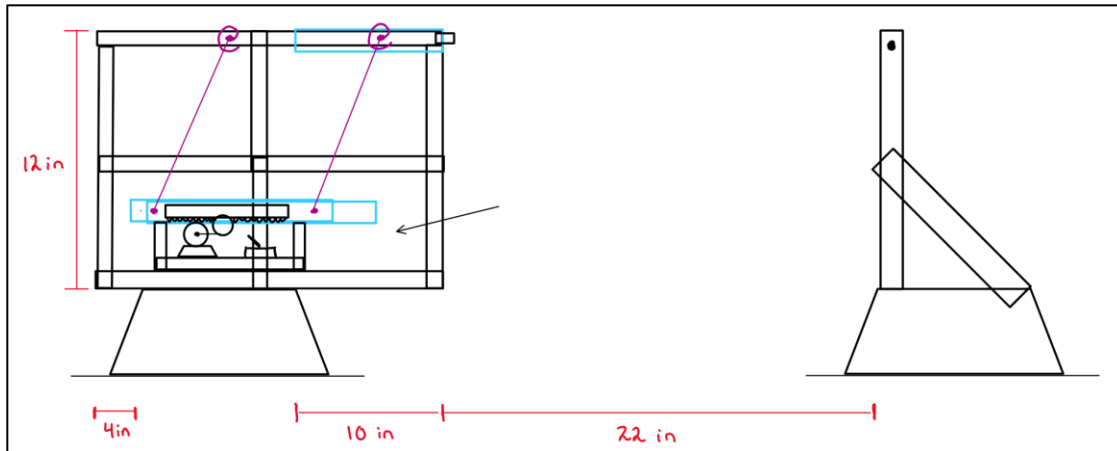


Figure 12: Concept C - Position 4

The release of the latch mechanism can be timed to the motion of road deck 3 using the gear train or released by direct physical contact with road deck 3 at the end of its stroke. A simple push to release latch mechanism [6] [7] would trigger the motion of the four-bar and allow the rack and pinion subassembly to swing into the fully retracted position.

This design offers several advantages that make it a strong candidate for further development and prototyping. Leveraging gravity as the primary energy storage mechanism streamlines the retraction process by enabling the bridge to utilize gravitational forces rather than opposing them. This design is also relatively simple, with fewer moving parts than other concepts, reducing costs and improving ease of actuation. Additionally, the large support structure provides a stable foundation capable of bearing the load of the steel-plated roadway and offers additional space for any counterweights, reinforced linkages, or revised mechanisms in future iterations.

Despite these advantages, there are several challenges that must be considered if this design is chosen. The telescoping action of road decks 2 and 3 inherently creates a small step in the roadway to allow for smooth linear motion. This step will need to be minimized to meet the deck flatness tolerance of $\leq 1/16$ in, ensuring both safety and a smooth roadway for vehicle traffic. Another significant concern is the load on the latch mechanism during the retraction of road deck 3. The

combined weight of the two road decks, motor system and geartrain places the latch under substantial stress. A robust locking solution capable of releasing easily while under high load will be required to prevent premature wear or failure to release. Additionally, the four-bar mechanism will require proper damping to control its motion during the retraction process. Without damping there could be damage to the road decks and/or motor system during the four-bar mechanism's portion of the retraction. Safety considerations also extend to the numerous pinch points that are created by the various components of the design; however, these concerns can be mitigated using guards or covers at the cost of additional weight and materials. Addressing these challenges early in the design process will ensure a robust solution that meets the required design objectives while remaining mechanically reliable and safe to operate.

Overall, the four-bar retracting bridge is a simple, effective way of meeting the design criteria and can be further refined through the iterative design process to create an efficient, reliable solution. Continued analysis, prototyping, and testing will be critical to validating the design's performance, optimizing its components and ensuring its compliance with the project requirements. Considering both the pros and cons of the design, it remains a promising candidate for further development.

7.0 Concept Selection Method

This section outlines the structured decision process used to evaluate and select the final bridge concept. The following approach includes evaluation scales for each design objective, ranking and weighing by importance, and a weighted decision table to compare each concept.

7.1 Evaluation Scale

Each key design objective was assigned a quantitative or qualitative scale that allowed for fair comparison between different bridge concepts. These scales were developed based on project requirements, physical limitations, and performance expectations set during concept development.

Table 2: Evaluation Scale for Load Capacity

Load Capacity	Num. Score
$\geq 7\text{-}8$ lb	10
6-7 lbs	8
4-6 lbs	6
< 4 lbs	0

Table 3: Evaluation Scale for Cost

Cost	Num. Score
$\leq \$150$	10
\$150 - \$200	8
\$200 - \$250	6
$> \$250$	0

Table 4: Evaluation Scale for Actuation.

Ease of Actuation	Num. Score
≤ 20 N	10
20 - 25 N	8
25 - 30 N	6
> 30 N	0

Table 5: Evaluation Scale for Safety.

Safety	Num. Score
Excellent	10
Good	7
Adequate	5
Poor	0

Table 6: Evaluation Scale for Aesthetics.

Aesthetic & Finish	Num. Score
Excellent	10
Good	8
Adequate	6
Poor	0

Table 7: Evaluation Scale for Durability.

Durability	Num. Score
≥ 30 cycles	10
20 - 30 cycles	8
10 - 15 cycles	6
< 10 cycles	0

These scales were selected to ensure consistent evaluation of mechanical and practical performance for each concept. For example, load capacity is scored according to the bridge's ability to support the standard steel plates, while ease of actuation is rated based on the estimated pulling force need to retract the bridge.

7.2 Pairwise Comparison Weighting Matrix

Each design objective was compared using a pairwise comparison method. A value of 1 was assigned when the row objective was considered more important than the column objective. Row totals were normalized to obtain relative weights for use in the final decision table.

Table 8: Pairwise Comparison Weighting Matrix

Criteria	Load Capacity	Durability	Ease of Actuation	Cost	Safety	Aesthetic & Finish	Row Total	Normalized Weights
Load Capacity		1	1	1	1	1	5	0.33
Durability	0		1	1	1	1	4	0.27
Ease of Actuation	0	0		1	1	1	3	0.20
Cost	0	0	0		1	1	2	0.13
Safety	0	0	0	0		1	1	0.07
Aesthetic & Finish	0	0	0	0	0		0	0.00

Load capacity and durability received the highest weights since they directly affect bridge reliability and performance under repeated use. Ease of actuation was ranked next, as the bridge must deploy with relatively minimal effort. Cost was given a moderate weight as it influences material and manufacturing choices but does not affect function. Safety and aesthetics were assigned lower weights because they are qualitative and generally satisfied across all designs.

7.3 Weighted Selection Table

Using the normalized weights from 7.2 and the evaluation scales from 7.1, a Weighted Selection Table was developed. Each concept received numeric scored based on its estimated

performance for every objective, which were then multiplied by their respective weights. The totals provide an overall performance score that reflects both importance and measured performance.

Table 9: Weighted Selection Table for Concept Comparison

		Design Options					
		Design Concept A		Design Concept B		Design Concept C	
Design Objective:	Rel. Weight	Numeric Score	Weighted Value	Numeric Score	Weighted Value	Numeric Score	Weighted Value
Load Capacity	0.33	10	3.33	6	2.00	8	2.67
Durability	0.27	6	1.60	6	1.60	8	2.13
Ease of Actuation	0.20	6	1.20	8	1.60	10	2.00
Cost	0.13	6	0.80	6	0.80	6	0.80
Safety	0.07	8	0.53	8	0.53	6	0.40
Aesthetic & Finish	0.00	7	0.00	10	0.00	7	0.00
		Total:	7.47	Total:	6.53	Total:	8.00

Based on the evaluation and the weighting of key design objectives, Concept C was selected for further development. This concept was chosen due to it performing the best amongst the three designs as per the weighted selection criteria. Although concepts A and B outperformed in certain criteria such as looks and load capacity, concept C was consistently highly ranked throughout the table and therefore was selected.

8.0 Iterative Design Process

The following section discusses the iterative process of reaching a final design. Throughout development, several adjustments and refinements were made to the chosen concept to improve performance and overall functionality.

8.1 First Iteration

The first iteration of the design incorporated several significant changes from the original concept. The motor was relocated to the main structure, and the rack and pinion mechanism was replaced with a pulley system to reduce cost and complexity. In addition, the positions of the four-bar linkage and retracting decks were revised, allowing deck 2 to retract into the main structure while

deck 3 swings downward under gravity. This modification eliminated the need for a push-to-release latch mechanism and avoided the potential issues associated with its operation. Figure 13 below shows a CAD model of this iteration.

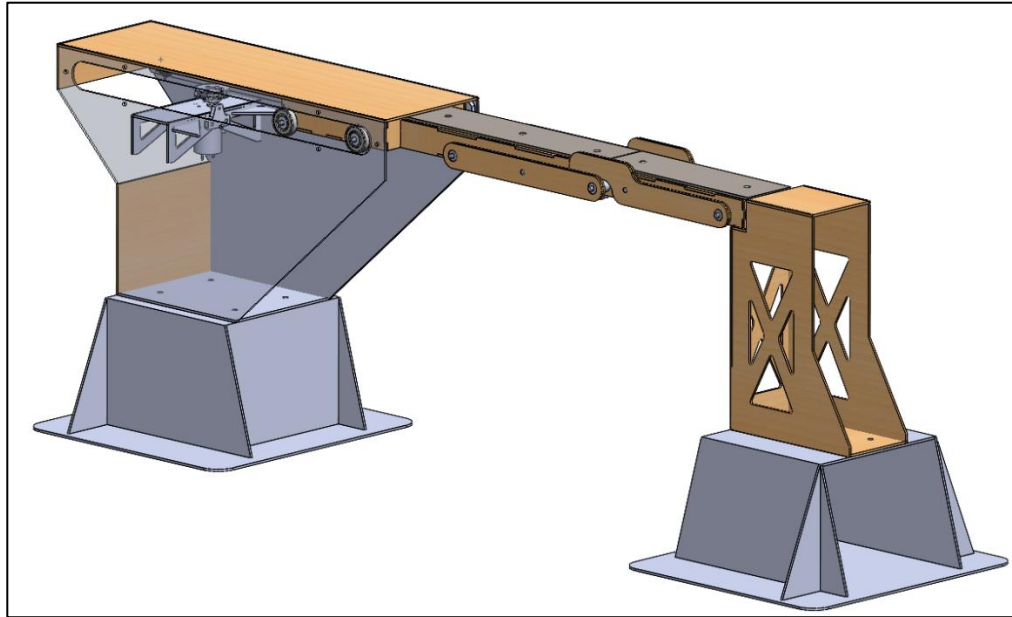


Figure 13: First Iteration of the Four-Bar Retracting Bridge

Upon building this design, several key issues became apparent. The main structure lacked sufficient rigidity and was noticeably flimsy during retraction. Additionally, the road decks tended to sag, resulting in an uneven roadway surface. With these challenges identified, the design progressed into a second iteration.

8.2 Second Iteration

To address the issues identified in the initial prototype, stiffening cross members were added to improve the overall rigidity of the main structure, and flanges were incorporated to counteract bending forces during the retraction process, resulting in smooth operation. The interface between the

two decks was also revised, making small geometry changes and implementing a sliding latch to eliminate sagging. This iteration is shown in Figure 14.

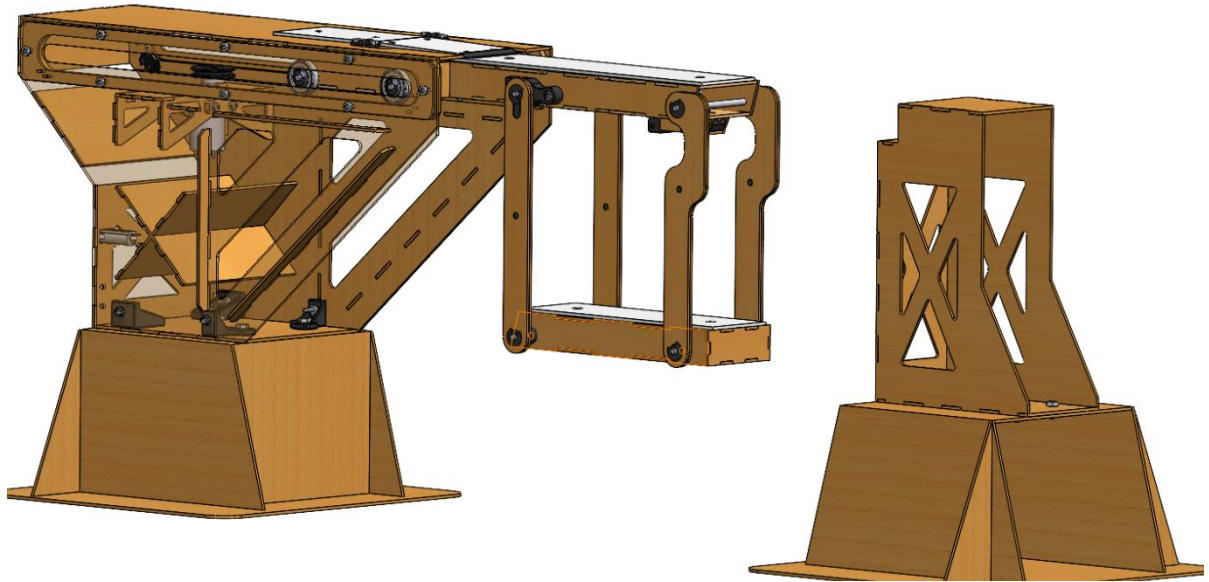


Figure 14: Second Iteration of the Four-Bar Retracting Bridge

Despite significant improvements to the overall stability of the bridge, the roadway continued to sag, and the decks occasionally bound during retraction. It was ultimately determined that these problems stemmed from the four-bar linkages being arranged in co-linear when fully extended. As the bridge retracted and the third deck was allowed to fall, play within the linkages caused the far edge of deck 3 to drop forward, causing the mechanism to bind rather than follow the intended path. In the final design, this issue was resolved by adjusting the links to no longer be co-linear when fully extended, ensuring the deck followed the desired path as predicted by calculations.

9.0 Final Design Description

After selecting the design that best fit the criteria, and iterating through initial prototypes, a final design was pursued.

9.1 Overview of Final Design

The final bridge design consists of a two-deck retracting system mounted on rigid upward vertical extensions from both pylons to near the maximum permitted height. From this elevated position, the bridge spans the gap by resting on the opposite pylon extension when fully extended. The system was designed to prioritize structural stability and reliable mechanical movement, while remaining within the design constraints. In the fully deployed state, the structure forms a continuous and smooth road surface composed of two main decks.

1. **Motor-side deck** – retracts horizontally via bearings within a track.
2. **Far deck** – connected via a four-bar linkage, allowing controlled motion during retraction.

The figures below show an overall view of the completed prototype in its fully extended configuration, as well as the retracted configuration with both decks vertically stowed.



Figure 15: Final bridge prototype in fully extended position



Figure 16: Final bridge prototype in fully retracted position

9.2 Operational Principles

The bridge is operated using a pulley system mechanism actuated by a single DC motor mounted on the left (motor-side) pylon extension. The actuation is built around a spool and string drive system, which pulls and retracts the motor-side deck along bearings riding inside a horizontal slotted track.

The motor-side deck has two parallel shafts inserted through the deck structure. These shafts sit inside the 3D printed bearings, which in turn sit inside the horizontal slotted rail of the pylon extension. This configuration allows the entire deck to slide backwards with very little friction when the motor is switched on. Once the motor is rotating, the spool begins to wind the string, and a horizontal pulling force is initiated. This force draws the deck backwards along the slotted track while staying aligned through the bearing-guided shafts.

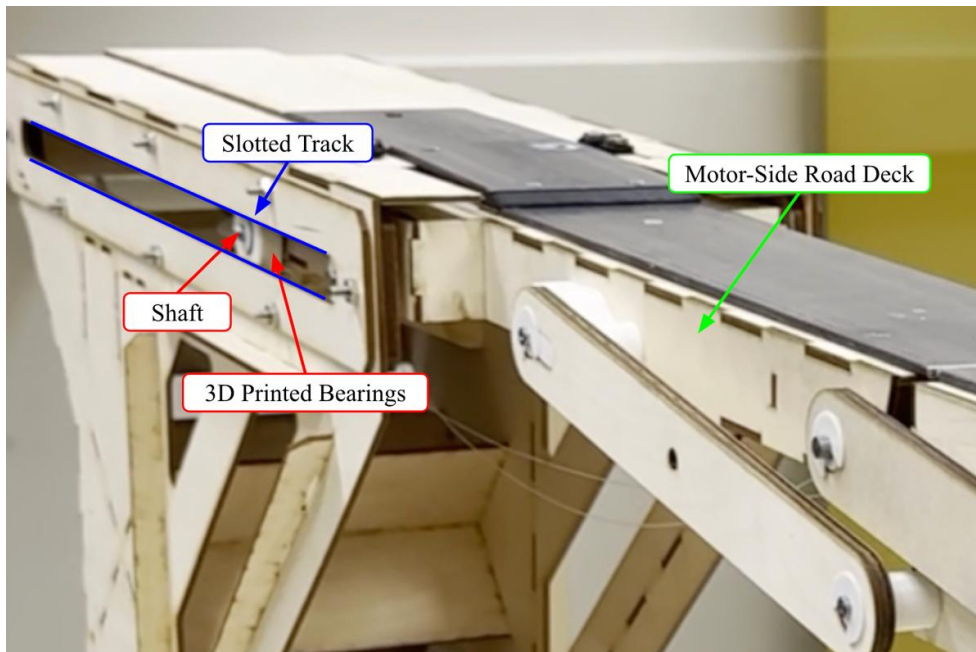


Figure 17: Motor-side deck showing shaft, bearings, and slotted track



Figure 18: Motor spool assembly with string routing to the deck

The far deck is connected to the motor-side deck through a four-bar linkage, which is what controls its rotation and folding motion during retraction. When the bridge is fully extended, the far deck rests on the opposite pylon and creates a level road deck. As soon as the motor-side deck begins to retract and the far deck slides off the support, gravity initiates the four-bar mechanism downwards. Friction dampers on the shafts of the four-bar mechanism control the rotation so that the deck does not swing aggressively, but instead in a controlled arc. Once rotated downward, the far deck hangs vertically beneath the motor-side deck as retraction continues.

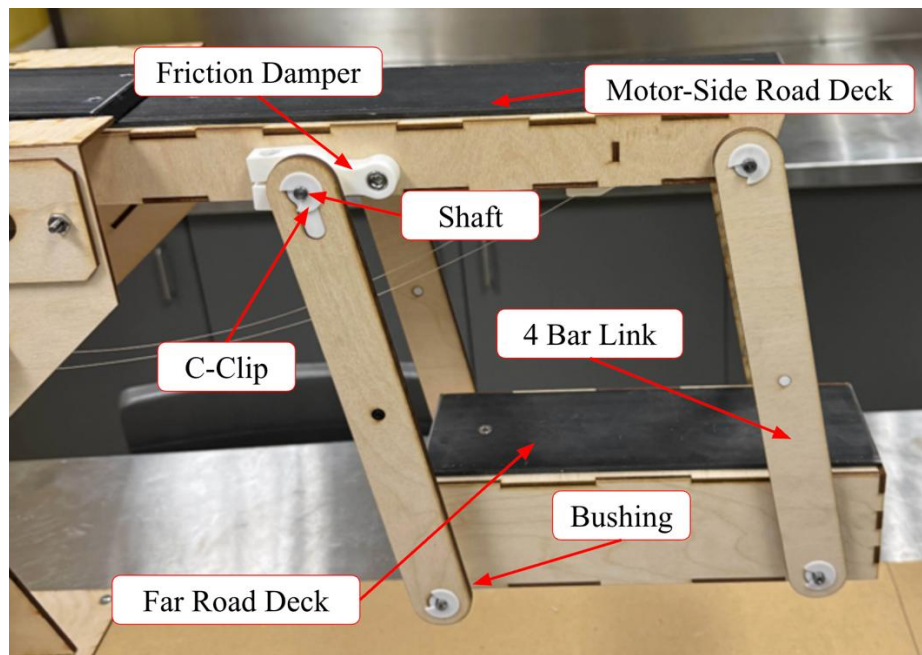


Figure 19: Four-bar mechanism connecting the two decks



Figure 20: Sequence of far deck rotation during retraction

As both decks retract into the main pylon structure, the motor-side deck eventually contacts the mounted limit switch of the motor circuit. This switch immediately cuts the power to the motor and prevents over-retraction.

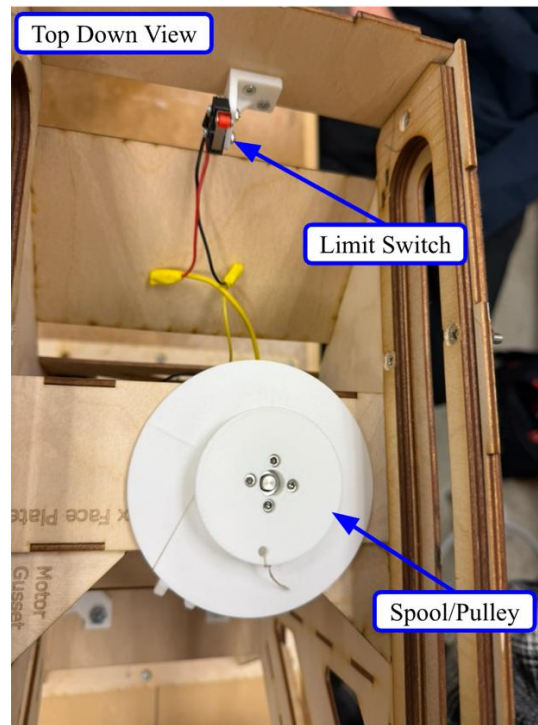


Figure 21: Limit Switch assembly used to end retraction motion

9.3 Prototype Construction Process

The final prototype was constructed using a combination of laser-cut plywood, 3D-printed components, off-the-shelf-fasteners, and machined steel shafts. The deck platforms, support pylons, and linkage arms were made from 3 mm plywood using a laser cutter to ensure dimensional accuracy and consistent fit across all components. These plywood parts were assembled using press-fit joints and reinforced with wood glue to increase structural rigidity.

Several components were 3D printed to support the systems described in Section 8.2. All rotational joints used 3D printed PLA bushings on steel shafts and retained with 3D printed C-clips,

allowing smooth pivoting and rotating. The same printing process was used to make the spool, pulley, and rolling bearings used within the motor-side deck's slotted track. The printed bearings were a cheap and low-friction option to achieve smooth sliding motion.

The friction-based dampers used in the four-bar mechanism were also 3D printed and assembled using a bolted clamping system. These dampers worked by clamping on a bushing that was keyed to the motion of the four-bar linkage. This simple design allowed the amount of damping to be easily tuned during construction by tightening the clamping fastener. The dampers were mounted directly onto the motor-side road deck.

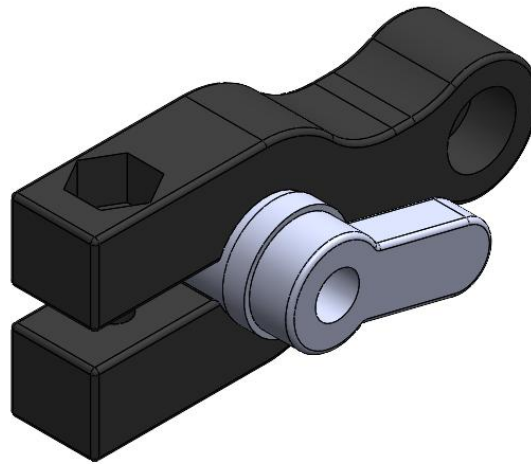


Figure 22: Friction-based Damper

Once the individual subassemblies were fabricated (decks, linkage arms, sliding track, pulley system, and dampers) the prototype was assembled inside the main pylon frame. The DC motor was mounted on the motor-side pylon extension, and the spool and string routing were installed. During assembly, there were two key challenges we faced. The joint between the two decks initially experienced minor buckling, which required tightening dimensional tolerances and adding

reinforcement to increase rigidity. The other challenge was that the first set of steel shafts were cut too short, leaving little room for the bushings and C-clips, requiring new shafts to be machined.

9.4 Engineering Analysis

The engineering analysis for the final design consisted of evaluation the kinematic behaviour of the four-bar mechanism, estimating structural and loading requirements for the shafts and other critical points, and assessing the motor torque required to retract the deck.

The graphical synthesis yielded a follower link length of approximately 9.5 inches and a coupler length of 7.5 inches parallel to the frame. The resulting mechanism geometry was modeled and verified using the PMKS+ Web tool to confirm the path met the design goal.

For our stress analysis, we wanted to ensure that two main points would hold up after prolonged use; the shafts, and the slot cut out on the pylon extension. The safety factor of the shaft was evaluated for shear as it experiences no axial loading and does not have a very large bending moment. The safety factor was determined to be 250, showing steel shafts were most likely not needed and a cheaper alternative could have been used. That being the case, it was the only material readily available from the machine shop for 1/4" shafts, making it the only choice. The top of the slot was another concern. Since the slot is so long and thin due to geometric constraints the most likely mode of failure is due to bending, thus making it the focus of analysis. The initial calculations, with the 3mm wide plywood, resulted in a safety factor of 1.875 which was too low. Increasing the width to 12mm resulted in a safety factor of 7.5 making it far more reliable. This was accomplished by bolting additional reinforcement, plywood slots, to the exterior and interior of the pylon.

The full calculation details are compiled in Appendix E.

10.0 Results & Performance

The following sections detail the testing results and overall performance of the four-bar retracting bridge and compare them to the objectives set at the beginning of the design process

10.1 Test Plan

A test plan was developed to determine performance of the bridge.

10.1.1 Scope

The goal of this test is to verify the performance of the completed retractable bridge prototype relative to the project's required functional, geometric, durability, and safety objectives. These objectives correspond directly to the constraints outlined in Table 1. The following aspects will be assessed:

1. **Span Performance:** Confirm the bridge spans 32 inches between pilons (Obj. #1)
2. **Deployed Height:** Confirm maximum height does not exceed 12 inches (Obj. #2)
3. **Deck Geometry:** Measure deck width ≤ 9 inches with a clear roadway ≥ 3 inches (Obj. #3). Measure deck flatness $\leq 1/16$ -inch deviation (Obj. #4).
4. **Load Capacity:** Confirm deck supports full steel plate mass without failure (Obj. #5).
5. **Durability:** Test whether the bridge can deploy and retract manually for ≥ 12 full cycles without noticeable damage (Obj. #6).
6. **Ease of Actuation:** Measure required pull force to initiate retraction; must be ≤ 20 N (Obj. #7).
7. **Safety:** Confirm absence of hazardous sharp edges or pinch points (Obj. #9).
8. **Retracted Geometry:**

- a. Retracted structure must not extend more than 4 inches behind the pylon (Obj. #11).
- b. Must maintain ≥ 12 inches of horizontal clearance in the center gap (Obj. #12).
- c. No part of the underside may extend below the pylon top (0") at any point (Obj. #13).

9. Aesthetic & Finish: Qualitative assessment of smooth finish, cleanliness, and organized construction (Obj. #14).

Hypothesis

It is expected that the bridge will fully retract successfully into a stowed configuration, and meet all major functional objectives related to motion and structural ability. The relevant parameters for the testing include the deck's travel distance, the four bar linkage motion, and any interference during retraction.

10.1.2 Design of the Test

- **Date of Testing:** Monday, November 24, 2025
- **Location:** ELW B118 (Design Lab)
- **Client/Organization:** MECH 350
- **Test Conducted By:** MECH 350 Group 21

Test Method

A combination of dimensional inspection, functional performance testing, and durability cycling will be used. This aligns with Lecture 15 recommendations for testing engineering prototypes [4].

Required Equipment

- Tape measure (± 1 mm accuracy)

- Steel road deck plates (provided)
- Straightedge or calipers for deck flatness
- Spring scale (0-50N) to measure pull force
- Camera – to document test runs

Independent Variables

- Starting condition of the bridge: Full extended
- Applied load: steel plate load

Dependent Variables

- Span length
- Deployed height
- Deck width
- Flatness deviation
- Load capacity
- Number of completed cycles
- Pull force required
- Retracted geometry
- Qualitative safety and aesthetics

Number of Trials

- 3 trials per measurement for dimensional checks
- 12 full cycles for durability (as required by Obj. #6)

10.1.3 Test Procedure

Dimensional & Geometric Testing:

1. Place bridge in full deployed position
2. Measure and record:
 - a. Span length across pilons
 - b. Maximum height of any component
 - c. Total deck width and clear roadway width
 - d. Flatness variation using a straightedge over steel plates
3. Place steel plates on the deck and observe any sagging or failure.

Functional Testing: Retraction

1. Reset the bridge into fully deployed configuration.
2. Trigger mechanism (turn motor on).
3. Use a spring scale to measure the required pull force.
4. Observe:
 - a. Whether mechanism initiates retraction
 - b. Speed/behavior of the four-bar folding
 - c. Whether bridge fully retracts without obstruction

Durability Testing (Cycle Life)

1. Manually reset the bridge to fully deployed position after each retraction.
2. Complete 12 full cycles:
 - a. Deploy (manual) → Retract (mechanical)
3. Record any wear, deformation, or performance degradation.

Safety & Aesthetic Evaluation

1. Visually inspect for sharp edges, pinch points, or dangerous features.
2. Evaluate cleanliness, organization, and finish quality of the bridge.

10.2 Observations & Data

This section summarizes the raw observational notes and recorded measurements from the testing process. Quantitative values are entered into the tables provided below, while qualitative observations are bullet-pointed in the space provided.

Table 10: Dimensional Measurements

Measurement	Target Value	Measured Value
Bridge Span	32"	32"
Max deployed height	$\leq 12"$	11.875"
Deck width (total)	$\leq 9"$	7.70"
Roadway width	$\geq 3"$	3"
Deck flatness	$\leq 1/16"$	1/32"
Retracted length behind pylon	$\leq 4"$	4"
Retracted center clearance	$\geq 12"$	12.25"
Underside clearance limit	0"	0"

Table 11: Load Capacity Data

Measurement	Target Value	Measured Value
Deck supports steel	$\geq \sim 4$ lbs	9 lbs

Table 12: Actuation Force Data

Measurement	Target Value	Measured Value
Pull force for retraction	≤ 20 N	5.34 N

Table 13: Durability Data

Cycle Number	Retracted Successfully?	Issues Observed
1	Yes	N/A
2	Yes	N/A
3	Yes	N/A
4	Yes	N/A
5	Yes	N/A
6	Yes	N/A
7	Yes	N/A
8	Yes	N/A
9	Yes	N/A
10	Yes	N/A
11	Yes	N/A
12	Yes	N/A

Table 14: Safety & Aesthetic Evaluation

Aspect	Criteria	Observed Result
Safety	No sharp edges present, pinch points minimized, controlled retraction.	Good
Aesthetic finish clean/organized	Smooth laser-cut surfaces, clean assembly, organized structure with no loose or unfinished components.	Excellent

10.3 Test Results Table

Table 15: Results Summary Table

Objective	Test Result	Objective Met?
32" bridge span	32"	Yes
Max height $\leq 12"$	11.875"	Yes
Deck width $\leq 9"$	7.70"	Yes
Roadway $\geq 3"$	3"	Yes
Deck flatness $\leq 1/16"$	1/32"	Yes
Load ≥ 4 lbs	9 lbs	Yes
Cycle life ≥ 12 cycles	30 Cycles	Yes
Actuation force ≤ 20 N	5.34 N	Yes
Safety (limited pinch/sharp)	Good	Yes
Retracted length $\leq 4"$	4"	Yes
Center clearance $\geq 12"$	12.25"	Yes
Underside clearance = 0"	0"	Yes
Aesthetic finish clean/organized	Excellent	Yes

10.4 Analysis & Interpretation

The testing results showed that the final prototype met all performance objectives and behaved as predicted. The bridge spanned the full 32" gap and remained below the 12" height limit, confirming that the design and geometry was accurately translated from the concept to the CAD, to the physical build. Also, the load capacity was measured more than double the requirement which shows the improved rigidity that was achieved through iterations. The prototype passed a 30-cycle reliability test which shows that the combinations of shafts, bushings, dampers, and other mechanisms proved to be consistent and reliable. The low actuation forces demonstrates that friction was minimized through aligned of the sliding bushings and main structure. Finally, safety and aesthetic objectives were all satisfied, showing that the physical assembly was well-finished, organized, and professional.

11.0 Future Work Recommendations

This section presents possible improvements to the overall design, as well as a reflection on what would be done differently if the project were to be done again.

11.1 Prototype Improvements

There are a few adjustments that we would make to improve the performance and reliability of the current prototype, without changing the overall design. First, increasing the rigidity in the deck and pylon extensions by tightening tolerances or adding more reinforcements might prevent any deflection during the retraction, which was something observed during our durability testing. Secondly, the sliding bushings in the motor-side track could be improved by slightly tightening the tolerance to reduce any play and improve the decks flatness. Lastly, managing the rope slack is another practical improvement; by adding small underside rope guides we could keep the line aligned during operation and prevent any slack from tangling in the moving parts.

11.2 Redesign Recommendation

If the bridge were to be redesigned from the ground up, there are a few conceptual changes that we would implement. The main structure pylon extension could be reconfigured using a thinner beam and column design rather than large plywood plates, reducing the material costs and waste, while still maintaining stiffness. Also, optimizing the overall bridge height relative to the four-bar linkage length would minimize the theoretical ramp height from a road and reduce material use.

12.0 Conclusions

As Group 21, we completed a three-month design process focused on meeting the client's goal of developing a retractable bridge. Our final design successfully met all design objectives and constraints set by the client. Through iterative design, analysis and testing, a final retractable bridge prototype was delivered and presented to the client on the 1st of December 2025. The presentation outlined the bridge's main mechanism, which includes a motor-driven pulley system and a four-bar mechanism that lowers the deck. While the prototype performs as intended, several recommendations are outlined in section 10 for future improvements. Overall, the completed design provides a strong foundation for future development if needed. Ultimately, this prototype can now be scaled and further refined for real-world applications.

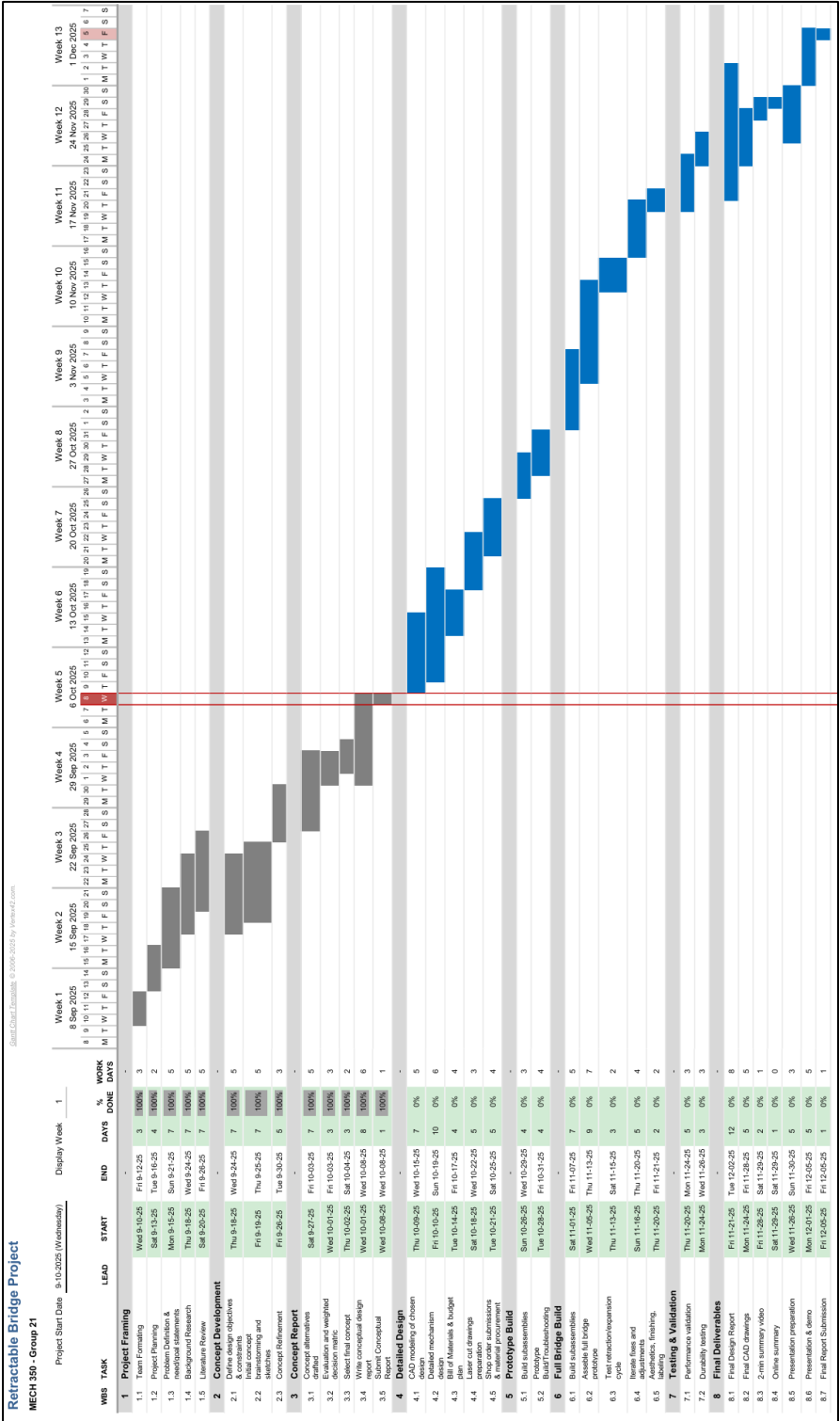
References

- [1] S. B. Partner, "Three-Segment Folding Bridge Kieler Hörn," [Online]. Available: <https://www.sbp.de/en/project/three-segment-folding-bridge-kieler-hoern/>.
- [2] Heatherwick, "Rolling Bridge," Heatherwick, [Online]. Available: <https://heatherwick.com/project/rolling-bridge>.
- [3] Atelier Kempe Thill, "Folding Bridge," Atelier Kempe Thill Studio, [Online]. Available: <https://atelierkempethill.com/projects/0028-folding-bridge>.
- [4] N. Dechev, "Major Design Project Description," MECH 350, 2025.
- [5] thang010146, "Stretch and contraction mechanism 1," YouTube, 25 March 2012. [Online]. Available: <https://www.youtube.com/watch?v=4UpjmxQ3900>.
- [6] Slant 3D, "Print in Place Latch Mechanism | Design for Mass Production 3D Printing," YouTube.com, 2 July 2024. [Online]. Available: <https://www.youtube.com/watch?v=4wRKmcWXEkW>. [Accessed 7 October 2025].
- [7] OffGrid Workshop, "3D Printed Latch Mechanism," YouTube.com, 8 September 2024. [Online]. Available: <https://www.youtube.com/watch?v=dAaK9dWrCCM>. [Accessed 7 October 2025].
- [8] P. Lumber, "6mm - 5x5 Baltic Birch," [Online]. Available: Peacock Lumber Ltd., "Baltic Birch." [Online]. Available: https://www.peacocklumber.ca/c/baltic-birch?srsId=AfmBOoo9OiV5R0nGue12bK1CU_Yr8c54x2BfvNzoNnHB0k4BMxg6E4Zx (accessed Dec. 8, 2025).. [Accessed 8 December 2025].

- [9] P. Lumber, "3mm - 5x5 Baltic Birch BB/BB," [Online]. Available: <https://www.peacocklumber.ca/p/3mm-5x5-baltic-birch-bbbb/PBB18>. [Accessed 8 December 2025].
- [10] OnlineMetals.com, "1/4" Carbon Steel Round Bar 12L14 Cold Drawn," [Online]. Available: https://www.onlinemetals.com/en/buy/carbon-steel/0-25-carbon-steel-round-bar-12l14-cold-drawn/pid/7575?srsltid=AfmBOop9maM_Wn1NWrgFuWXRGSjhAXh3LBIhZhR_Jjg2xmm8IRL7f705. [Accessed 8 December 2025].
- [11] OnlineMetals.com, "1/8" x 3" Carbon Steel Rectangle Bar Commercial Quality Hot Rolled," [Online]. Available: https://www.onlinemetals.com/en/buy/carbon-steel/0-125-x-3-carbon-steel-rectangle-bar-commercial-quality-hot-rolled/pid/10007?srsltid=AfmBOor_2M2OC82K7f1kl5KPbMCTf7NvaPLaDYXrd_0YvCAcFhvT5IH8. [Accessed 8 December 2025].
- [12] DigiKey, "W28-XQ1A-13," [Online]. Available: https://www.digikey.ca/en/products/detail/te-connectivity-potter-brumfield-relays/W28-XQ1A-13/1128429?gclsrc=aw.ds&gad_source=1&gad_campaignid=17336435733&gbraid=0AAAAADrbLlj2jy5JJ8q97eX1B_impYi36&gclid=Cj0KCQiA6NTJBhDEARIsAB7QHD32cuNzZkd1VBmSUZhKAS2RGLaw. [Accessed 8 December 2025].
- [13] S. Plus, "2.1mm Female DC (DC14) Power Jack Plug Adapter," [Online]. Available: <https://www.shopperplus.ca/p-311009-ada-2-1mm-dcjack-f-21mm-dc-jack-female-adaptor-21mm-dc-plug-dc14-primecables#sku311009>. [Accessed 8 December 2025].

- [14] DigiKey, "SS-5GL2-F-3T," [Online]. Available:
<https://www.digikey.com/en/products/detail/omron-electronics-inc-emc-div/SS-5GL2-F-3T/5236958>. [Accessed 8 December 2025].
- [15] DigiKey, "30SQ045," [Online]. Available:
https://www.digikey.ca/en/products/detail/anbon-semiconductor-int-l-limited/30SQ045/18801010?gclsrc=aw.ds&gad_source=1&gad_campaignid=17336435733&gbraid=0AAAAADrbLlj2jy5JJ8q97eX1B_impYi36&gclid=Cj0KCQiA6NTJBhDEARIsAB7QHD1uTH8YhdcOPCSM-250T04lQhn8lPw0ylexK. [Accessed 8 December 2025].
- [16] DigiKey, "SW-T2-4B-B-A2-MA2," [Online]. Available:
https://www.digikey.ca/en/products/detail/adam-tech/SW-T2-4B-B-A2-MA2/15284424?gclsrc=aw.ds&gad_source=1&gad_campaignid=17336435733&gbraid=0AAAAADrbLlj2jy5JJ8q97eX1B_impYi36&gclid=Cj0KCQiA6NTJBhDEARIsAB7QHD2ljUzm0Kx7B1qiM4Mu4LvtYeHMSRtLx8_YeFMRbZomtxQ0DhN. [Accessed 8 December 2025].
- [17] RobotShop, "12V 50RPM 694 oz-in Brushed DC Motor," [Online]. Available:
<https://www.robotshop.com/products/12v-50rpm-694-oz-in-brushed-dc-motor>. [Accessed 8 December 2025].

Appendix A – Gantt Chart



Appendix B – Bill of Materials

This section contains the complete bill of materials for the final iteration of the bridge design.

Item #	Part Name	Description	Qty.
1	Mech350_Pylon	Pylon	2
2	Mech350_SidePlate	Static Structure Side Plate	2
3	Mech350_bottomPlate	Static Structure Bottom Plate	1
4	Mech350_BackPlate	Static Structure Rear Bottom Plate	1
5	Mech350_TopPlate	Static Structure Top Plate	1
6	Mech350_Deck2SideRear	Deck 2 Rear Side Plate	2
7	Mech350_Deck2Side	Deck 2 Front Side Plate	2
8	Mech350_Deck2Front	Deck 2 Front Plate	1
9	steel plate	Deck 2 Road Surface	1
10	Mech350_Deck2Deck2	Deck 2 Top Plate	1
11	Mech350_Deck2FrontSlot	Deck 2 Front Stiffener	1
12	Mech350_Deck2Bottom	Deck 2 Bottom Plate	1
13	92141A009	18-8 Stainless Steel Washer	14
14	90480A009	Low-Strength Steel Hex Nut	30
15	91253A196	Black-Oxide Alloy Steel Hex Drive Flat Head Screw	20
16	Mech350_Deck2CrossMember	Deck 2 Support Plate	1
17	Mech350_BackPlate2	Static Structure Top Rear Plate	1
18	Mech350_4BarThin	4-Bar Linkage Bar	4
19	Mech350_4BarBushing	4-Bar Keyed Linkage Bushing	2
20	Mech350_4BarBushingRound	4-Bar Long Linkage Bushing	2
21	Mech350_4BarBushingRound	4-Bar Linkage Dowel Pin	4
22	Mech350_BendyLinkThin	4-Bar Linage Far	4
23	Mech350_4BarBushingShort	4-Bar Short Linkage Bushing	4
24	Mech350_MedShaft	4-Bar Short Shaft	2
25	Mech350_LongShaft	4-Bar Long Shaft	2
26	Mech350_StructureBasePlate	Static Pylon Base Plate	1
27	Mech350_StructurePlate	Static Pylon Side Plate	2
28	Mech350_StructurePlateTop	Static Pylon Top Plate	1
29	Mech350_StructurePlateShort	Static Pylon Face Plate	1
30	Mech350_StructurePlateShort	Static Pylon Rub Plate	1
31	Mech350_Deck3Bottom	Deck 3 Bottom Plate	1
32	Mech350_Deck3Side	Deck 3 Side Plate	2
33	Mech350_Deck3End	Deck 3 End Plate	2
34	Mech350_Deck3Top	Deck 3 Top Plate	1
35	steel plate 10	Deck 3 Road Deck	1
36	BEARING BUSH	Deck 2 Roller	4
37	Mech350_SlotReinforcement	Static Structure Slot Reinforcement Inside	2
38	Mech350_BearingBushing	Deck 2 Roller Bushing	4
39	Mech350_SlotReinforcementThin	Static Structure Slot Reinforcement Outside	2
40	Mech350_BearingShaft	Deck 2 Roller Shaft	2
41	Mech350_MotorBoxFacePlat	Motor Support Face Plate	1

42	Mech350_Motor	12V DC Motor	1
43	Motor_Coupling	DC Motor Shaft Adapter	1
44	Mech350_MotorSupport	Motor Support Plate	2
45	Mech350_MotorPulley	Main Pulley	1
46	Mech350_TieOff	Rope Tie Point	1
47	Mech350_MotorGusset	Motor Support Gusset	2
48	MX11 Limit Switch	Limit Switch	1
49	Mech350_SwitchSpacer	Limit Switch Mount	1
50	7347K57	Miniature Rounded Toggle Switch	1
51	Mech350_SnapRing	Snap Ring	8
52	Mech350_SnapRingSmall	Small Snap Ring	4
53	Mech350_HingeFemale	Angle Deck Female Hinge	2
54	Mech350_HingeMale	Angle Deck Male Hinge	2
55	Mech350_AngleDeck	Angle Deck Plate	1
56	Mech350_AngledSteel	Angle Deck Plate	1
57	Mech350_SteelMain	Static Structure Road Deck	1
58	Mech350_HingePin	Angle Deck Hinge Pin	2
59	Mech350_DeckDowel	Road Deck Dowel Pin	4
60	90480A007	Low-Strength Steel Hex Nut	4
61	91253A146	Black-Oxide Alloy Steel Hex Drive Flat Head Screw	4
62	91290A103	Alloy Steel Socket Head Screw	4
63	91251A197	Black-Oxide Alloy Steel Socket Head Screw	4
64	91251A540	Black-Oxide Alloy Steel Socket Head Screw	2
65	95462A029	Medium-Strength Steel Hex Nut	9
66	92865A537	Medium-Strength Grade 5 Steel Hex Head Screw	8
67	91274A103	Zinc-Flake-Coated Alloy Steel Socket Head Screw	6
68	Mech350_Wedge	Angle Deck Wedge	1
69	Mech350_Stiff1	Static Structure X Brace Center	1
70	Stiff2	Static Structure X Brace Side	2
71	Mech350_Stiffener3	Static Structure Side Stiffener	2
72	Mech350_Stiffener4	Static Structure Side Stiffener	2
73	Mech350_Stiffener5	Static Structure Side Stiffener	2
74	Mech350_RearPlate	Static Structure Angled Rear Plate	1
75	Mech350_Bolt Support	Static Structure Bolt Support	2
76	91251A199	Black-Oxide Alloy Steel Socket Head Screw	4
77	Mech350_BoltSupport2	Static Structure Bolt Support RH	1
78	Mech350_BoltSupport3	Static Structure Bolt Support LH	1
79	90294A108	Phillips Flat Head Screws for Wood	4
80	Mech350_DeckRiser	Main Structure Wood Deck	1
81	1393250_Circuit_Breaker.stp	Circuit Breaker	1
82	Part3^1. Mech350_Bridge	Power Supply Jack	1
83	Mech350_Damper	Linkage Damper	2
84	Mech350_StaticPylonRear	Static Pylon Back Plate	1
85	Mech350_RearPlate2	Static Pylon Small Back Plate	1
86	Mech350_Latch	Secondary Pulley Mount	1
87	Mech350_LatchPulley	Secondary Pulley	1
88	Mech350_LatchPin	Secondary Pulley Shaft	1
89	Mech350_WedgeRiser	Wedge Riser	1

Appendix C – Costing Breakdown

The costing of the final bridge design has been based on online prices for available parts and does not include machine time or man hours for manufactured parts. Prices are in Canadian dollars. There were five major classes of parts: 3D printed, 3mm plywood, 6mm plywood, hardware, and miscellaneous parts. The costs of the materials [8] [9] [10] [11] were estimated using the values in Table 16.

Table 16: Material Costs

Material Prices		
Material	Unit	Price/Unit
PLA	mL	0.011
3mm Plywood	in ²	0.008
6mm Plywood	in ²	0.012
1/4" 12L14 Shaft	in	0.199
1/8" x 3" Steel Bar	in	0.560

Starting with the 3D printed parts, the cost estimate was based on the Bambu labs software cost estimate feature along with the price per volume of PLA for parts that were not costed during printing. The cost of the 3D printed parts were found to be as outlined in Table 17. The total volume of the printed parts came out just under 195mL.

Table 17: 3D Printed Part Costing

3D Prints				
Part:	mL:	Cost/unit:	QTY:	Total Cost:
Mech350_4BarBushing	3.93	0.08	2	\$0.160
Mech350_4BarBushingRound	3.61	0.08	2	\$0.160
Mech350_4BarDowel	0.16	0.01	4	\$0.040
Mech350_4BarBushingShort	1.31	0.04	4	\$0.160
BEARING BUSH	6.68	0.07	4	\$0.298
Mech350_BearingBushing	2.46	0.06	4	\$0.240
Mech350_MotorPulley	34.90	0.50	1	\$0.500
Mech350_SwitchSpacer	2.95	0.11	1	\$0.110
Mech350_SnapRing	0.16	0.01	8	\$0.080
Mech350_SnapRingSmall	0.16	0.01	4	\$0.040
Mech350_HingeFemale	0.66	0.03	2	\$0.060
Mech350_HingeMale	0.49	0.03	2	\$0.060
Mech350_HingePin	0.16	0.01	2	\$0.020
Mech350_DeckDowel	0.16	0.01	4	\$0.040
Mech350_Wedge	4.10	0.05	1	\$0.046
Mech350_Bolt Support	16.87	0.23	2	\$0.460
Mech350_BoltSupport2	8.85	0.21	1	\$0.210
Mech350_BoltSupport3	8.85	0.21	1	\$0.210
Mech350_Damper	13.11	0.15	2	\$0.292
Mech350_Latch	9.83	0.11	1	\$0.110
Mech350_LatchPulley	0.82	0.03	1	\$0.030
Mech350_LatchPin	0.33	0.01	1	\$0.010

The laser cut plywood parts were costed next using the divided cost of a 5'x5' sheet of plywood. The true surface area of the parts was used for the purpose of costing, not considering the surface area of the inside cutouts. The 3mm plywood parts made up most of the bridge. Using the above cost per square inch yields the cost in Table 18.

Table 18: 3mm Part Costing

3mm Plywood				
Part:	in^2:	Cost/unit:	QTY:	Total Cost:
Mech350_SidePlate	134.7	1.046	2	\$2.09
Mech350_bottomPlate	50.2	0.390	1	\$0.39
Mech350_BackPlate	25.7	0.200	1	\$0.20
Mech350_TopPlate	120.8	0.938	1	\$0.94
Mech350_Deck2SideRear	16	0.124	2	\$0.25
Mech350_Deck2Side	8.6	0.067	2	\$0.13
Mech350_Deck2Front	6.9	0.054	1	\$0.05
Mech350_Deck2Deck2	80.6	0.626	1	\$0.63
Mech350_Deck2FrontSlot	6.7	0.052	1	\$0.05
Mech350_Deck2Bottom	70.8	0.550	1	\$0.55
Mech350_Deck2CrossMember	3.9	0.030	1	\$0.03
Mech350_BackPlate2	13.9	0.108	1	\$0.11
Mech350_4BarThin	12.4	0.096	4	\$0.39
Mech350_BendyLinkThin	12.6	0.098	4	\$0.39
Mech350_StructureBasePlate	32.4	0.252	1	\$0.25
Mech350_StructurePlate	57.7	0.448	2	\$0.90
Mech350_StructurePlateTop	14.9	0.116	1	\$0.12
Mech350_StructurePlateShort	26.3	0.204	1	\$0.20
Mech350_StructurePlateRub	3.3	0.026	1	\$0.03
Mech350_Deck3Bottom	26.2	0.203	1	\$0.20
Mech350_Deck3Side	25.4	0.197	2	\$0.39
Mech350_Deck3End	8.9	0.069	2	\$0.14
Mech350_Deck3Top	32.3	0.251	1	\$0.25
Mech350_SlotReinforcementThin	35.1	0.273	2	\$0.55
Mech350_MotorBoxFacePlat	15	0.116	1	\$0.12
Mech350_MotorSupport	7.3	0.057	2	\$0.11
Mech350_MotorGusset	2.2	0.017	2	\$0.03
Mech350_AngleDeck	5.3	0.041	1	\$0.04
Mech350_Stiff1	28.5	0.221	1	\$0.22
Stiff2	15.8	0.123	2	\$0.25
Mech350_Stiffener3	5.7	0.044	2	\$0.09
Mech350_Stiffener4	5.7	0.044	2	\$0.09
Mech350_Stiffener5	5.7	0.044	2	\$0.09
Mech350_RearPlate	24	0.186	1	\$0.19
Mech350_DeckRiser	35	0.272	1	\$0.27
Mech350_StaticPylonRear	12	0.093	1	\$0.09
Mech350_RearPlate2	5.3	0.041	1	\$0.04
Mech350_WedgeRiser	2.2	0.017	1	\$0.02

Although lesser used, 6mm plywood parts were also used in the final design for high load areas. The same costing process was used as the 3mm plywood and yielded the results of Table 19.

Table 19: 6mm Plywood Costing

6mm Plywood				
Part:	in^2	Cost/unit:	QTY:	Total Cost:
Mech350_SlotReinforcement	20.4	0.249	2	\$0.50
Mech350_TieOff	1.1	0.013	1	\$0.01

The hardware used in the bridge was a considerable portion of the final cost. These parts were acquired from the UVic machine shop but were costed using divided McMaster Carr prices as pricing was not available from the shop. The prices are outlined in Table 20.

Table 20: Hardware Costing

Hardware				
Part:	McMaster #	Cost/unit:	QTY:	Total Cost:
No. 8 Flat Washer	92141A009	0.0207	14	0.290
8-32 Hex Nut	90480A009	0.0202	30	0.606
8-32 x 5/8 FHCS	91253A196	0.208	20	4.160
6-32 Hex Nut	90480A007	0.0157	4	0.063
6-32 x 3/8 FHCS	91253A146	0.1278	4	0.511
M2.5 x 0.4mm x 10mm SHCS	91290A103	0.123	4	0.492
8-32 x 3/4" SHCS	91251A197	0.1868	4	0.747
1/4"-20 x 3/4" SHCS	91251A540	0.2276	2	0.455
1/4"-20 Hex Nut	95462A029	0.0632	9	0.569
1/4"-20 x 1/2" HHCS	92865A537	0.1277	8	1.022
4-40 x 1/2" SHCS	91251A110	0.123	6	0.738
M3 x 0.5mm x 8mm SHCS	91274A103	0.14	4	0.560
No. 4 Flat Head Wood Screw	90294A108	0.049	4	0.196

Finally, miscellaneous parts were considered as a group. This includes the provided electronic components along with the shafts and steel plates manufactured by the machine shops. Electronics were costed using both provided prices and estimating based on available part numbers [12] [13] [14] [15] [16] [17]. The shafts and steel plates were costed using the price per length of the material. Table 21 outlines the miscellaneous part cost.

Table 21: Misc. Parts Costing

Misc. Parts				
Part:	Part Number/in:	Cost/unit:	QTY:	Total Cost:
Circuit Breaker	W28-XQ1A-13	8.73	1	8.73
Power Cord Jack	Ada-2-1mm-DCJack-F	1.49	1	1.49
Limit Switch	Z5083-ND	2.82	1	2.82
Diode	30SQ045	2.03	1	2.03
Rocker Switch	SW-T2-4B-B-A2-MA2	1.92	1	1.92
Motor	FIT0492-A	23.06	1	23.06
Wire	N/A	0.25	12	3
Static Deck Steel	5.25	2.940	1	2.94
Angled Steel	4.44	2.486	1	2.49
Deck 2 Steel	13.5	7.560	1	7.56
Deck 3 Steel	10	5.600	1	5.60
Medium Shaft	4.56	0.906	2	1.81
Long Shaft	5.54	1.101	2	2.20
Bearing Shaft	6.78	1.347	4	5.39

Combining these costs a total cost was obtained of \$121.14. The final breakdown of each type of part, along with the total, can be found in Table 22.

Table 22: Total Bridge Cost

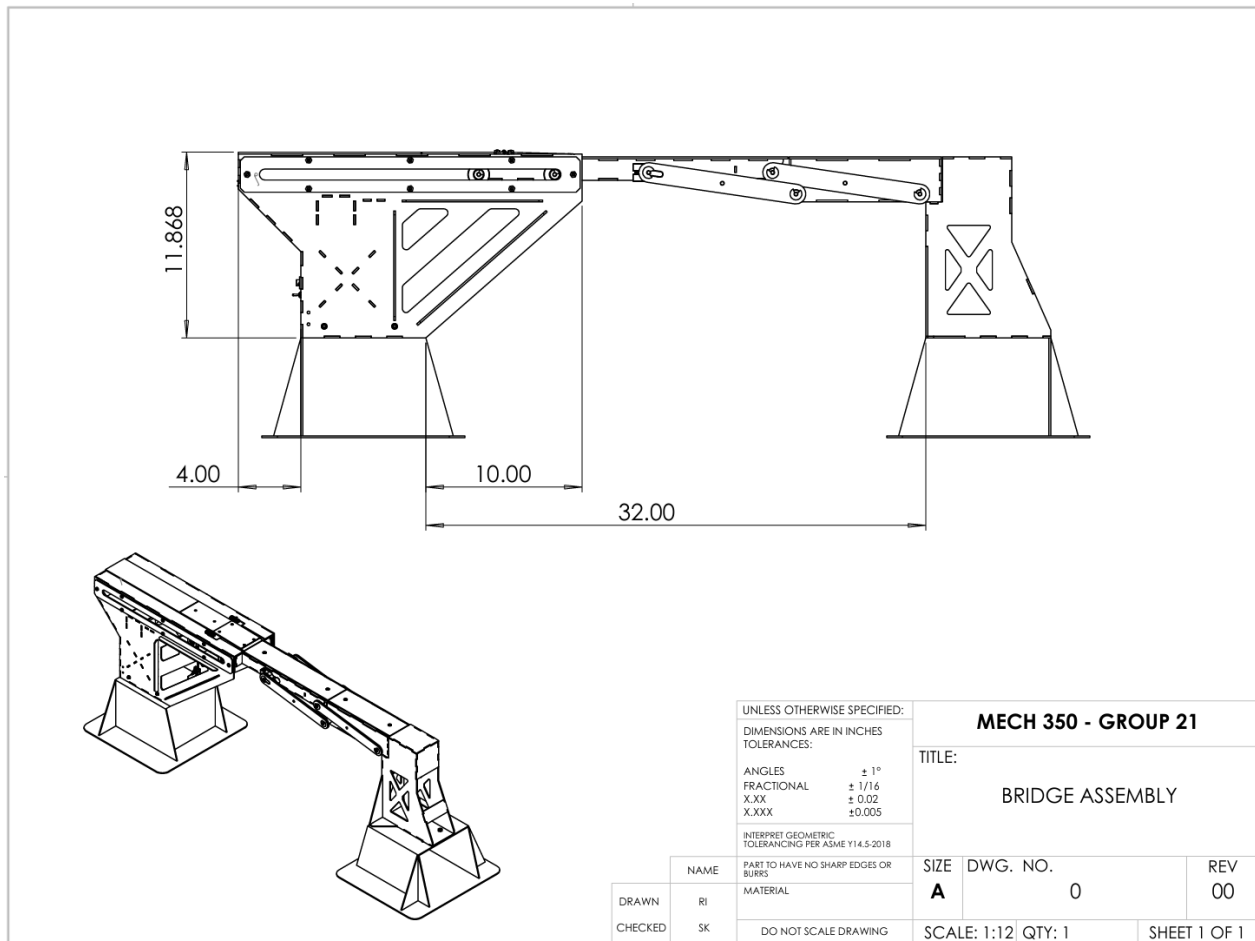
Cost Totals				
3D Prints	3mm Plywood	6mm Plywood	McMaster Parts	Misc. Parts
\$3.34	\$10.87	\$0.51	\$35.38	\$71.04
Total:				\$121.14

With a budget constraint of \$250, this left us well under the maximum dollar amount for the bridge when considering the final design.

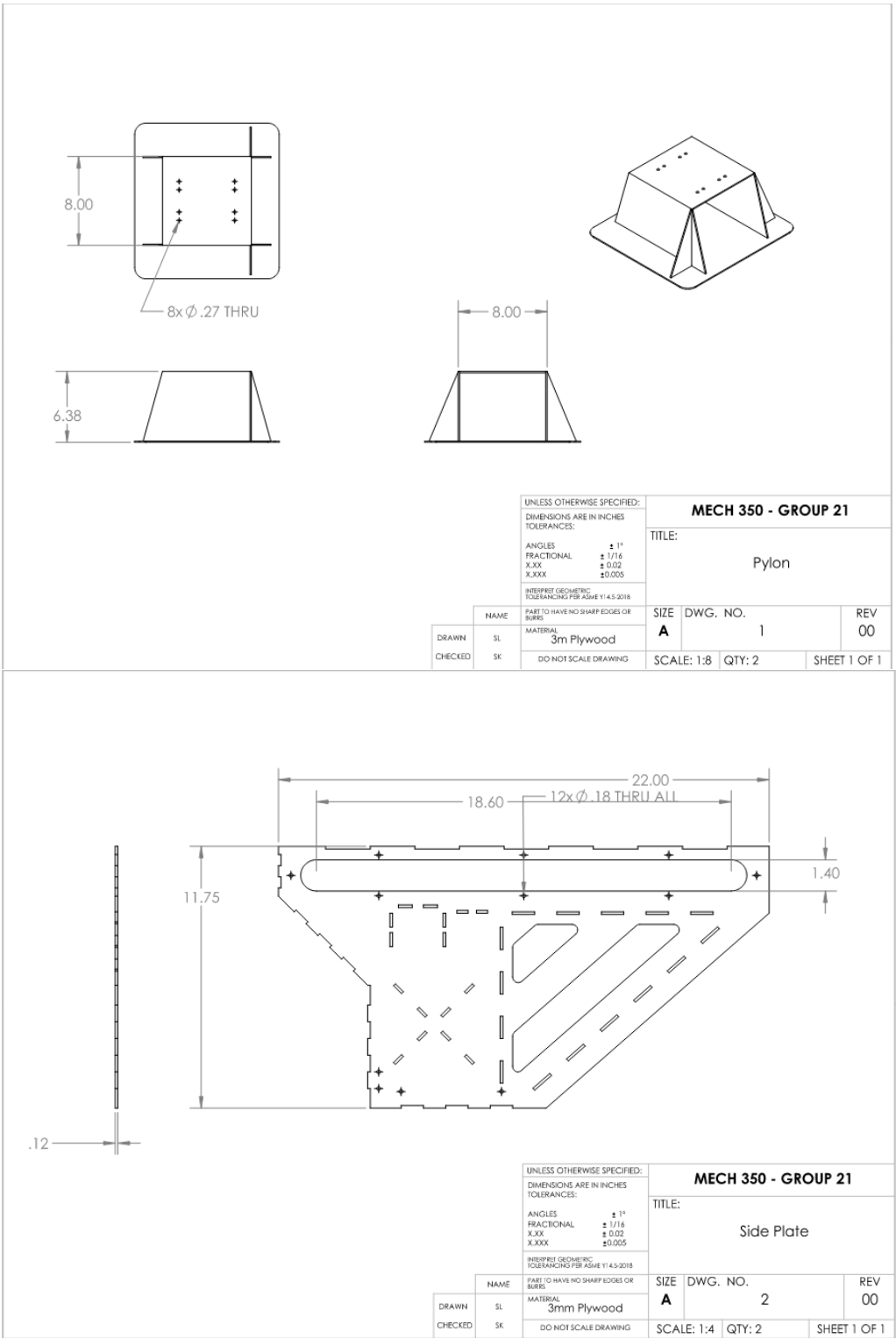
Appendix D – Drawings

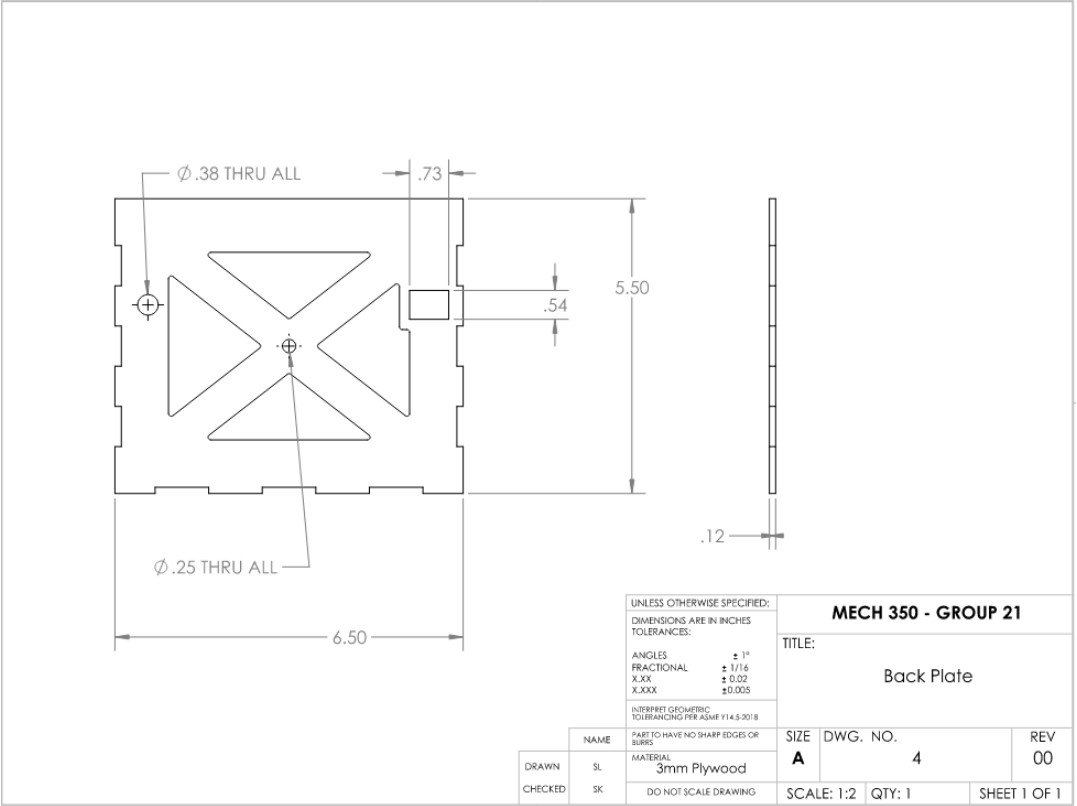
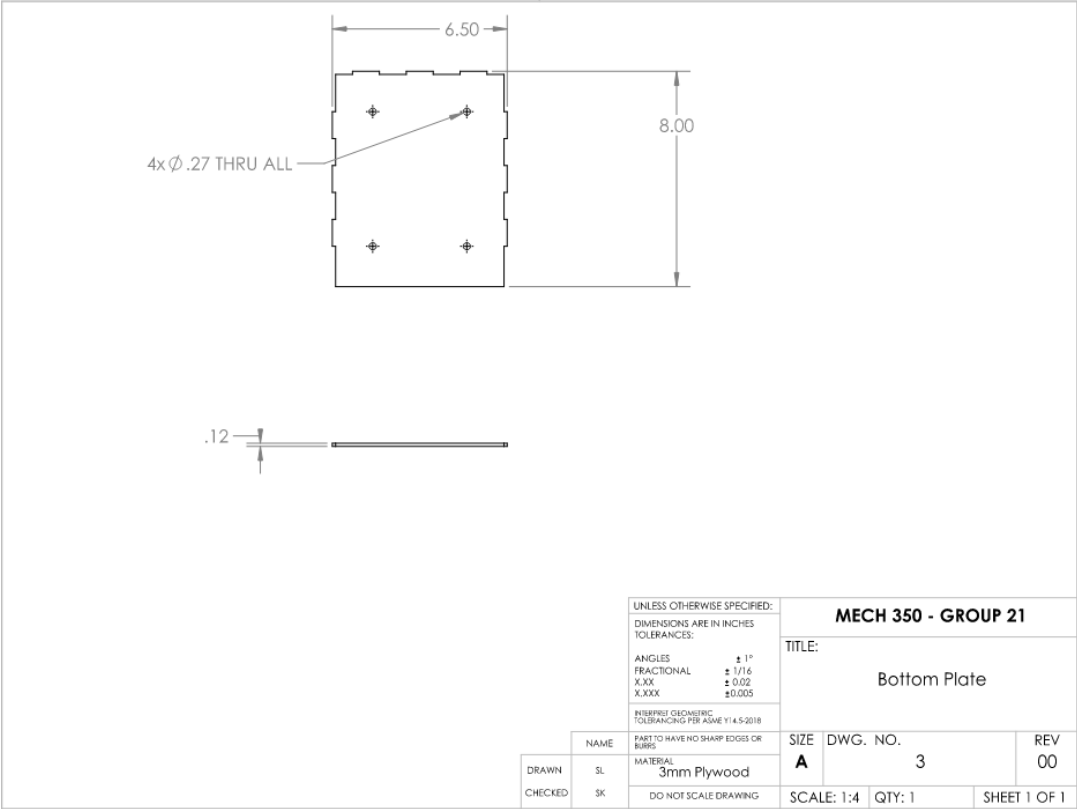
Drawings of the bridge and parts are outlined in this appendix. Drawing numbers correspond to item numbers in the BOM.

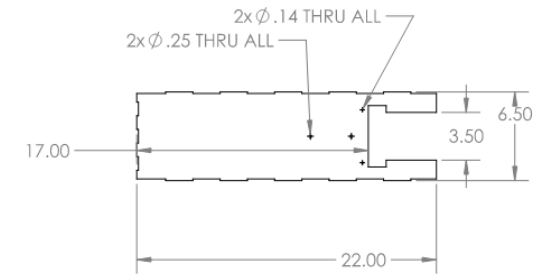
D.1 Assembly Drawings



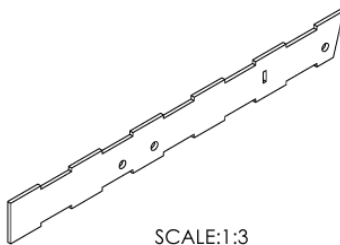
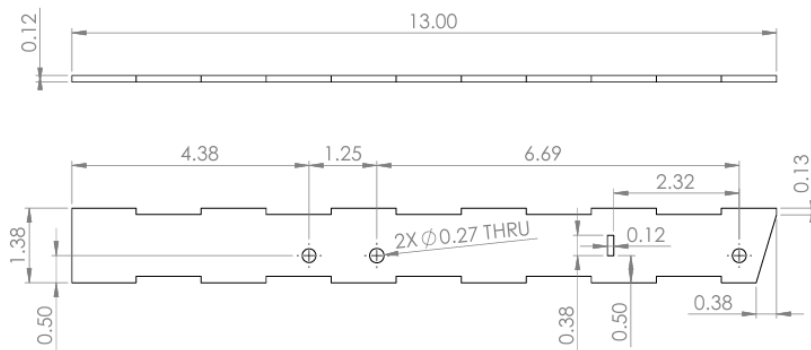
D.2 Part Drawings





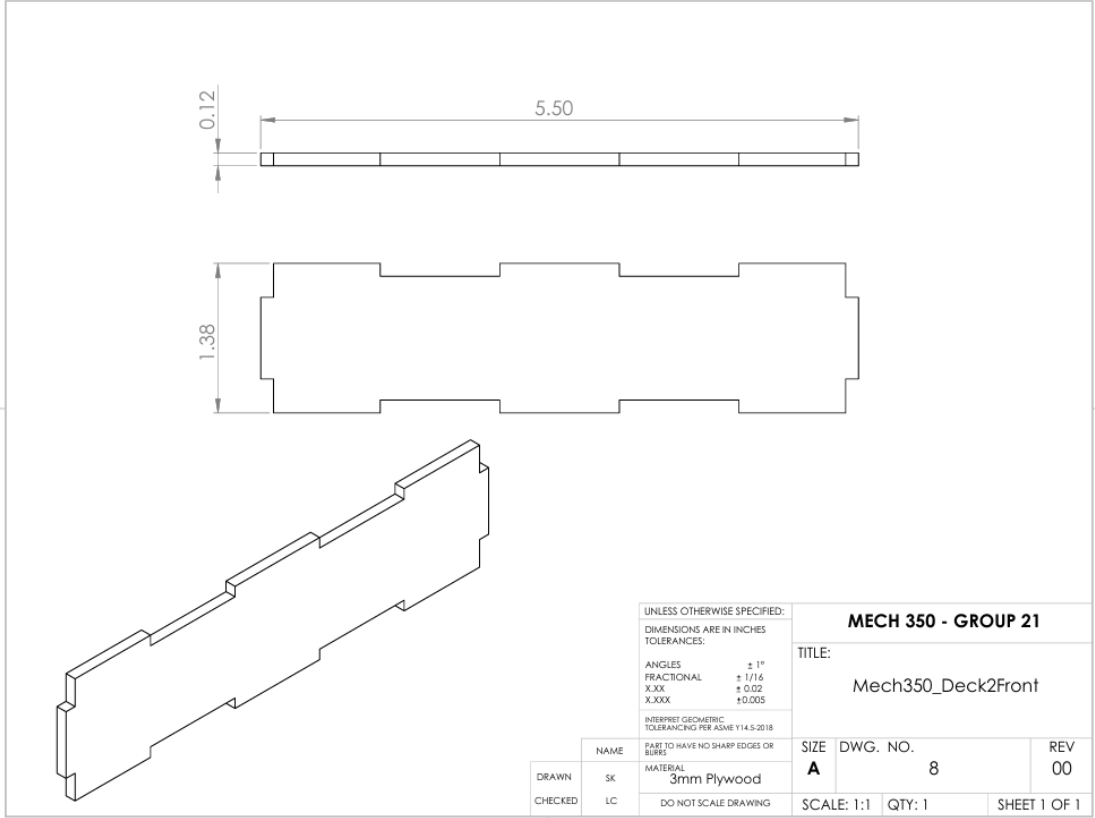
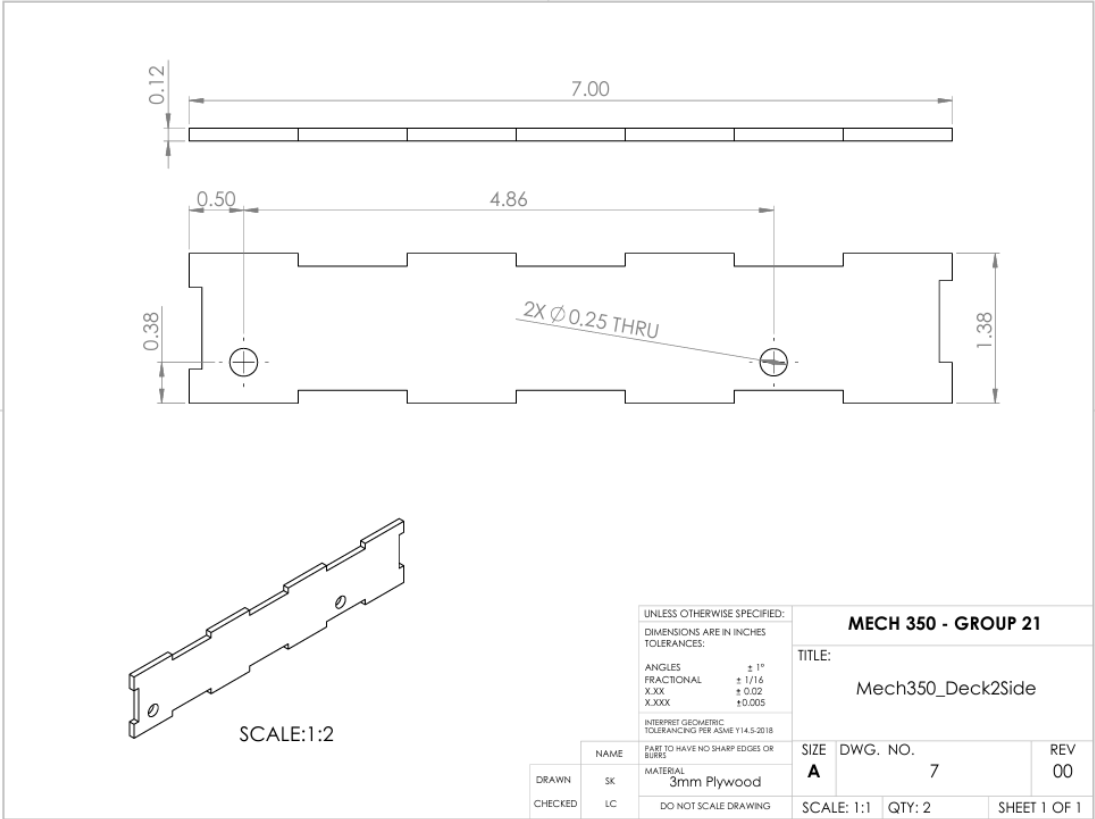


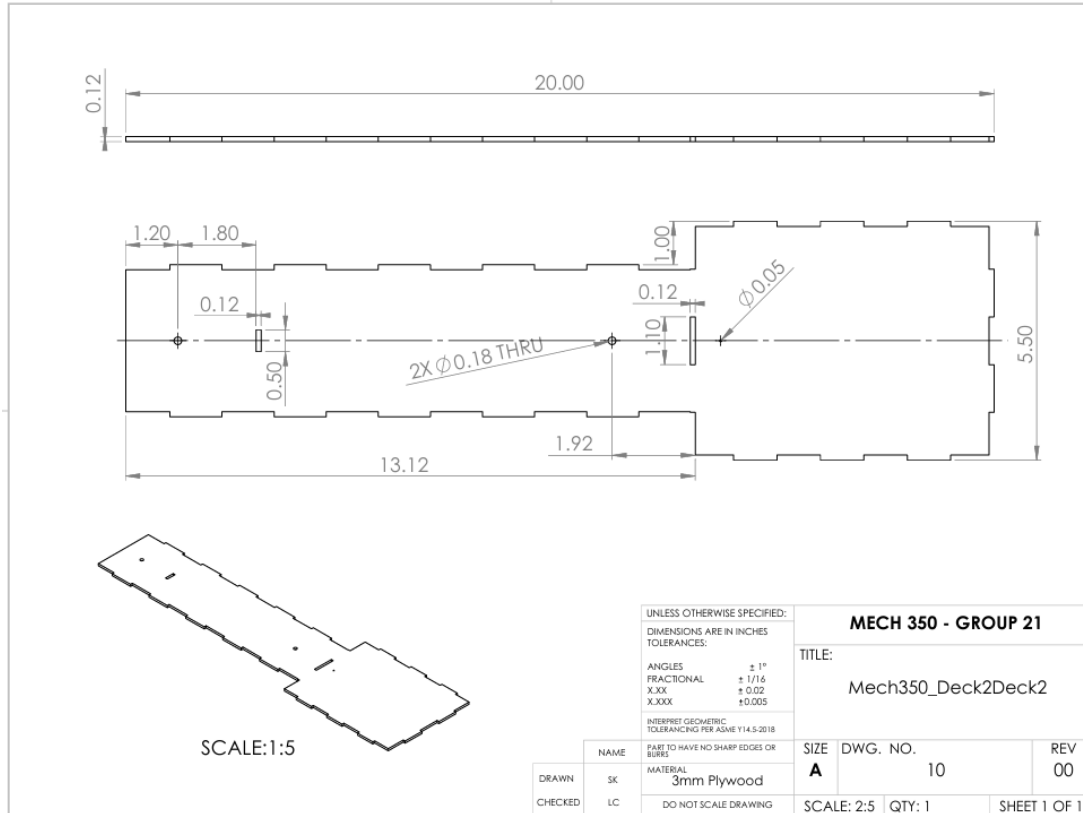
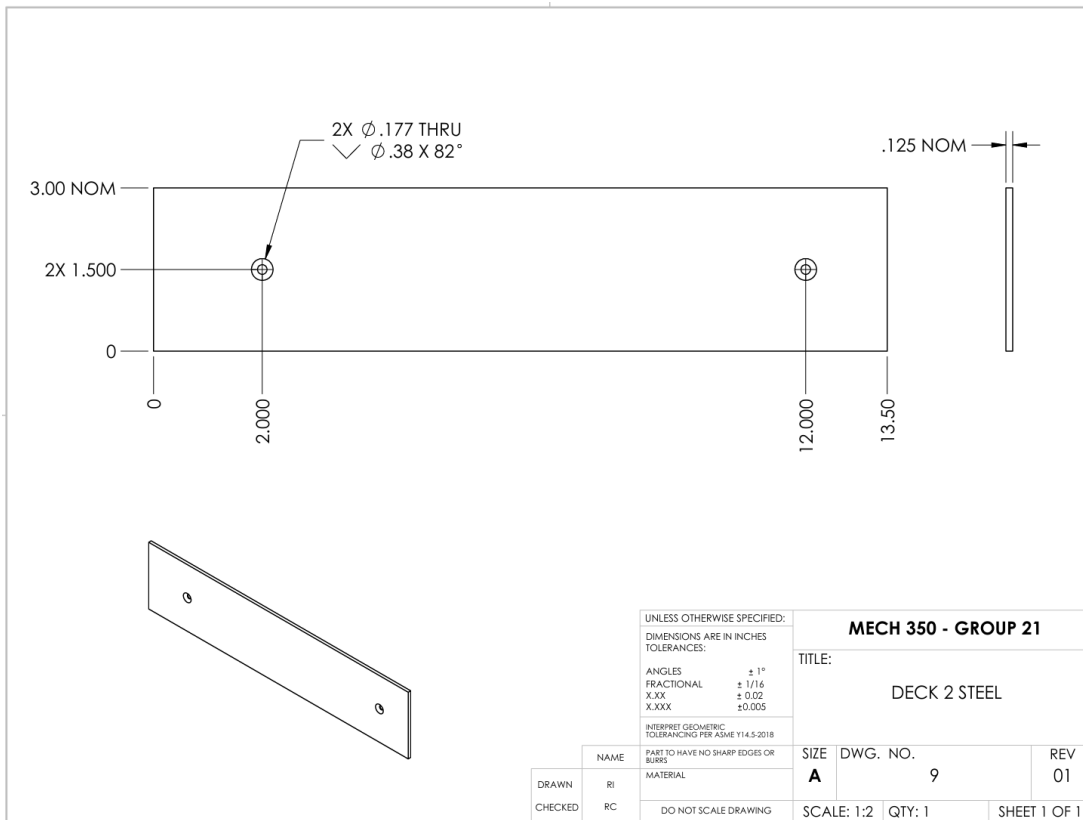
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PART TO HAVE NO SHARP EDGES OR BURRS		A	5	00
MATERIAL 3mm Plywood		SCALE: 1:8	QTY: 1	SHEET 1 OF 1
DO NOT SCALE DRAWING				

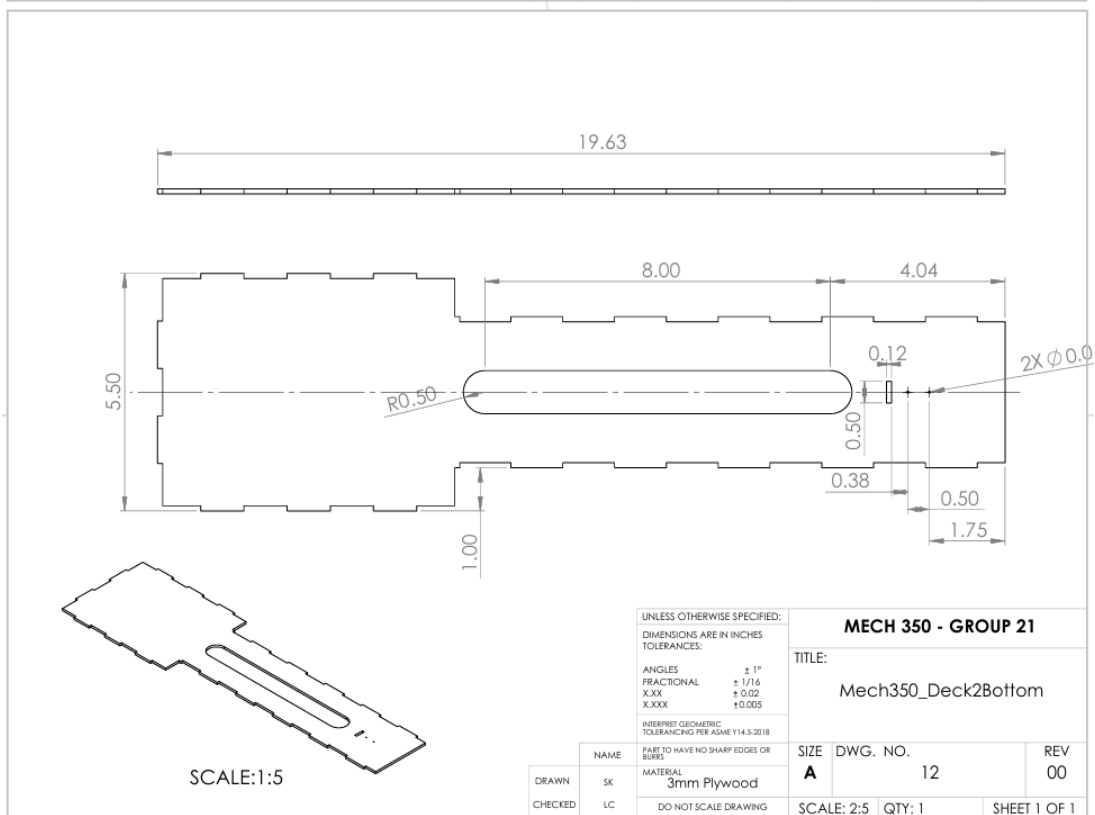
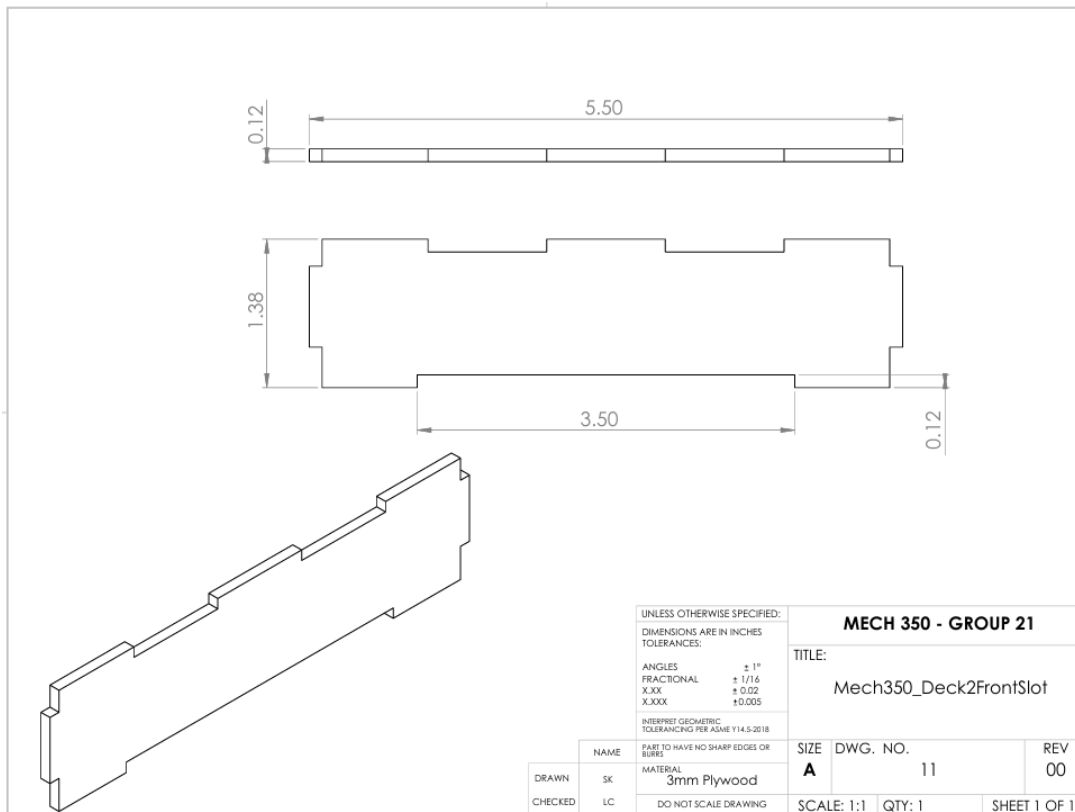


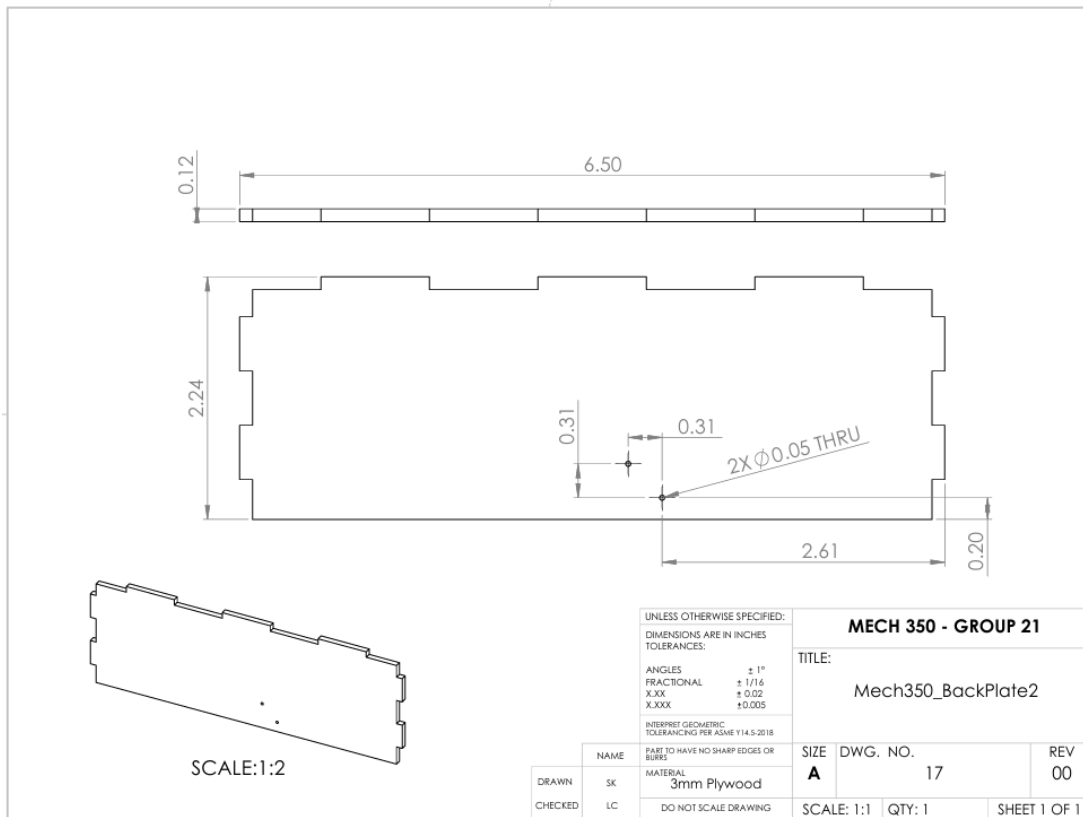
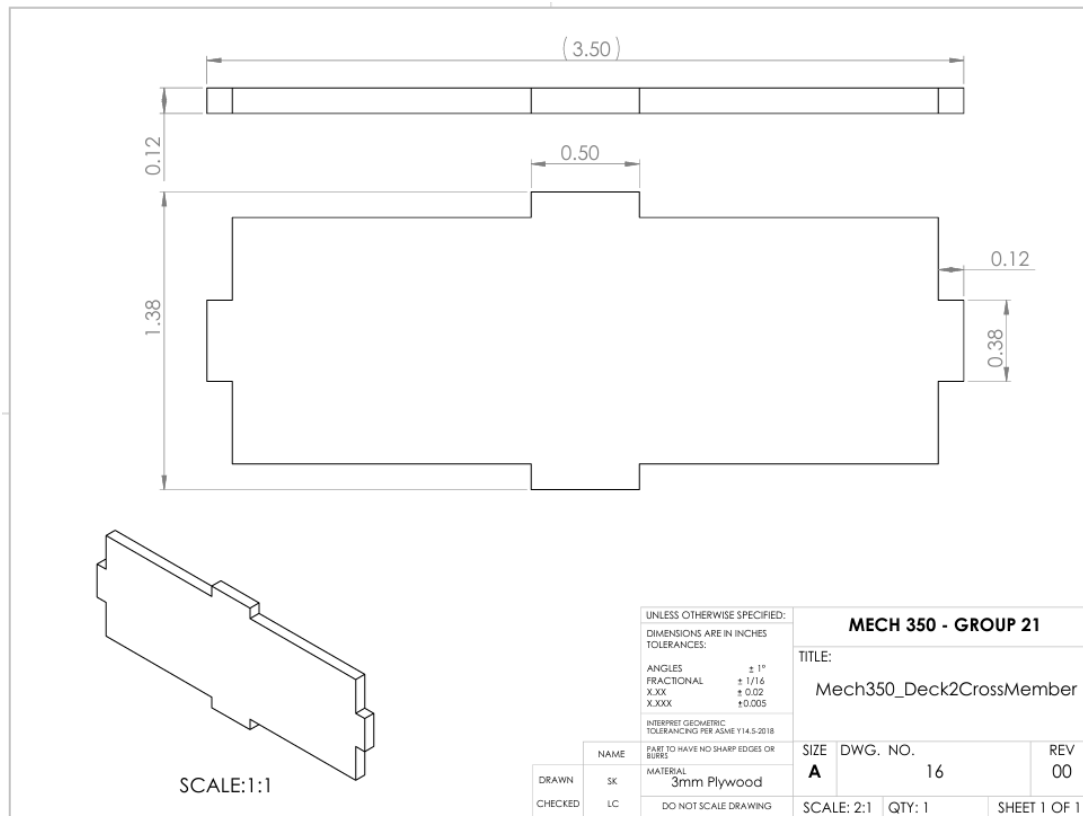
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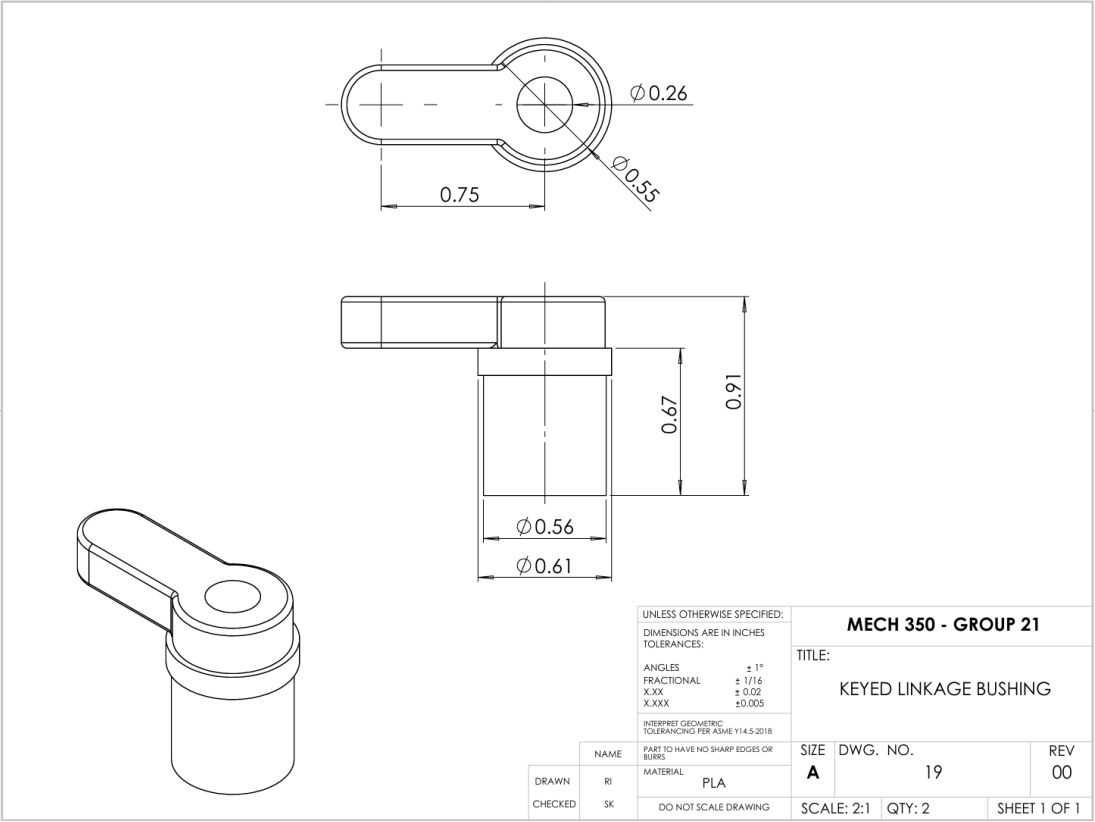
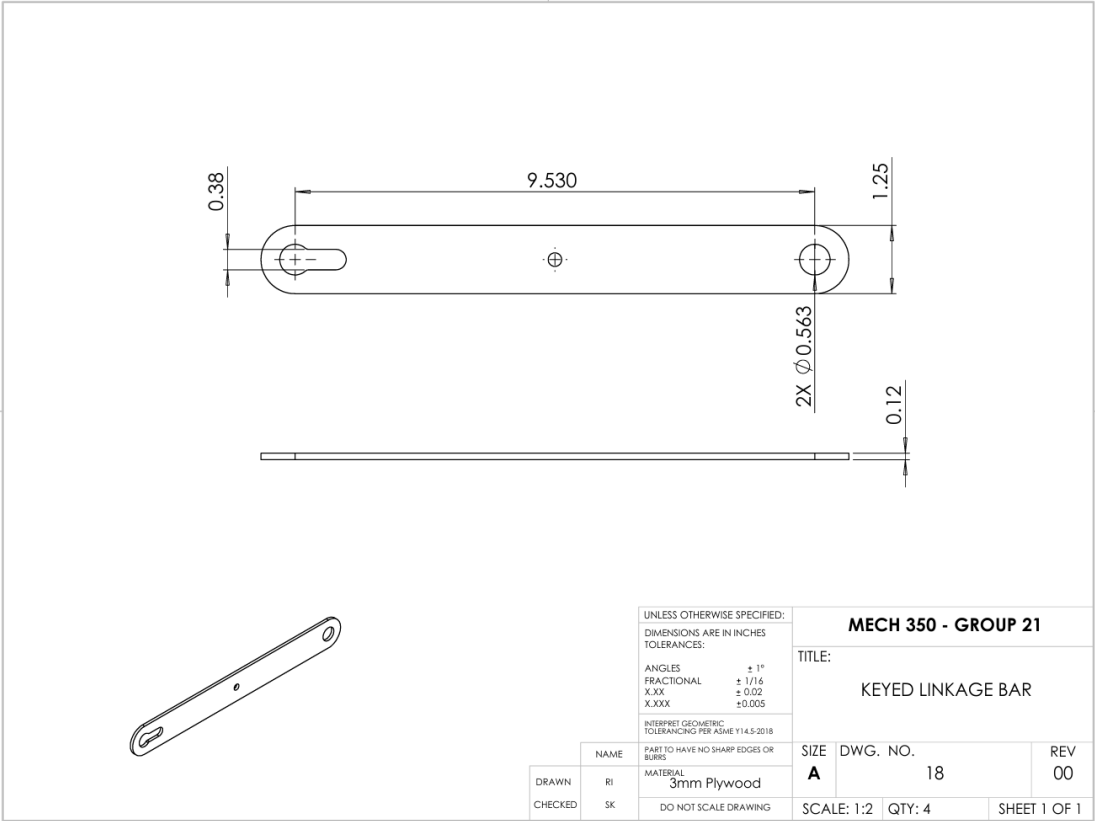
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DO NOT SCALE DRAWING				

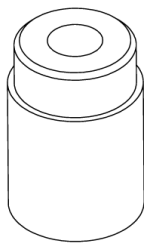
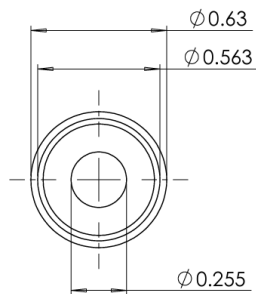
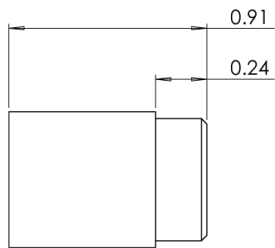




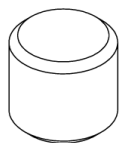
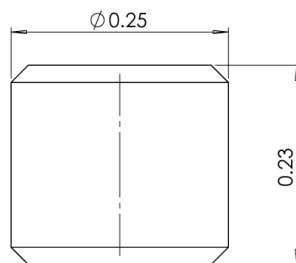




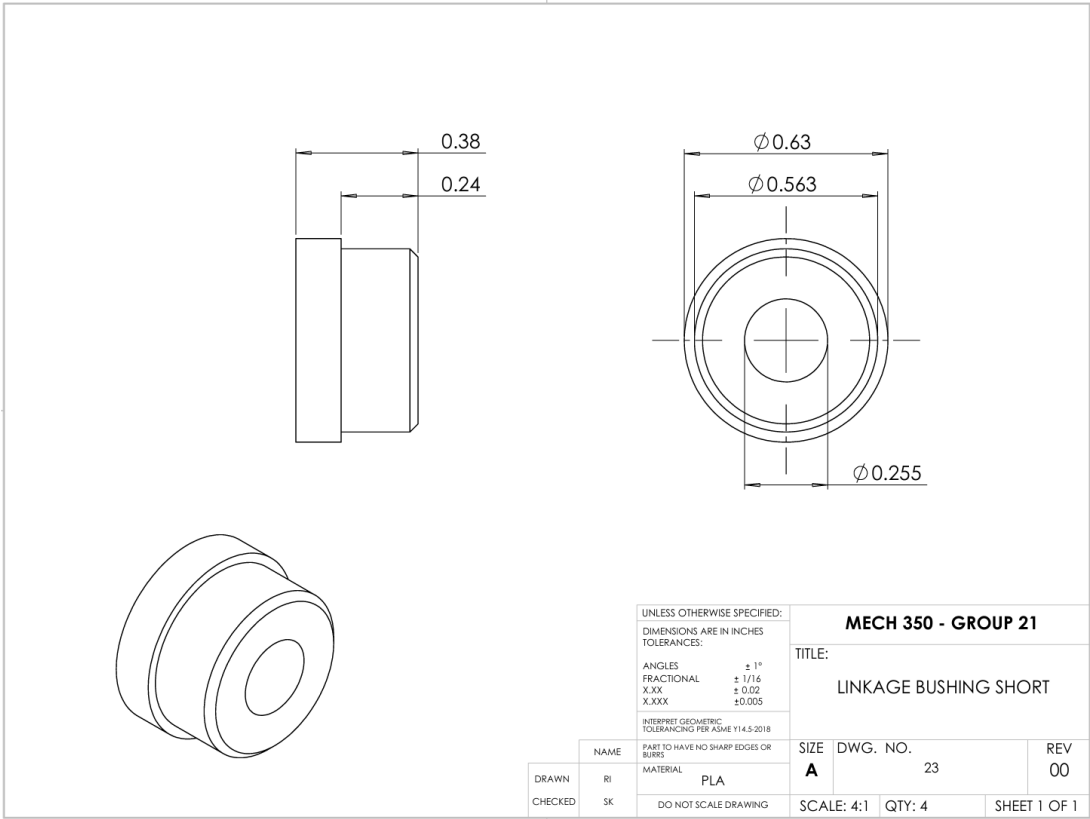
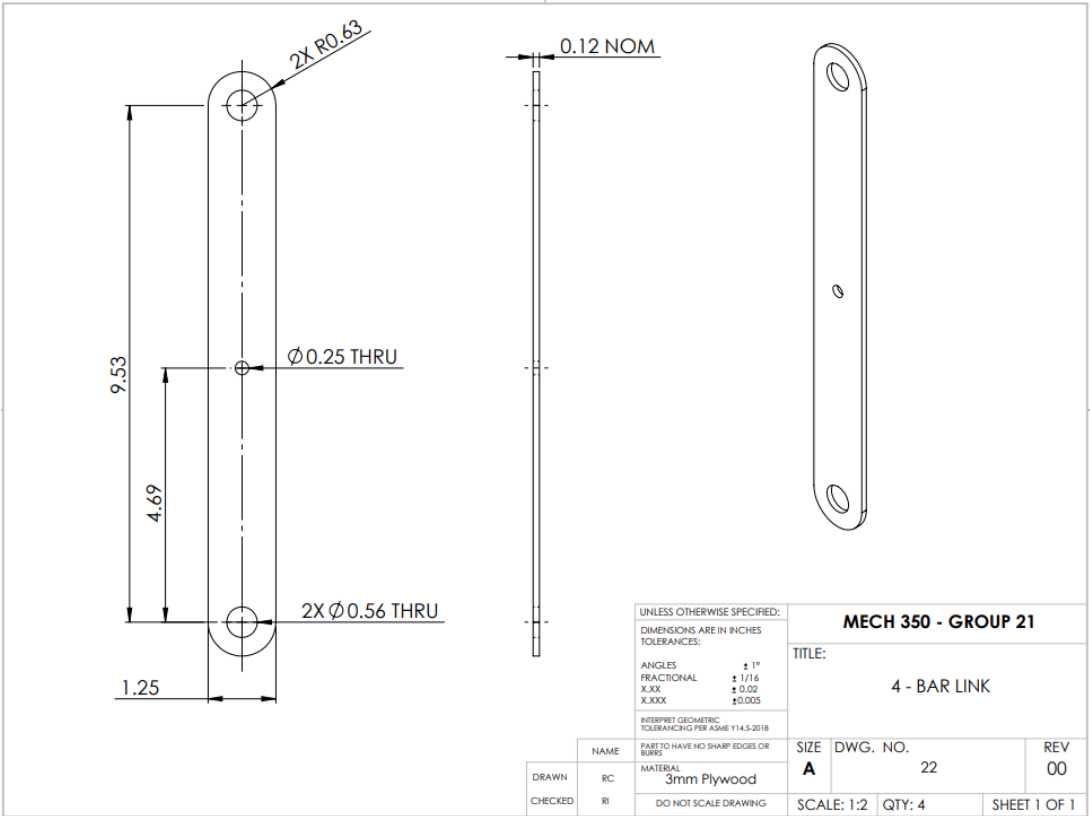


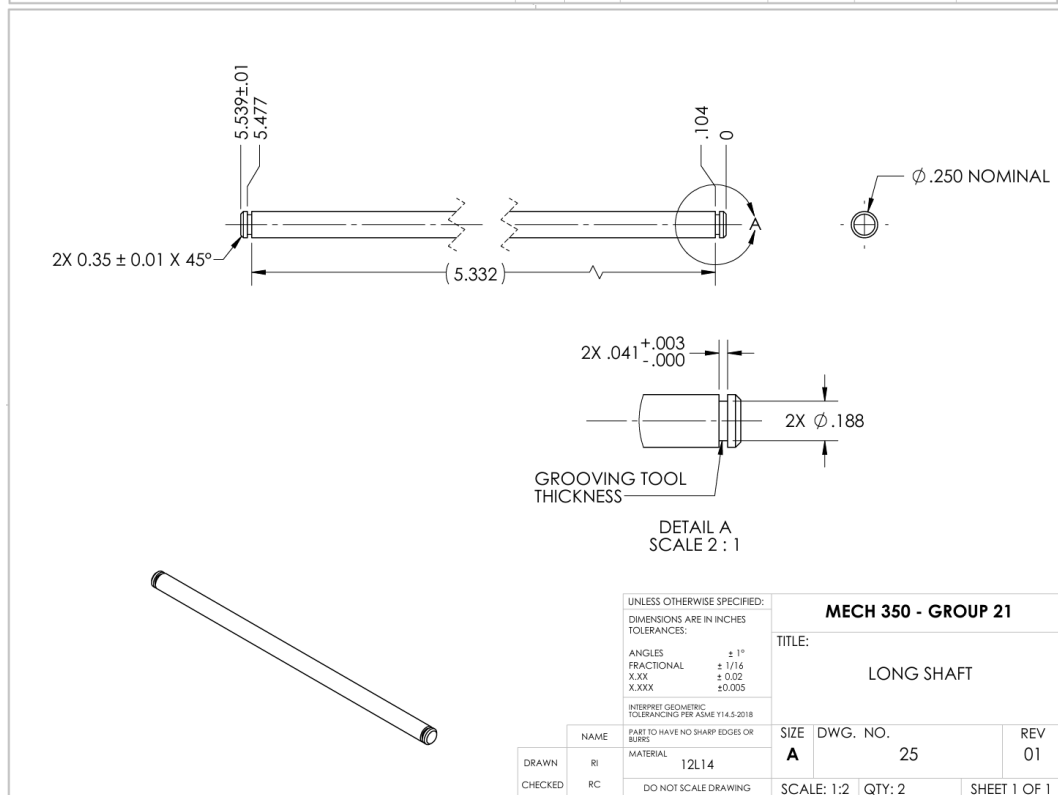
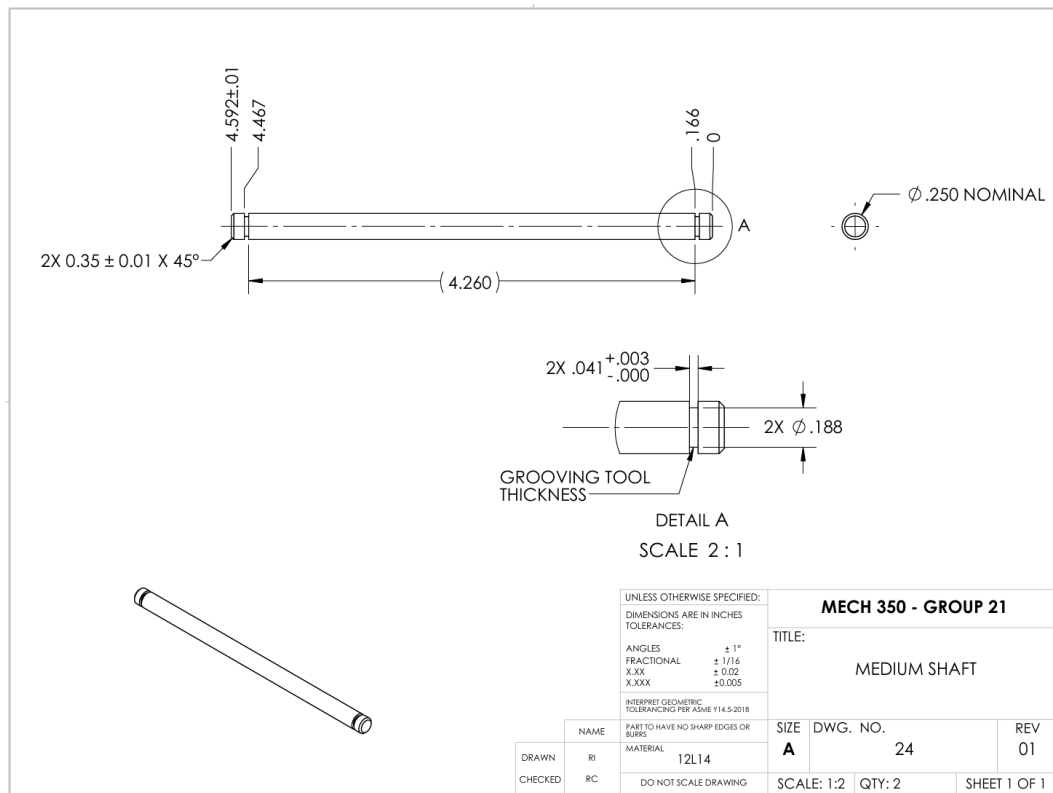


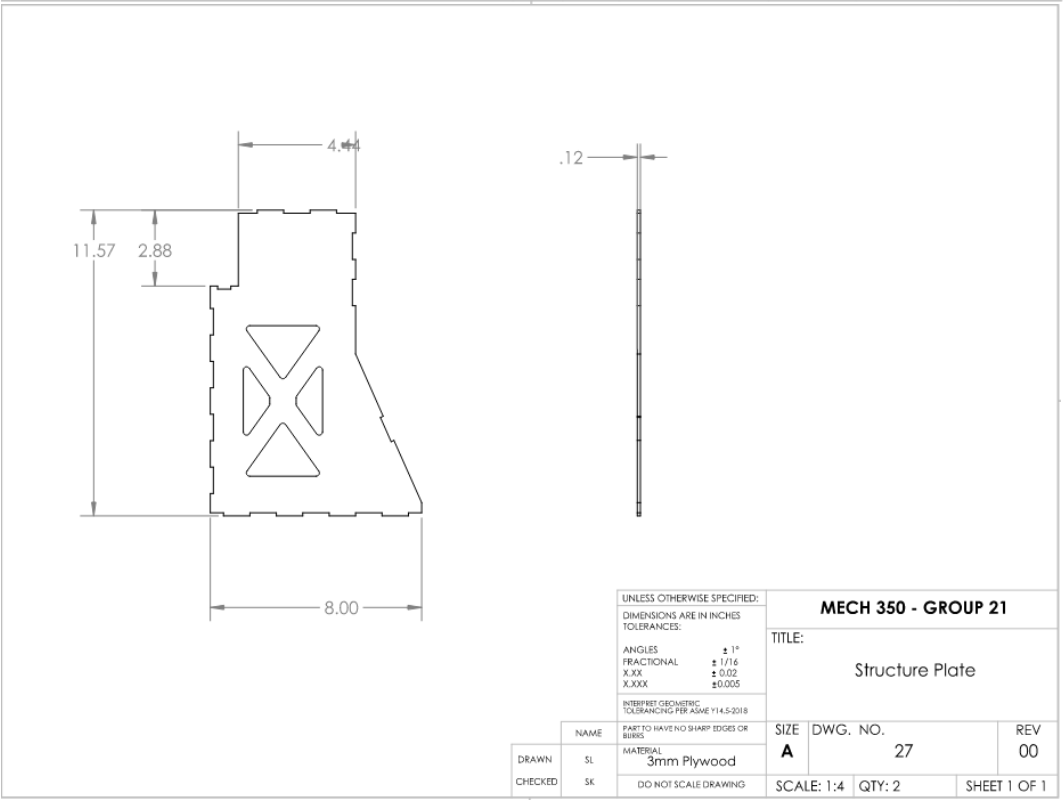
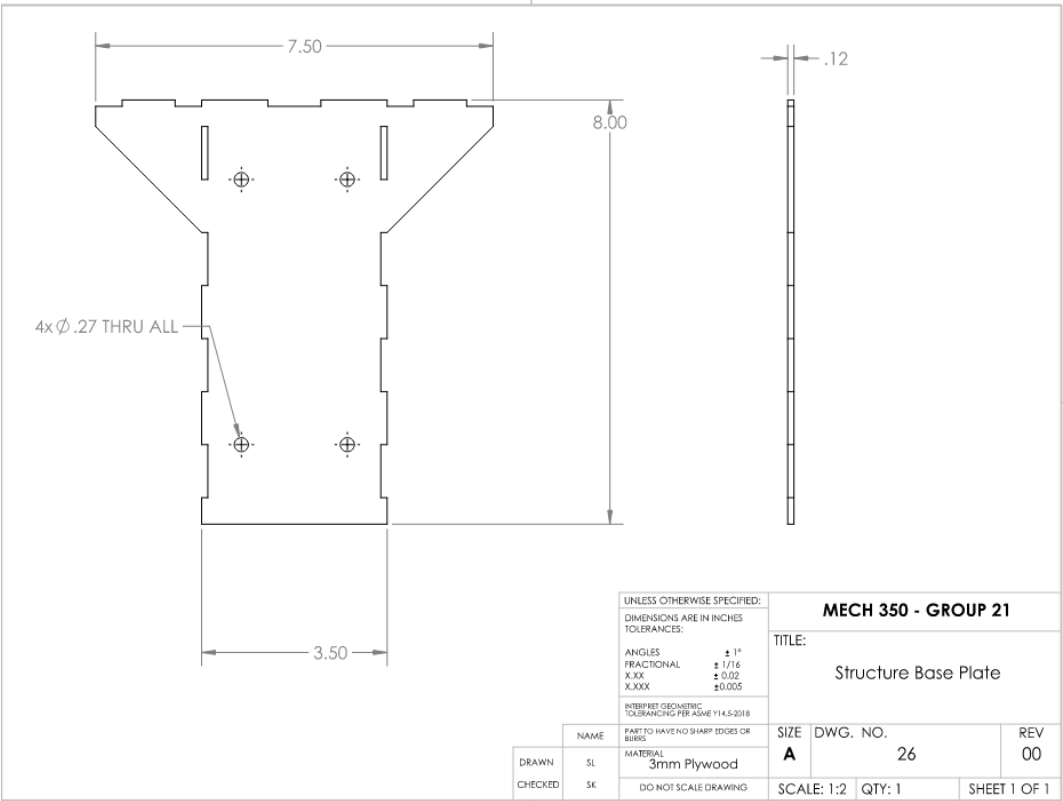
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		TITLE: LINKAGE BUSHING LONG		
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DRAWN				
CHECKED	SL			
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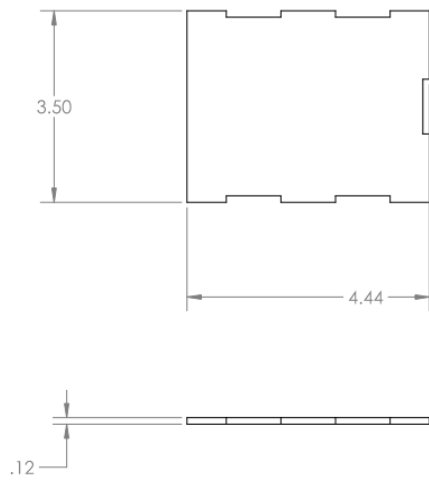


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		TITLE: LINKAGE DOWEL		
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	JP	A	21	00
DRAWN				
CHECKED	JP			
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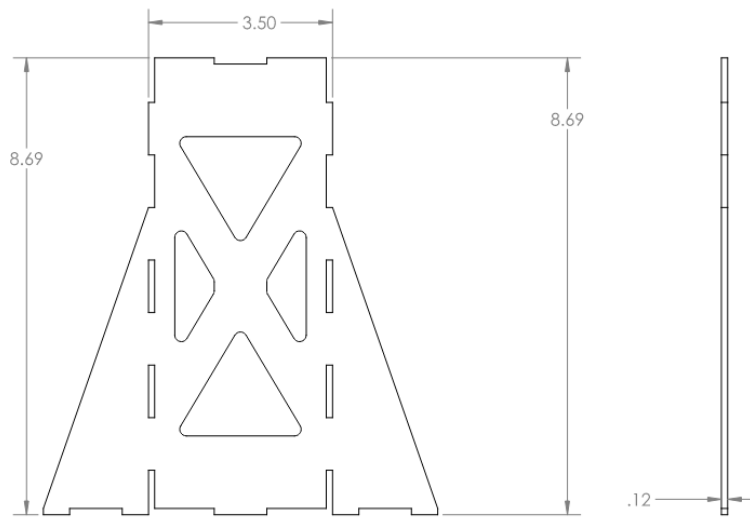




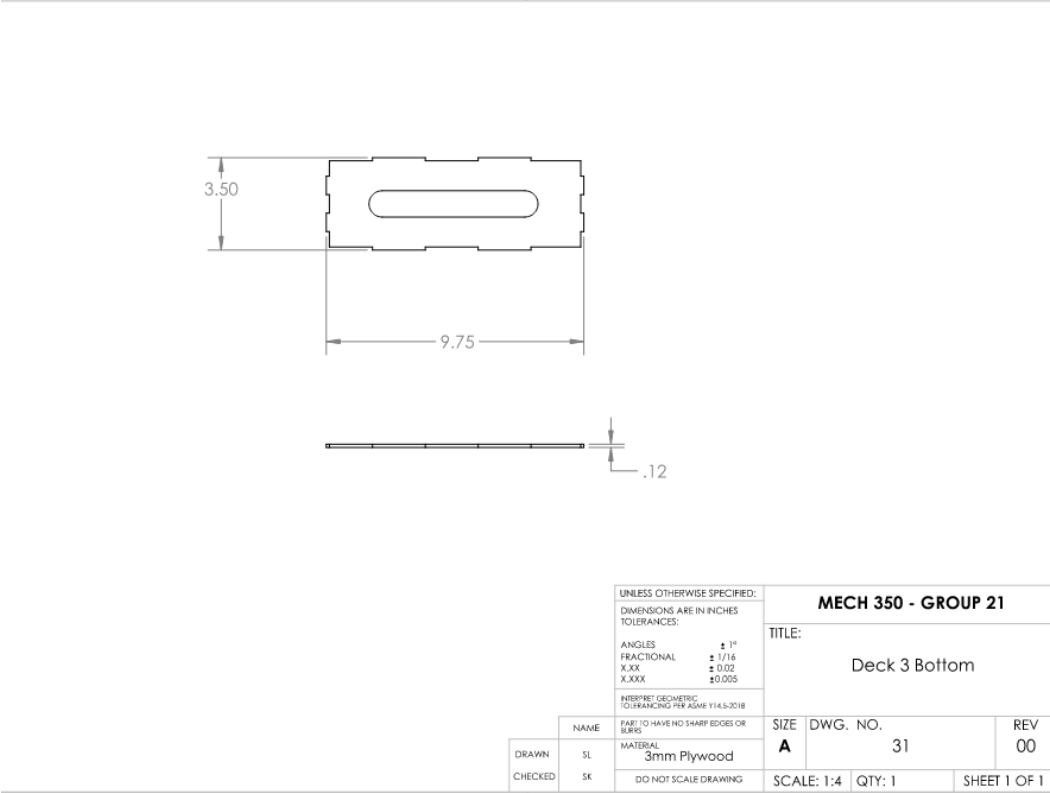
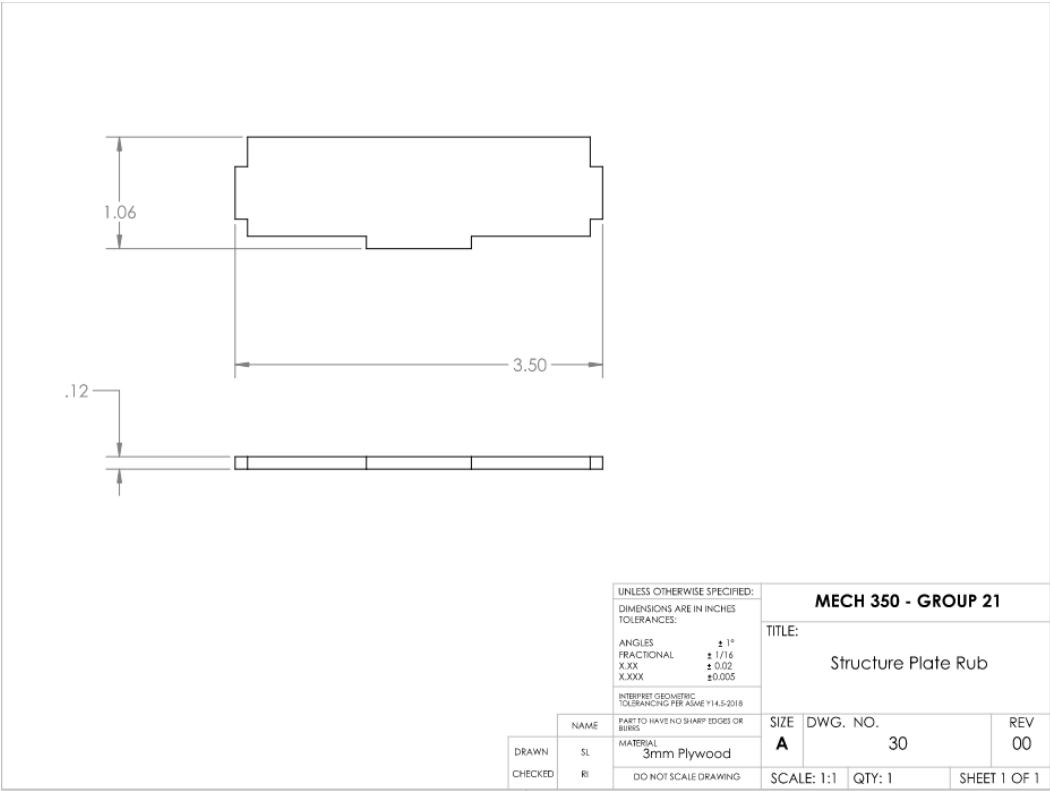


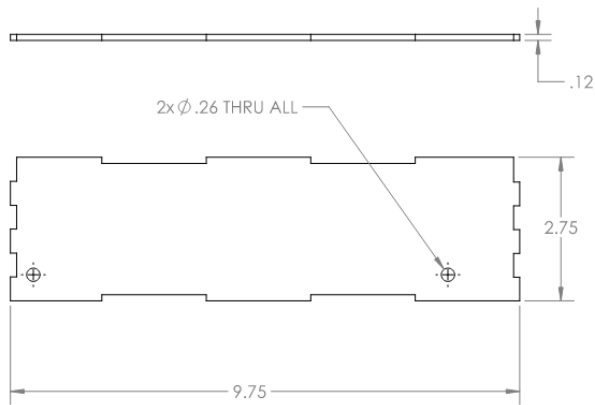


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				TITLE: Structure Plate Top		
				SIZE	DWG. NO.	REV
				A	28	00
DRAWN	SL	MATERIAL 3mm Plywood				
CHECKED	SK	DO NOT SCALE DRAWING		SCALE: 1:2	QTY: 1	SHEET 1 OF 1



UNLESS OTHERWISE SPECIFIED:				MECH 350 - GROUP 21		
DIMENSIONS ARE IN INCHES TOLERANCES:				TITLE:		
ANGLES FRACTIONAL ± 1° X.XX ± 1/16 X.XXX ± 0.02 ±0.005				Structure Plate Short		
INTERPRET GEOMETRIC TOLERANCING PER ASME Y14.5-2018				SIZE A	DWG. NO. 29	REV 00
PART TO HAVE NO SHARP EDGES OR BURRS						
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CHECKED		RI	3mm Plywood			
DO NOT SCALE DRAWING				SCALE: 1:2 QTY: 1 SHEET 1 OF 1		





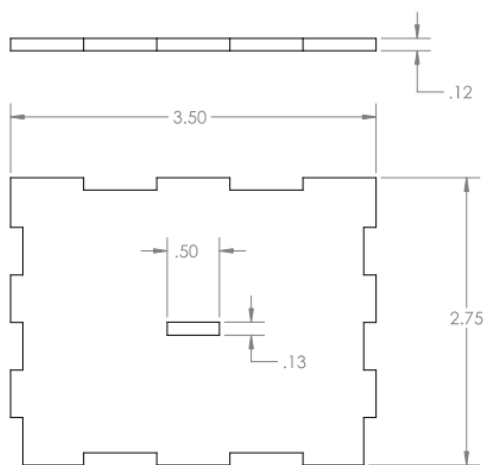
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 FRACTIONAL $\pm 1/16$
 X.XX ± 0.02
 X.XXX ± 0.005
 INTERPRET GEOMETRIC
 TOLERANCING PER ASME Y14.5-2018

MECH 350 - GROUP 21

TITLE:

Deck 3 Side

	NAME	PART TO HAVE NO SHARP EDGES OR BURS	SIZE	DWG. NO.	REV
DRAWN	SL	MATERIAL	A	32	00
CHECKED	SK	DO NOT SCALE DRAWING	SCALE: 1:2	QTY: 2	SHEET 1 OF 1



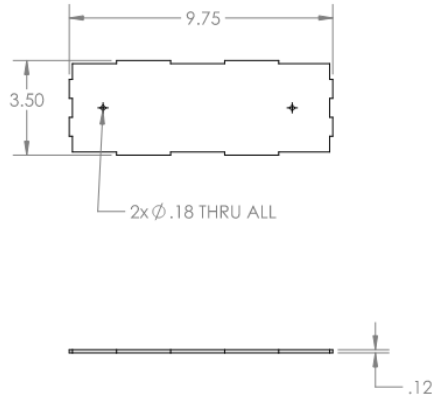
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 TOLERANCES:
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 FRACTIONAL $\pm 1/16$
 X.XX ± 0.02
 X.XXX ± 0.005
 INTERPRET GEOMETRIC
 TOLERANCING PER ASME Y14.5-2018

MECH 350 - GROUP 21

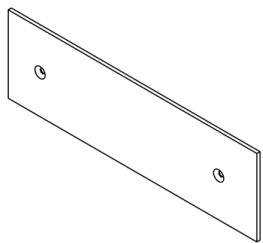
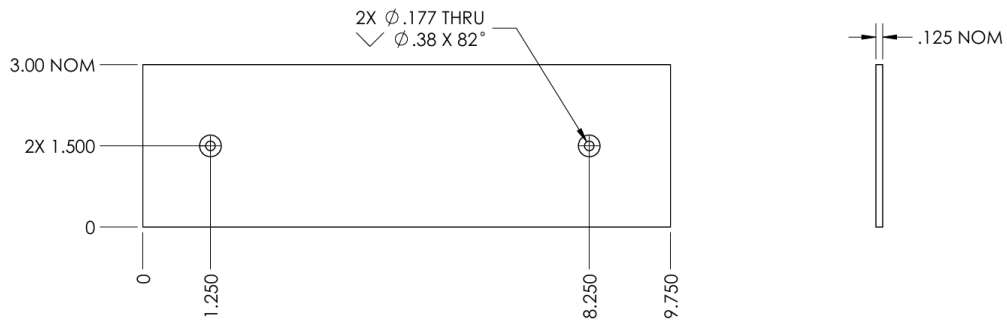
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Deck 3 End

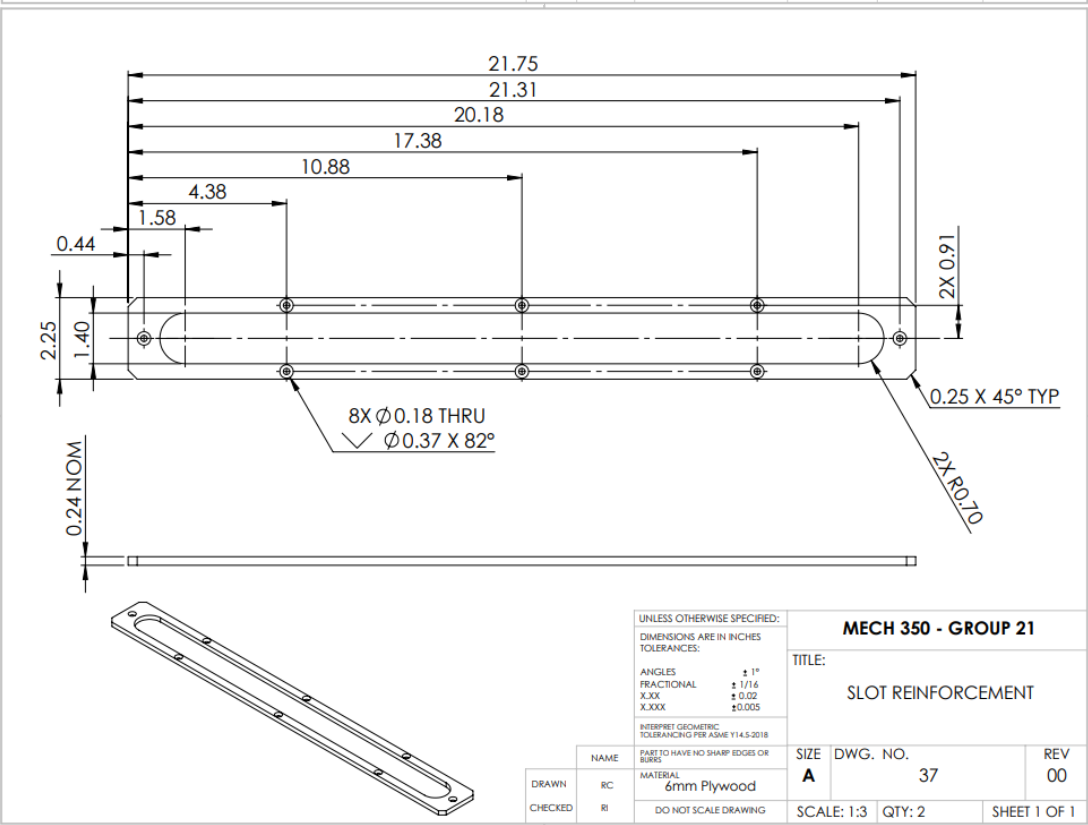
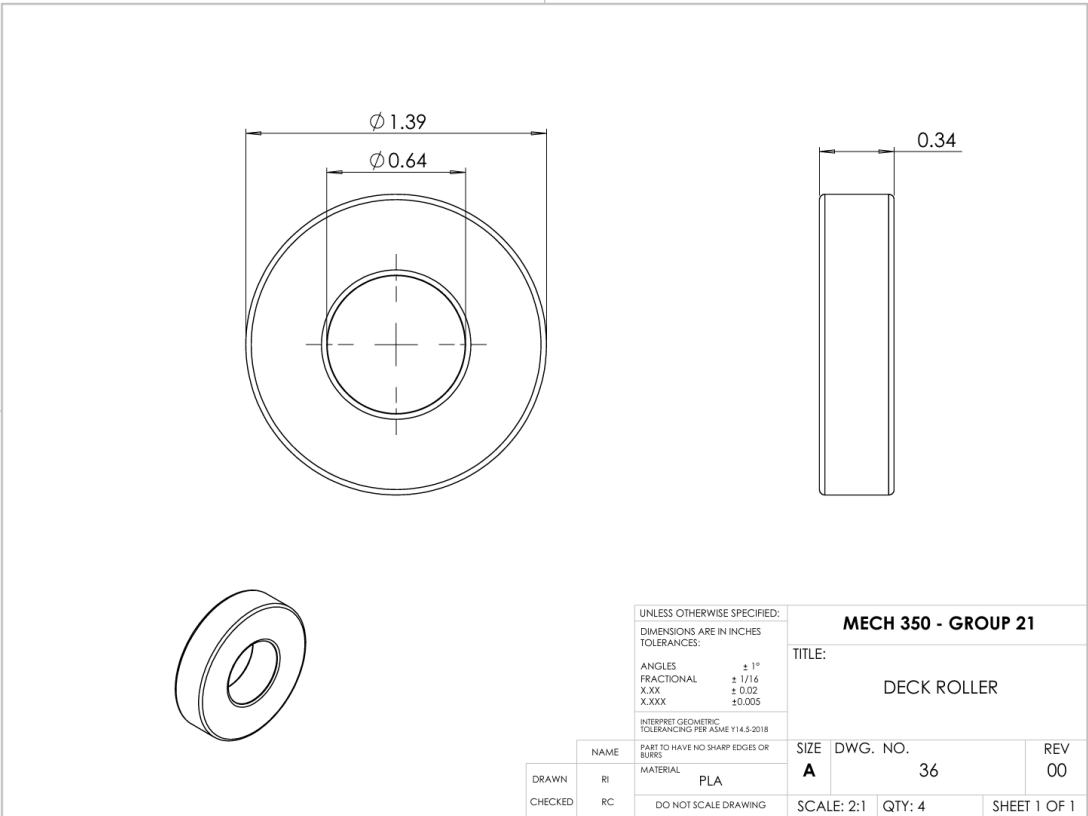
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DRAWN	SL	MATERIAL	A	33	00
CHECKED	SK	DO NOT SCALE DRAWING	SCALE: 1:1	QTY: 2	SHEET 1 OF 1

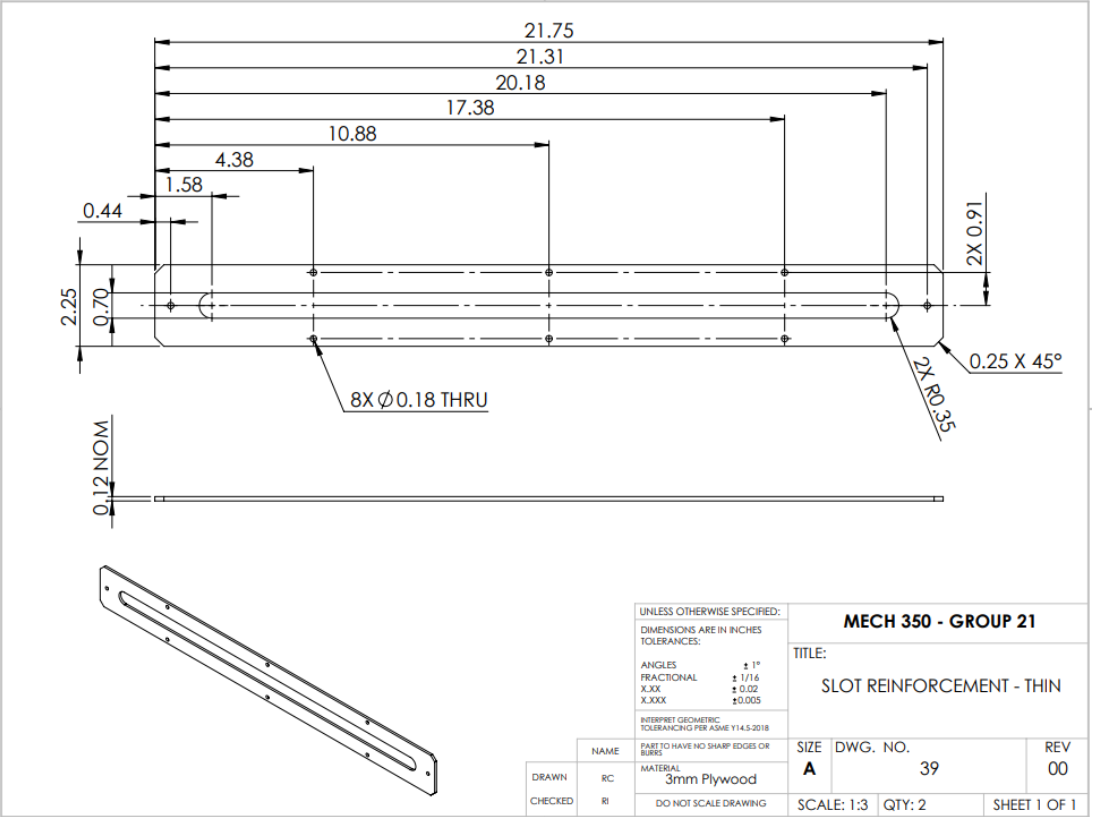
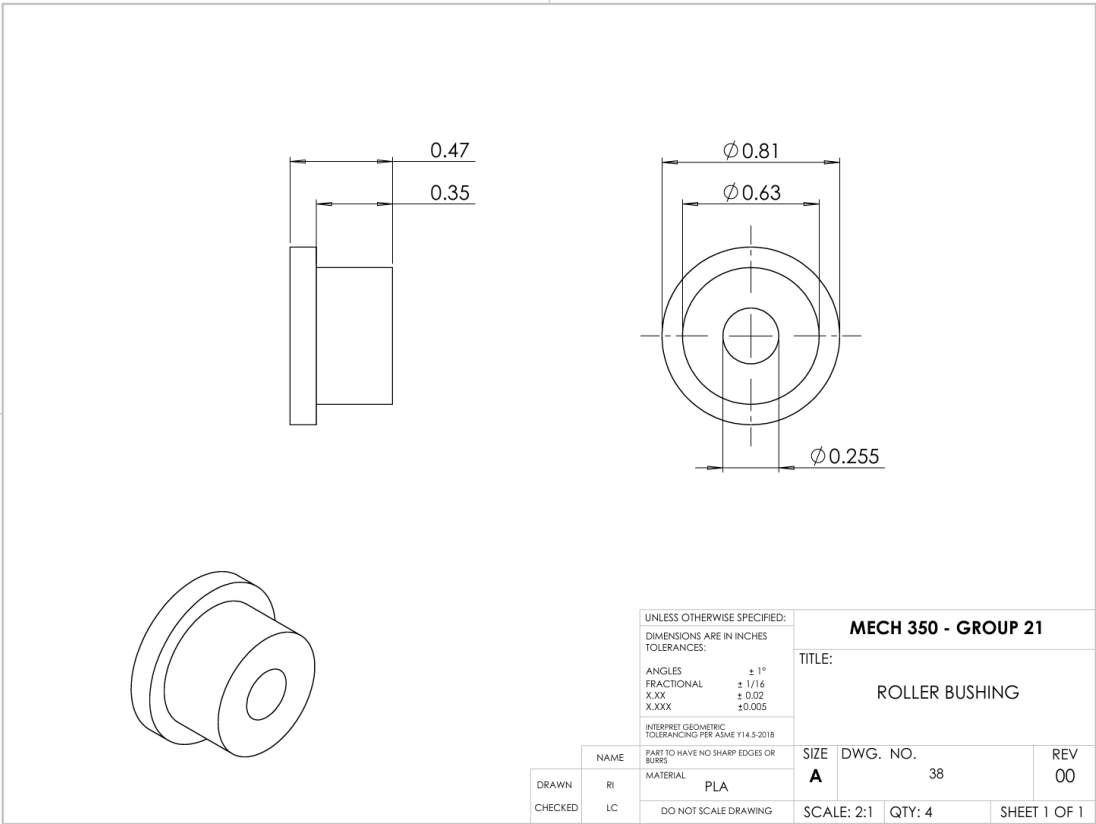


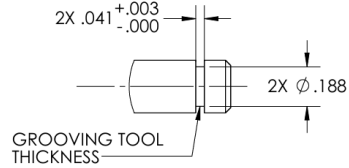
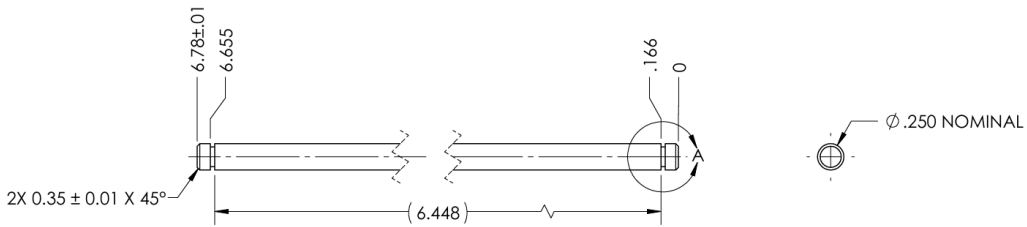
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		TITLE: Deck 3 Top		
DRAWN	SL	SIZE	DWG. NO.	REV
CHECKED	SK	A	34	00
DO NOT SCALE DRAWING		SCALE: 1:4	QTY: 1	SHEET 1 OF 1



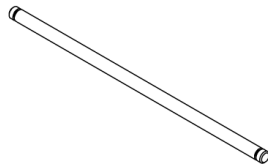
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		TITLE: DECK 3 STEEL		
DRAWN	RI	SIZE	DWG. NO.	REV
CHECKED	RC	A	35	01
DO NOT SCALE DRAWING		SCALE: 1:2	QTY: 1	SHEET 1 OF 1







DETAIL A
SCALE 2 : 1



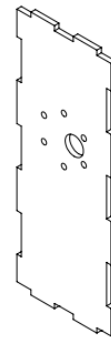
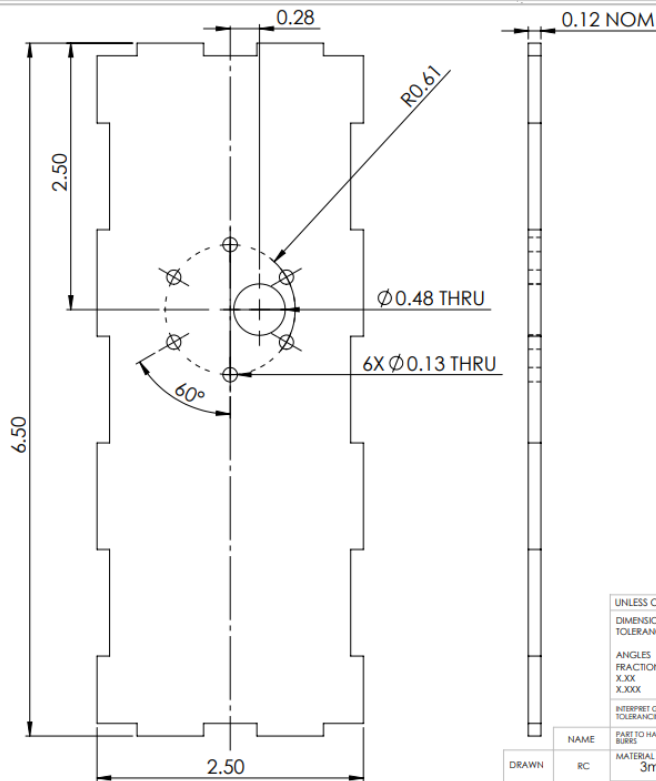
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DIMENSIONS ARE IN INCHES
TOLERANCES:
ANGLES ± 1°
FRACTIONAL ± 1/16
X.XX ± 0.02
X.XXX ± 0.005
INTERPRET GEOMETRIC
TOLERANCING PER ASME Y14.5-2018

MECH 350 - GROUP 21

TITLE:

BEARING SHAFT

DRAWN	RI	NAME	SIZE	DWG. NO.	REV
CHECKED	RC	MATERIAL	A	40	01
		DO NOT SCALE DRAWING	SCALE: 1:2	QTY: 2	SHEET 1 OF 1



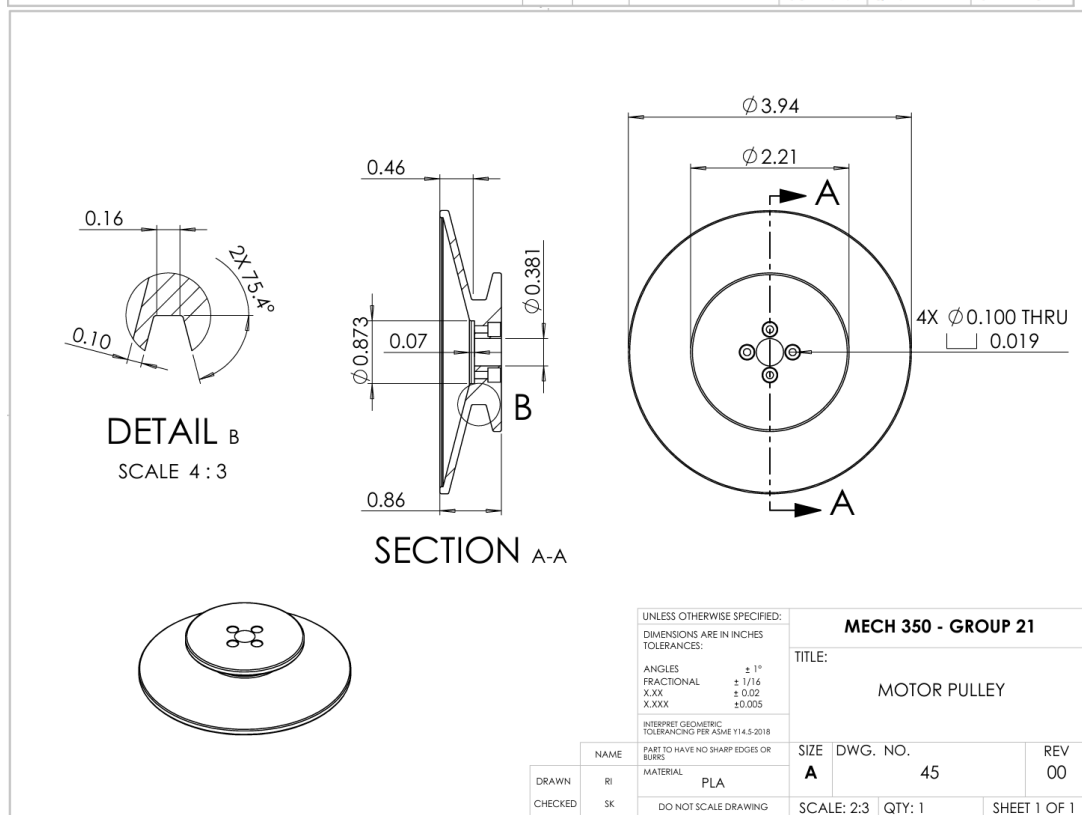
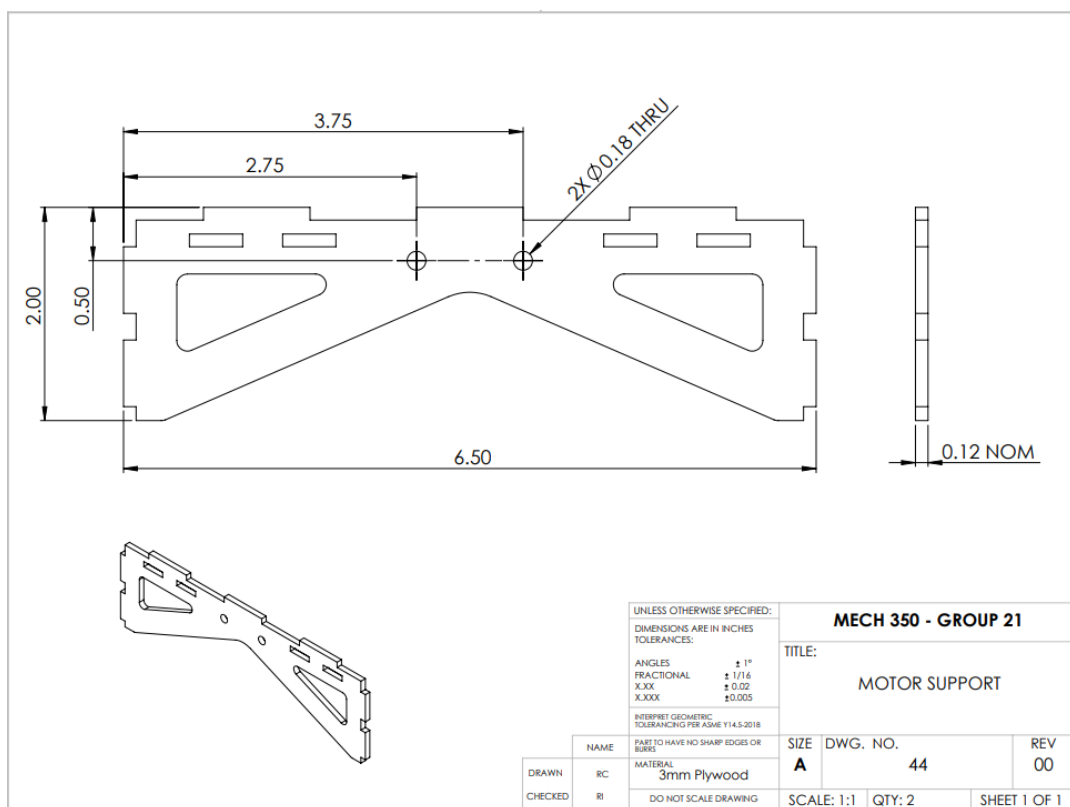
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DIMENSIONS ARE IN INCHES
TOLERANCES:
ANGLES ± 1°
FRACTIONAL ± 1/16
X.XX ± 0.02
X.XXX ± 0.005
INTERPRET GEOMETRIC
TOLERANCING PER ASME Y14.5-2018

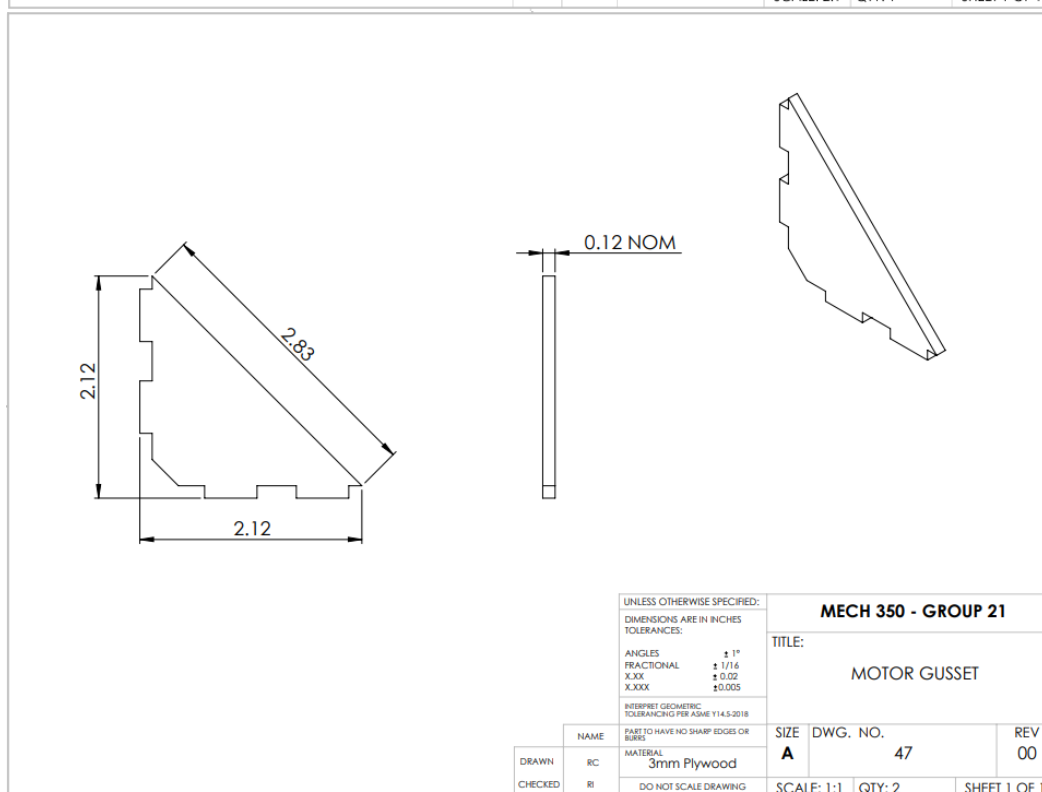
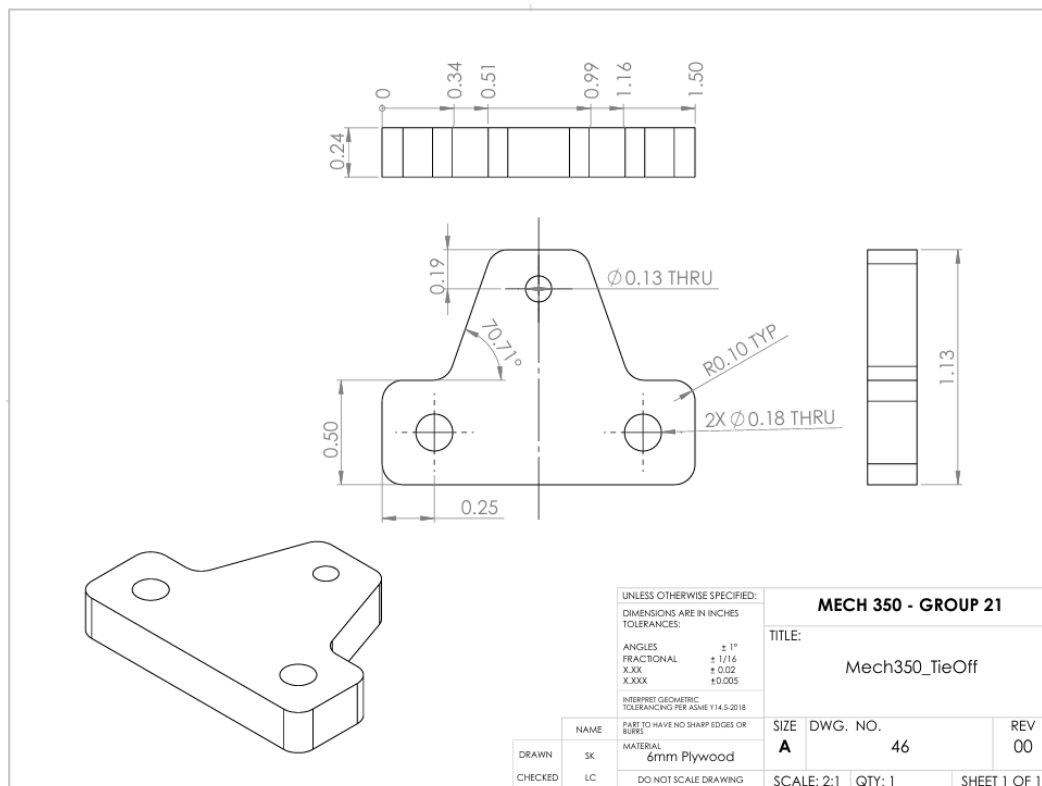
MECH 350 - GROUP 21

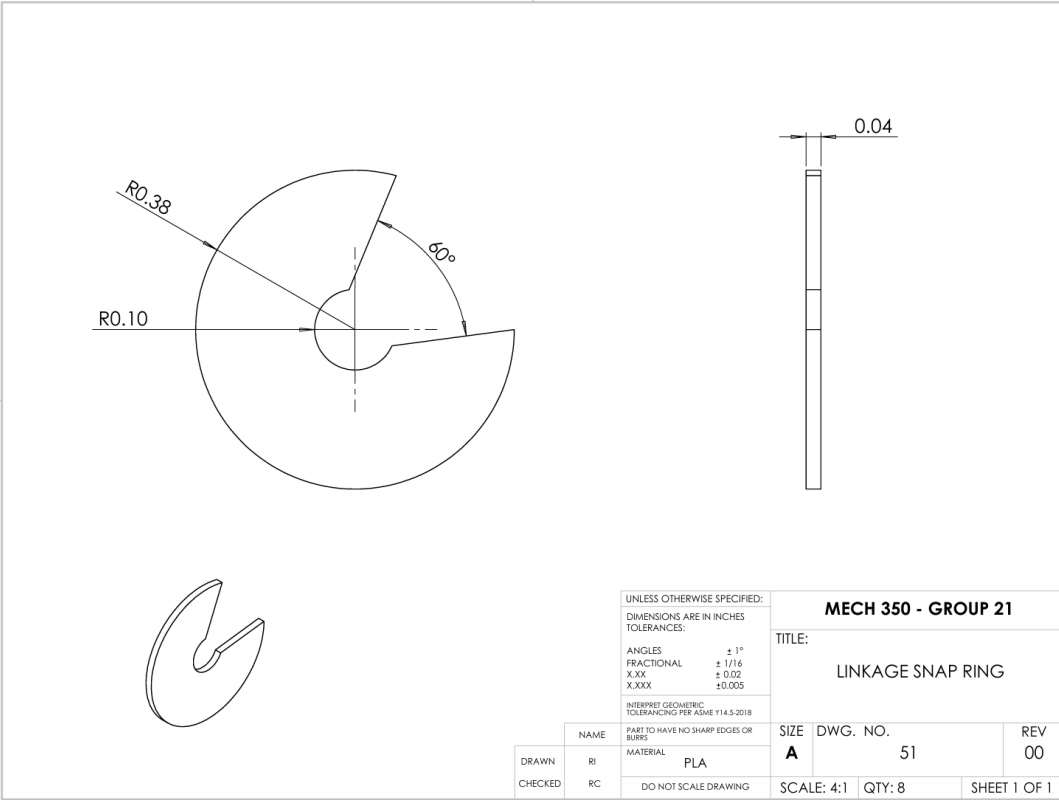
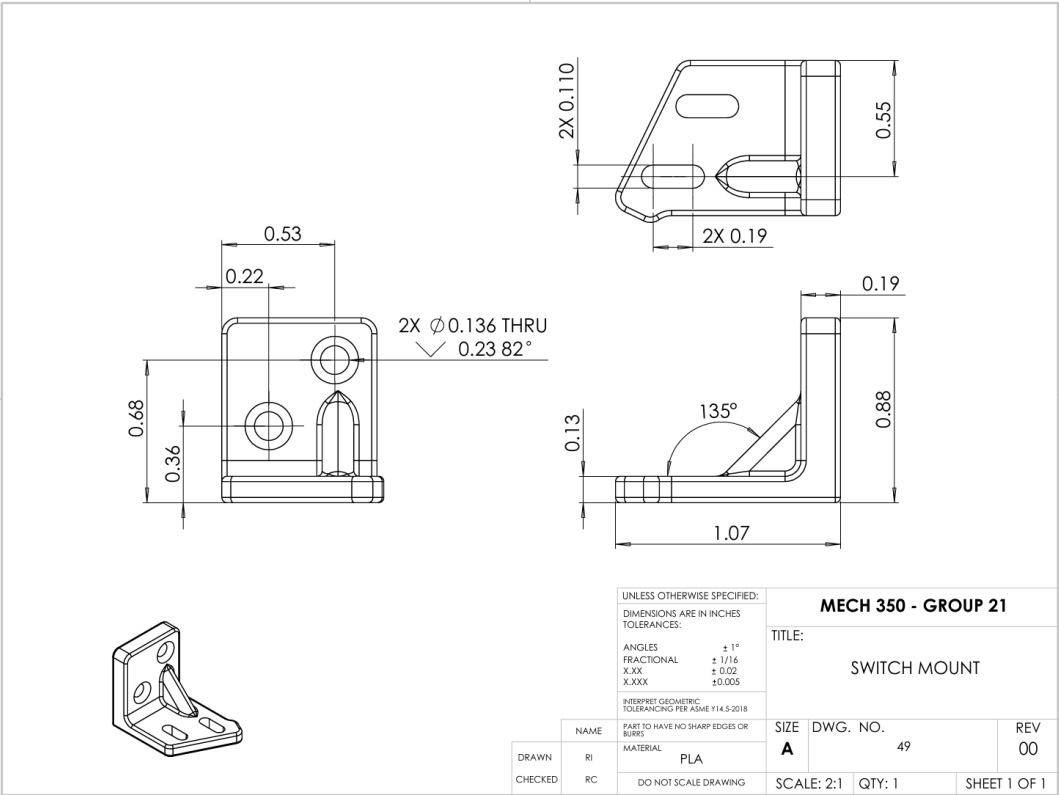
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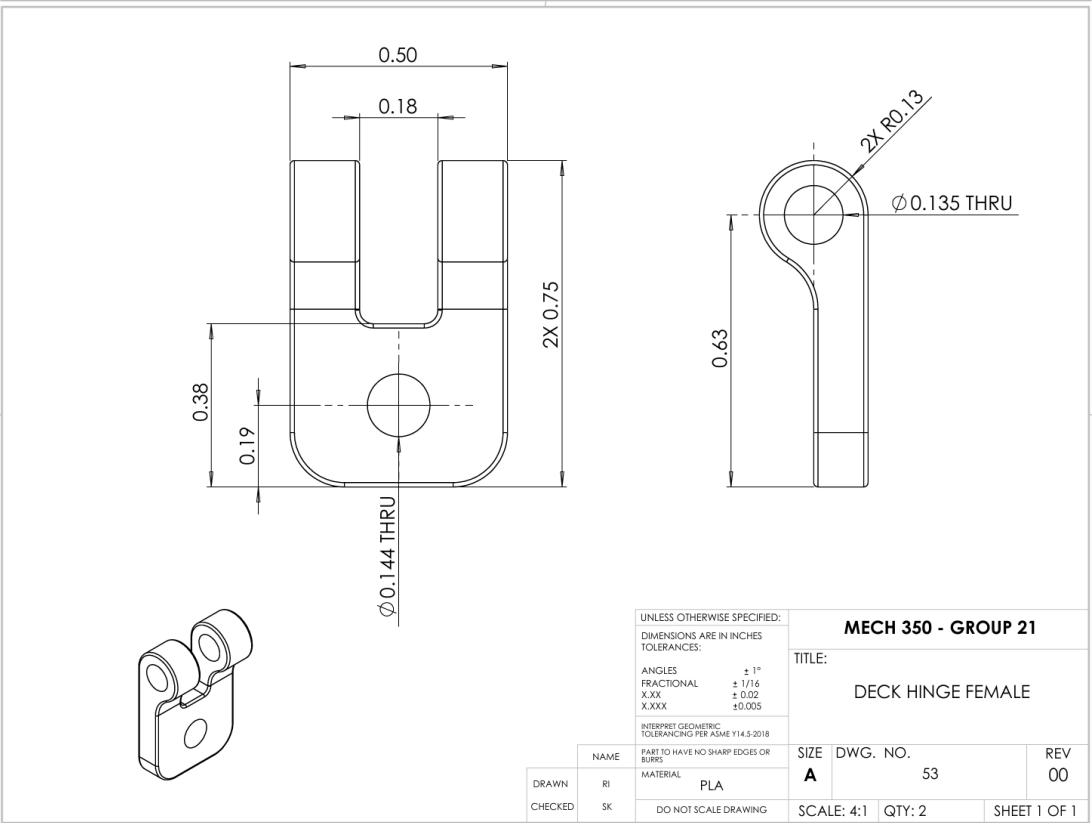
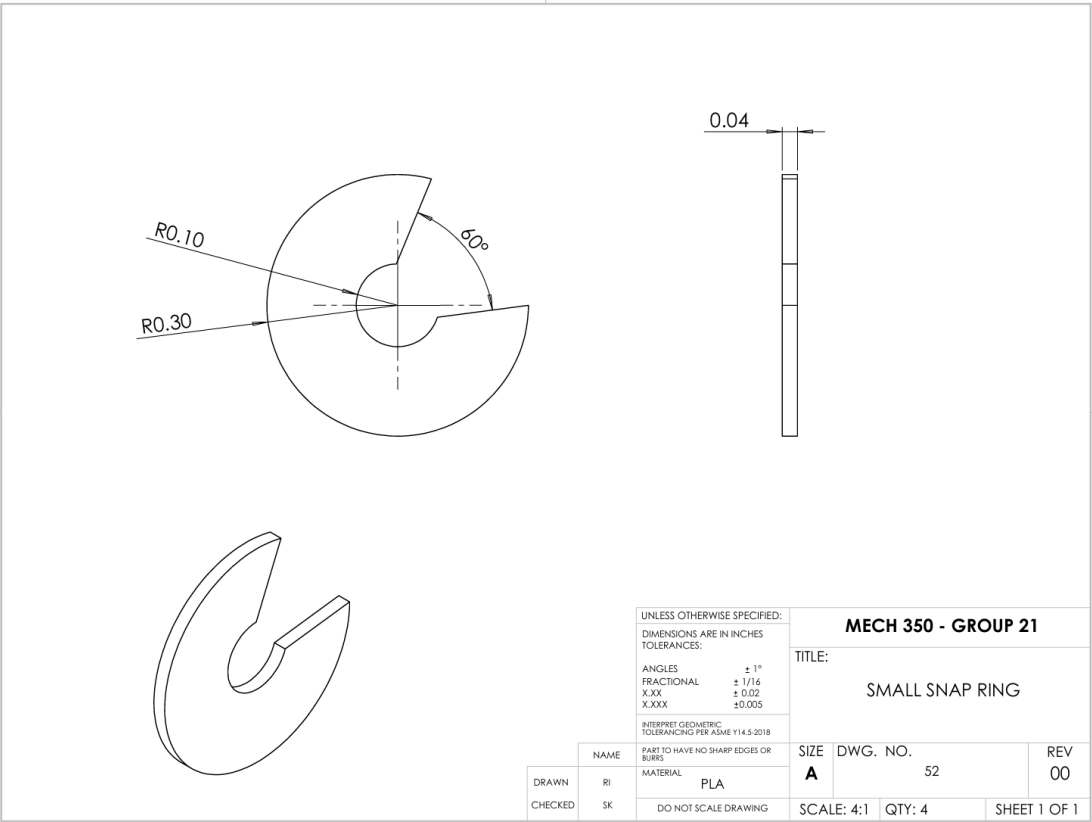
MOTOR BOX FACE PLATE

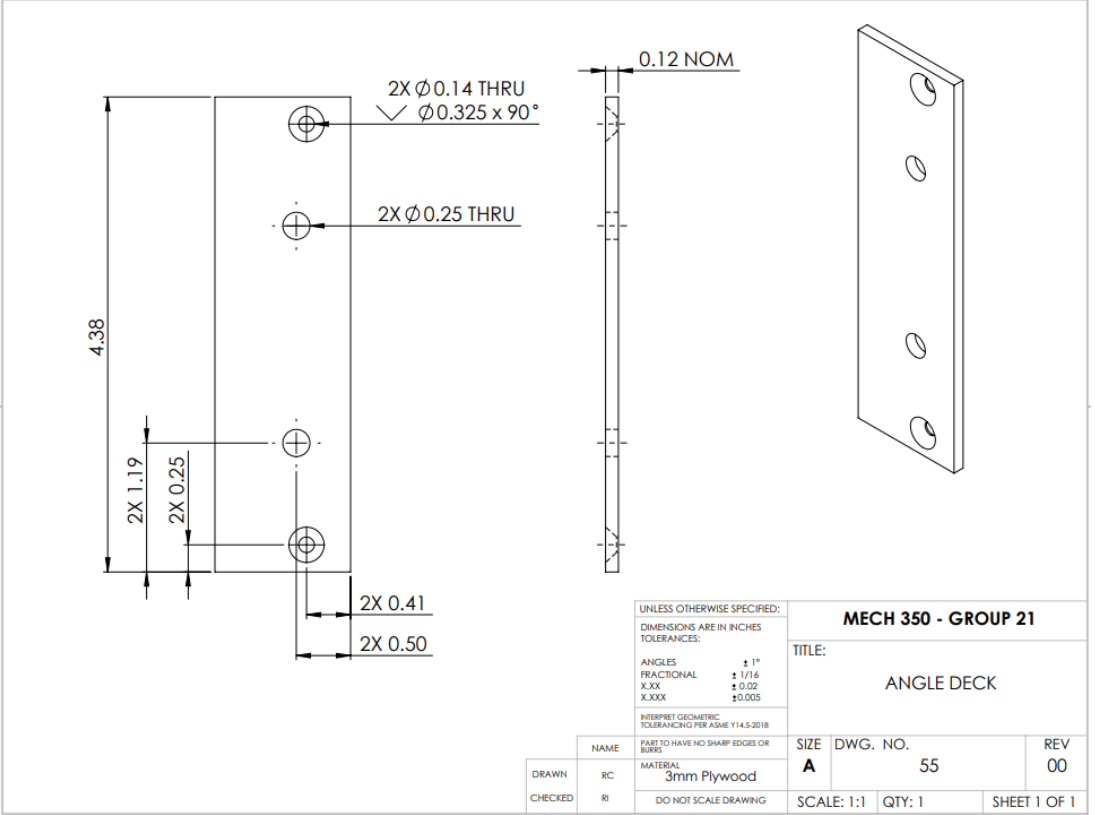
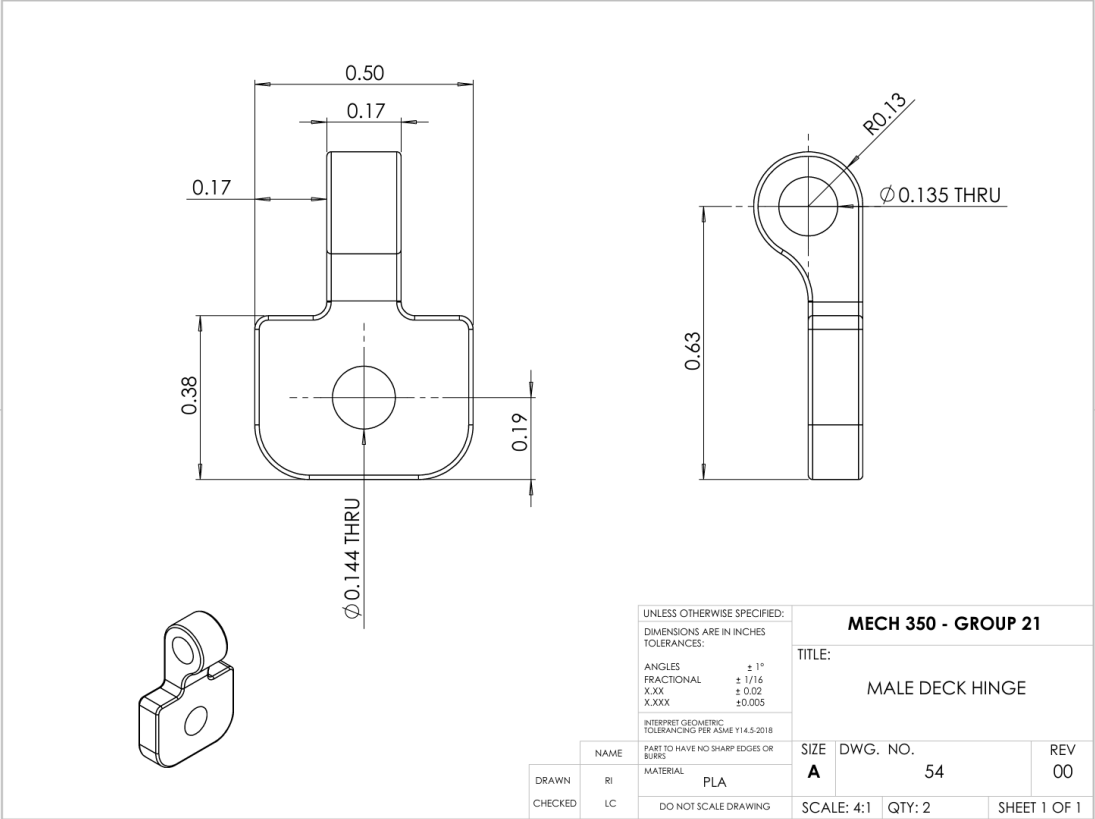
DRAWN	RC	NAME	SIZE	DWG. NO.	REV
CHECKED	RI	MATERIAL	A	41	00
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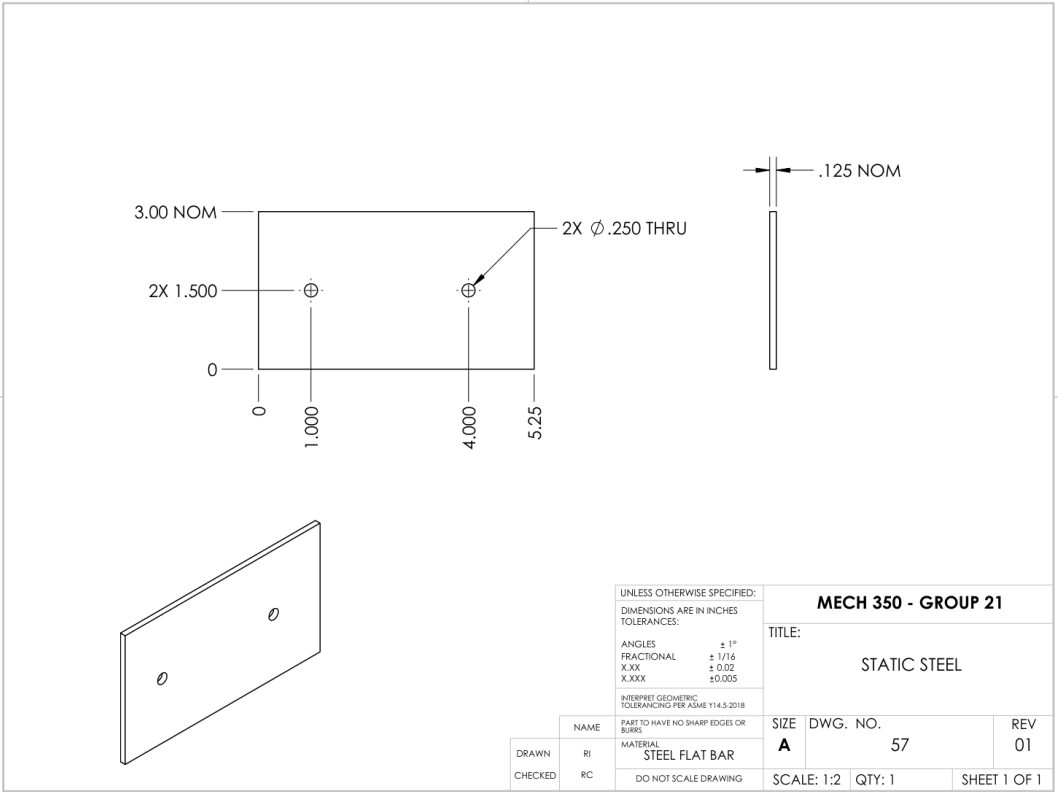
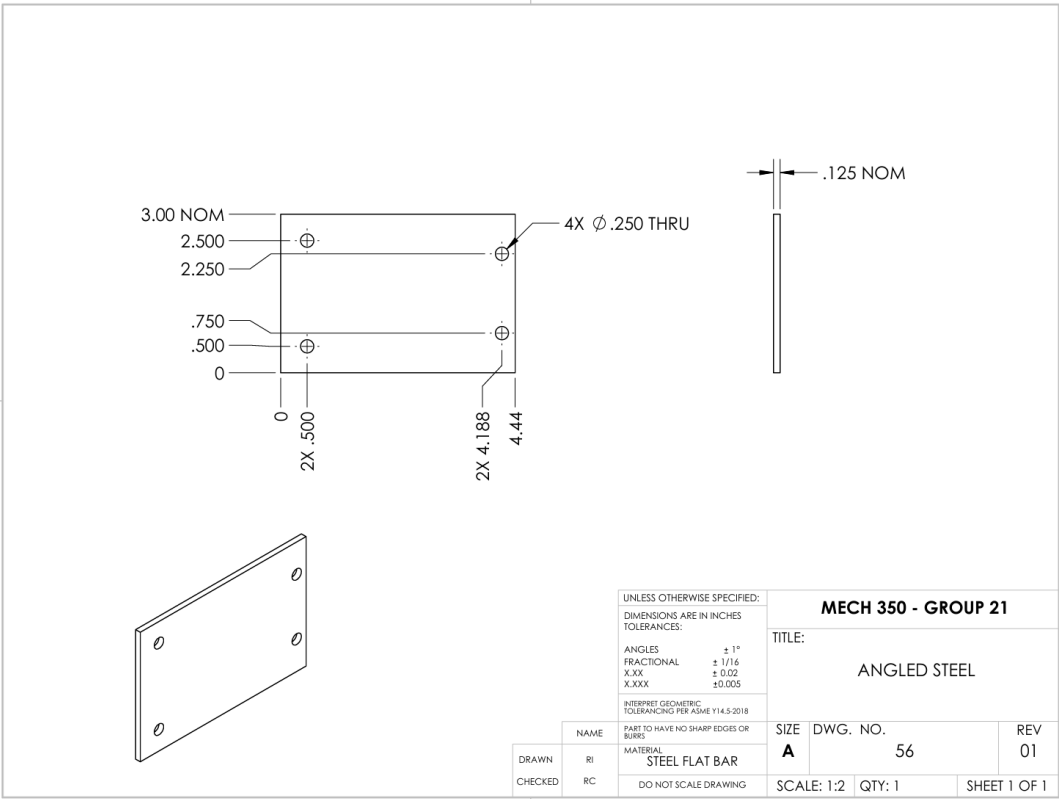


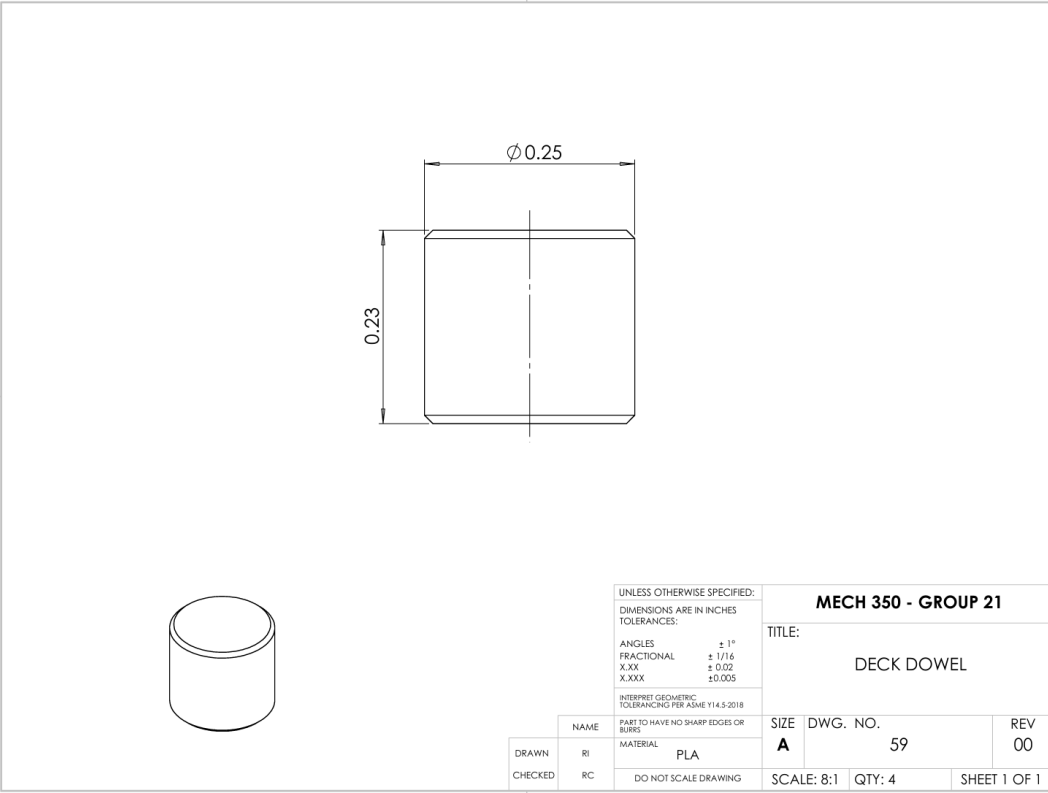
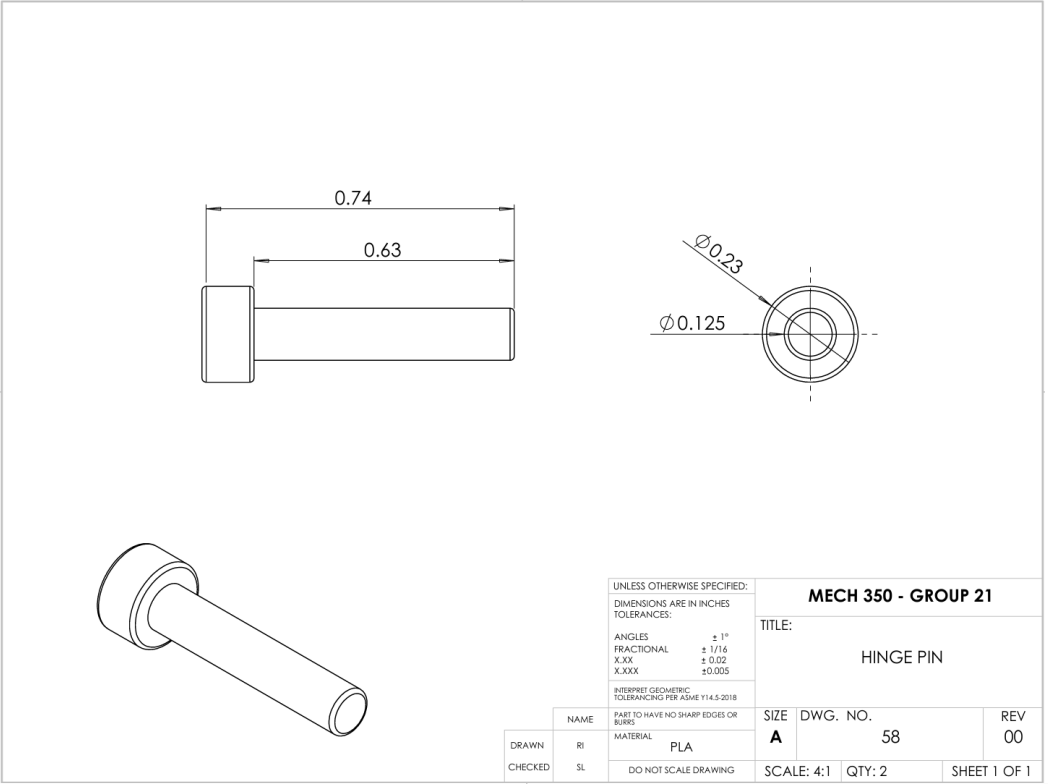


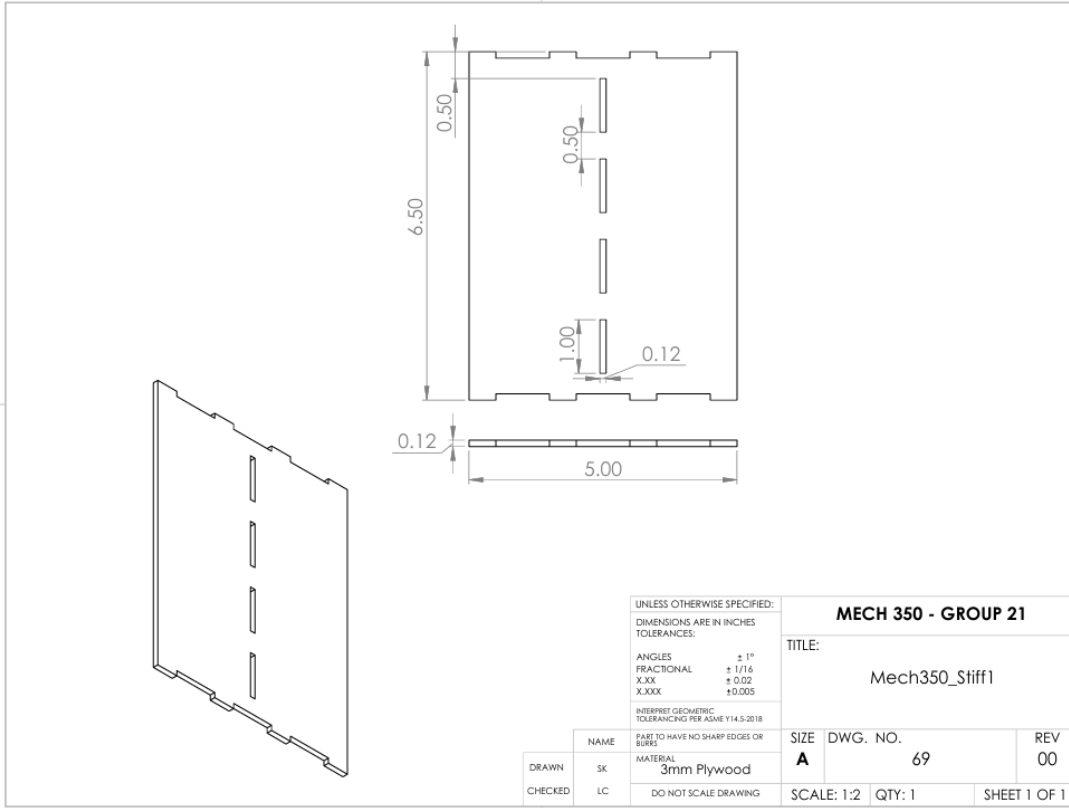
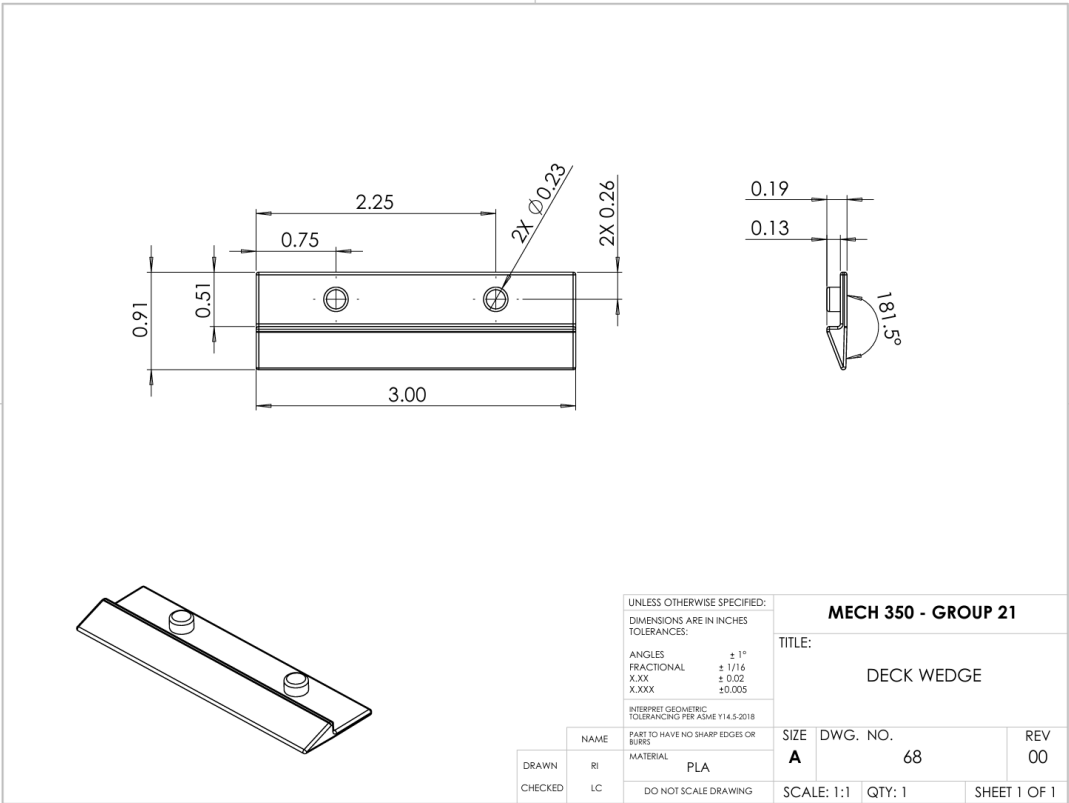


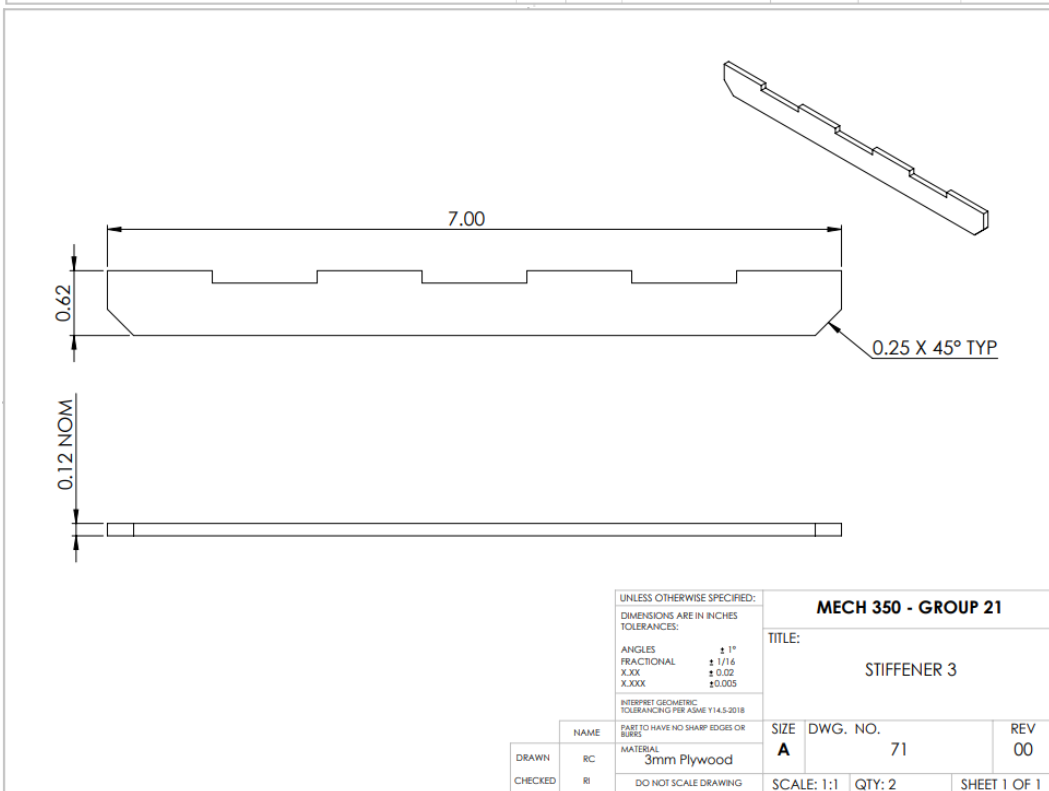
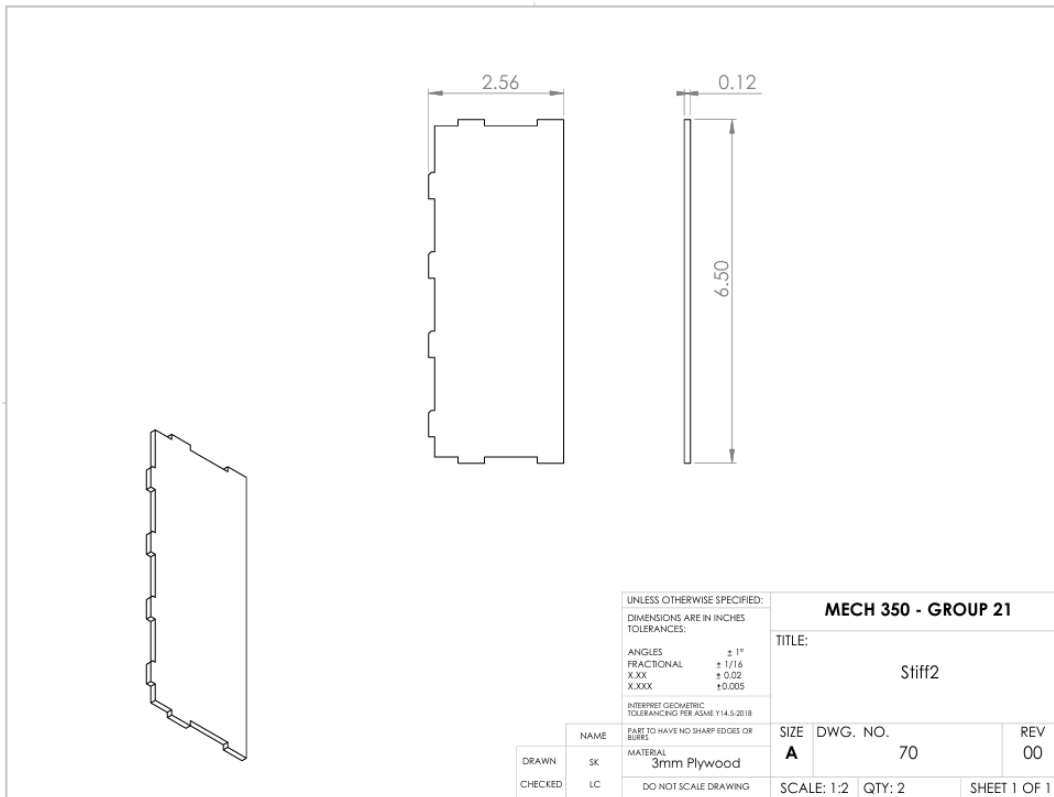


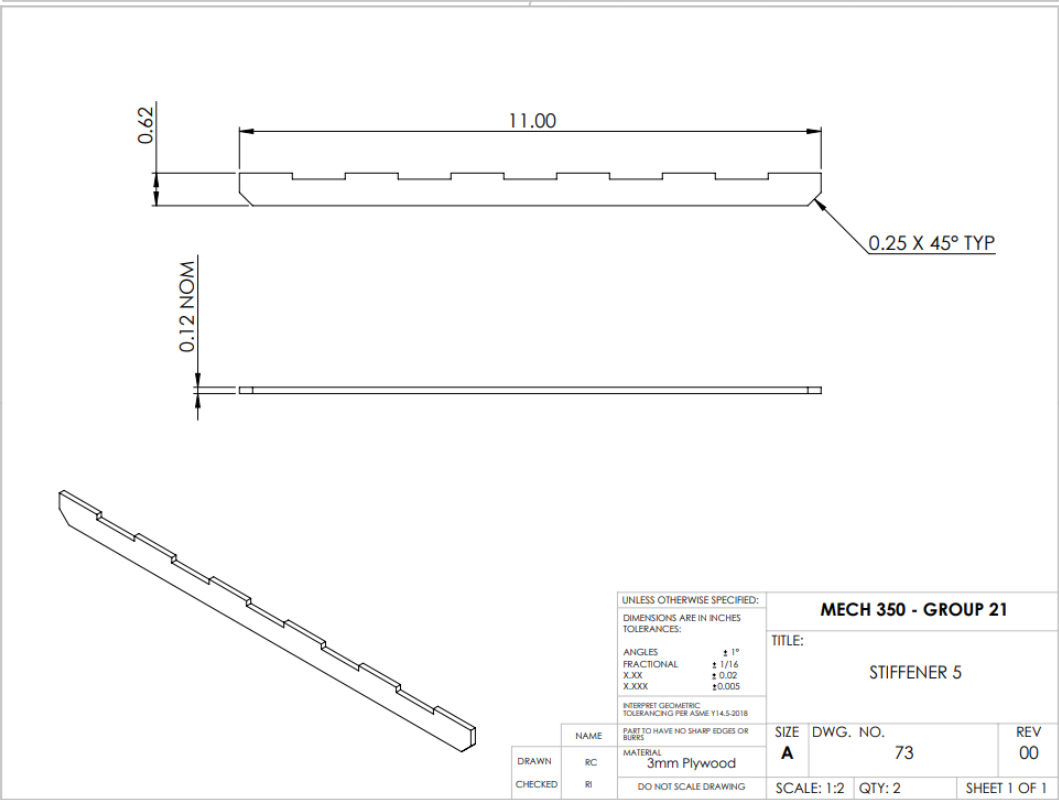
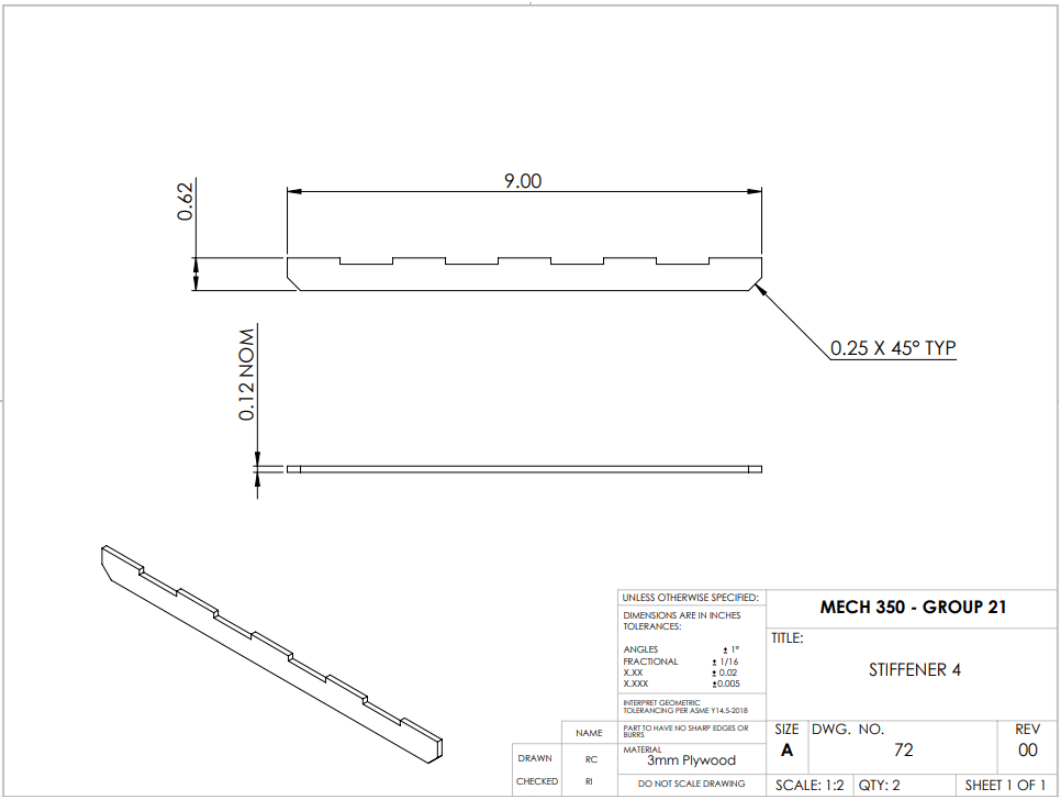


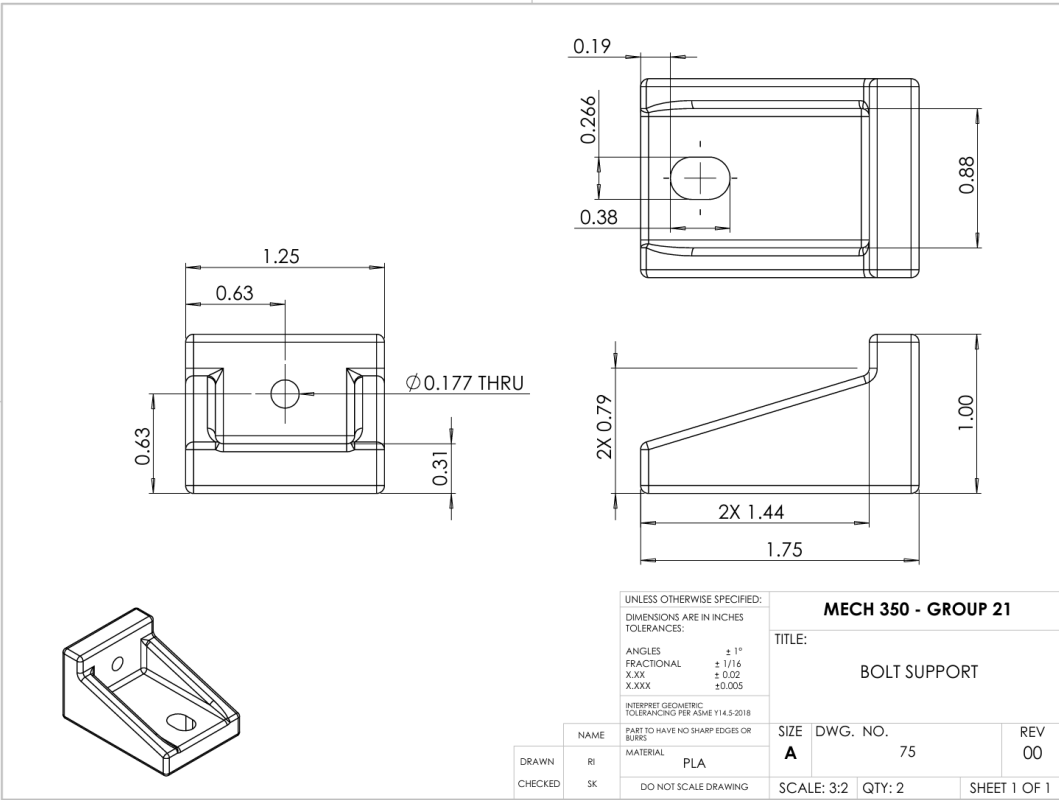
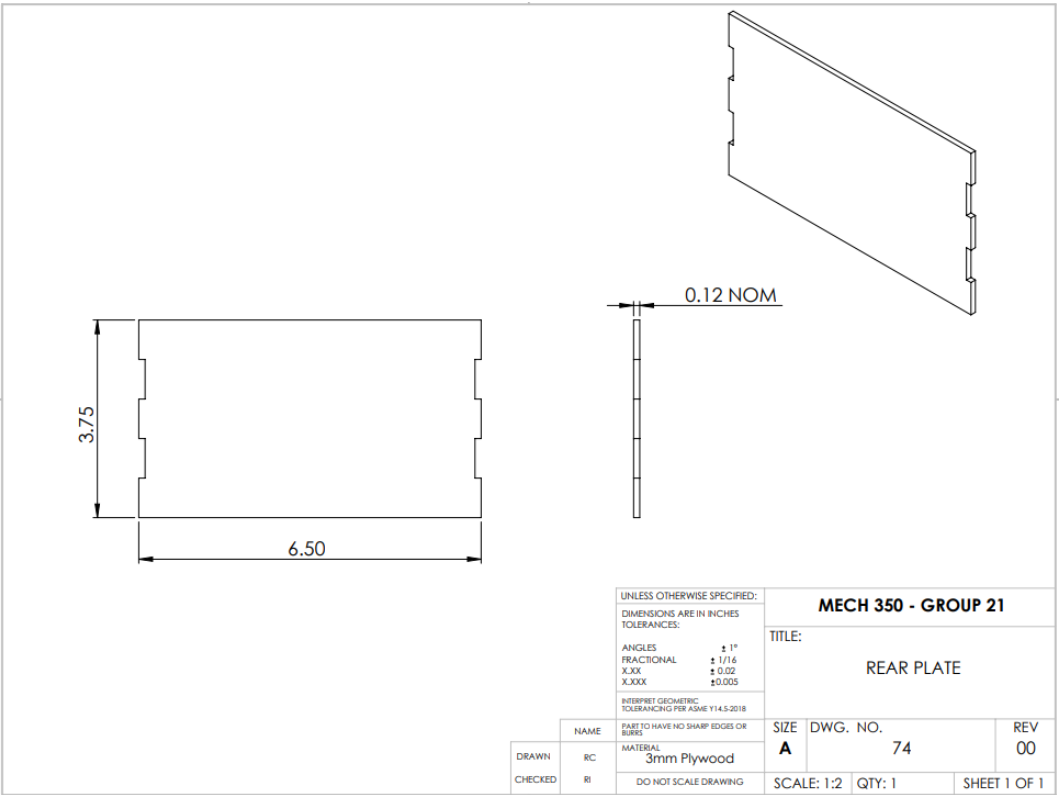


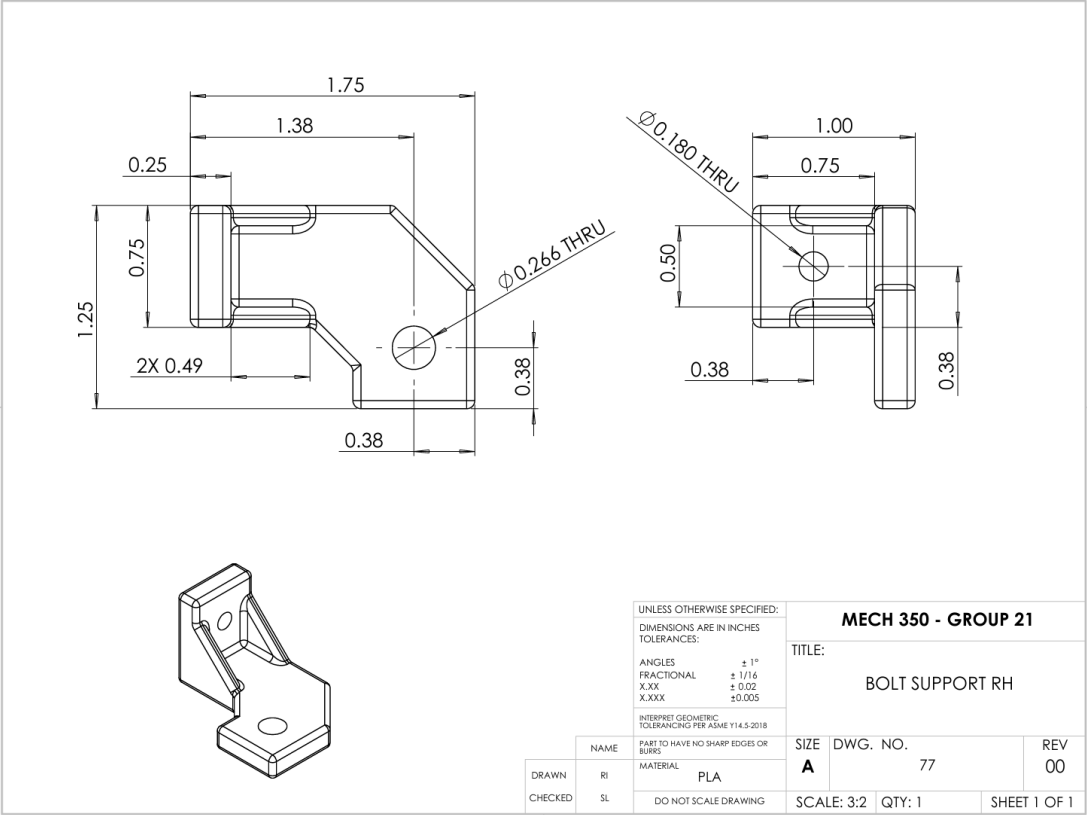


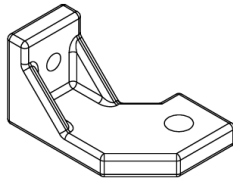
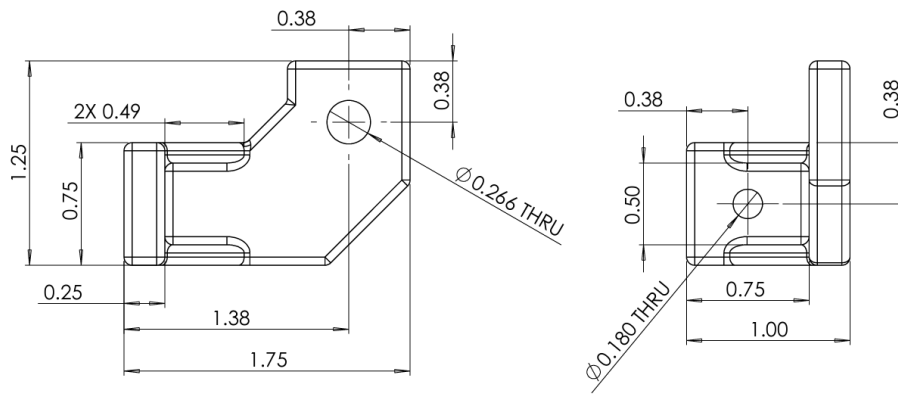




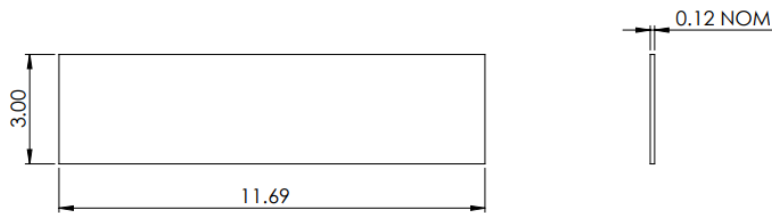
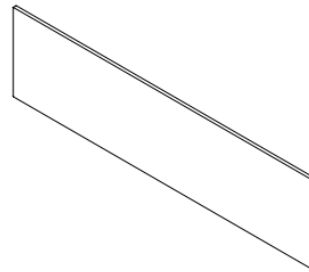




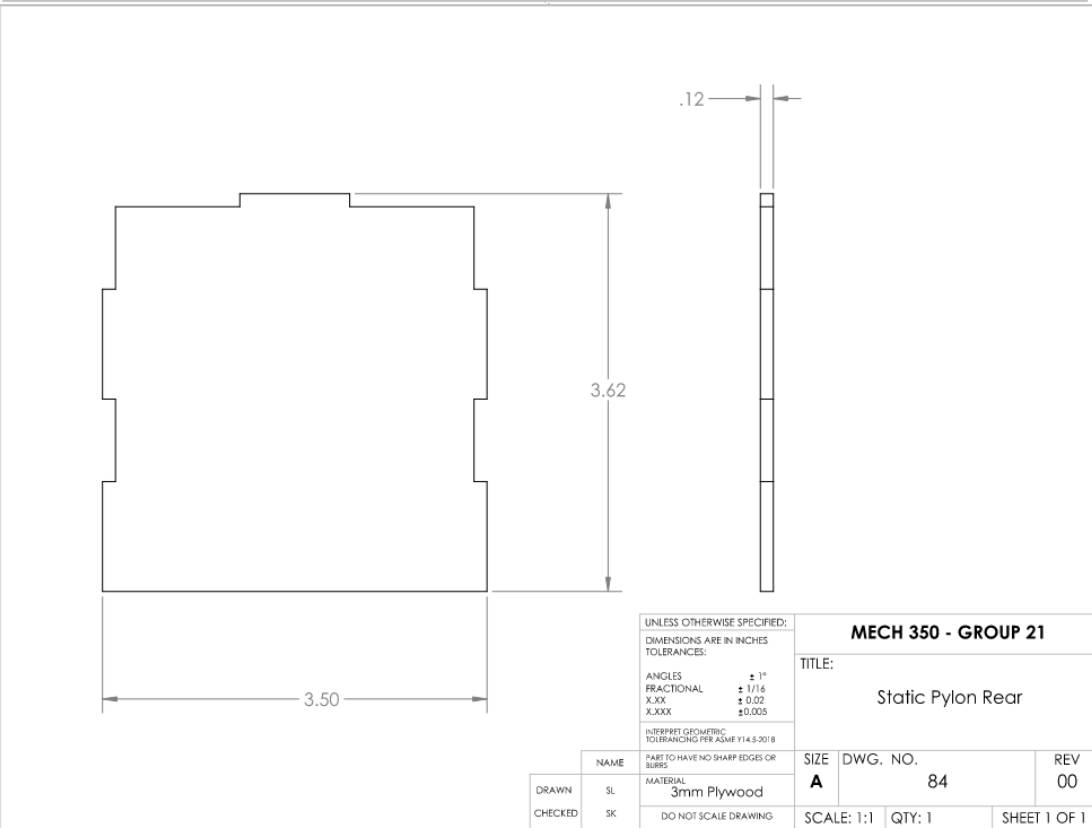
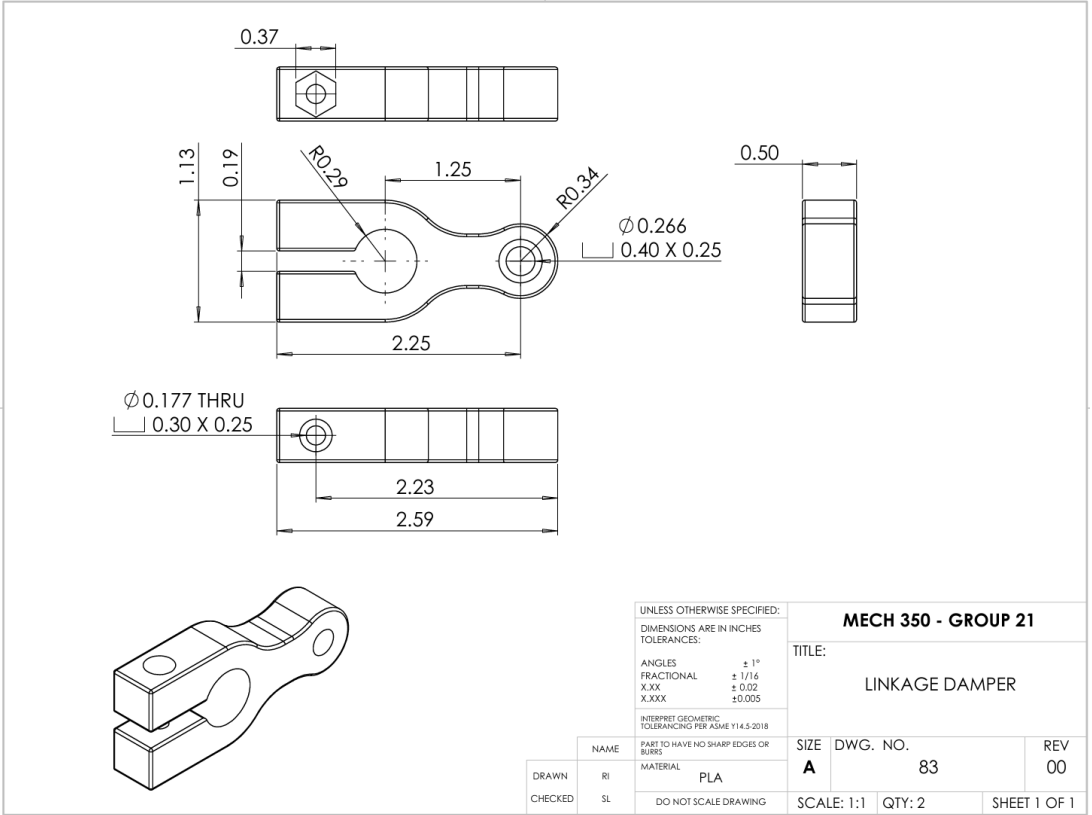


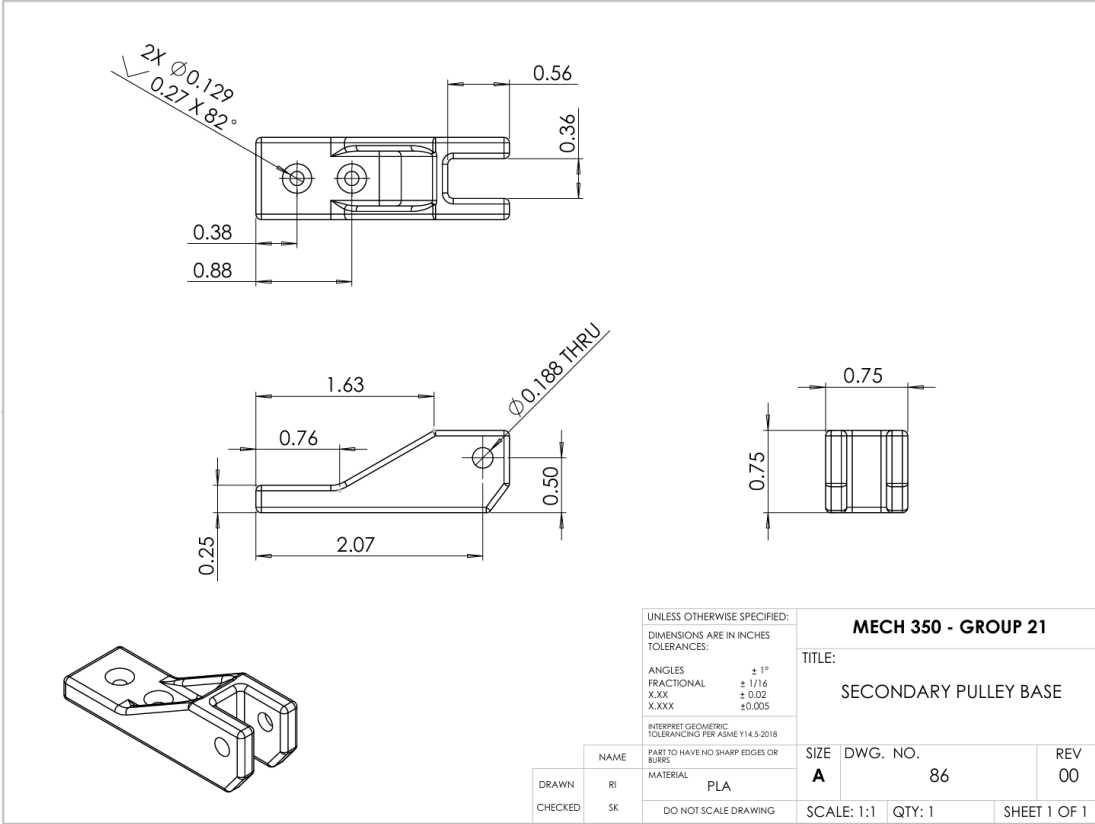
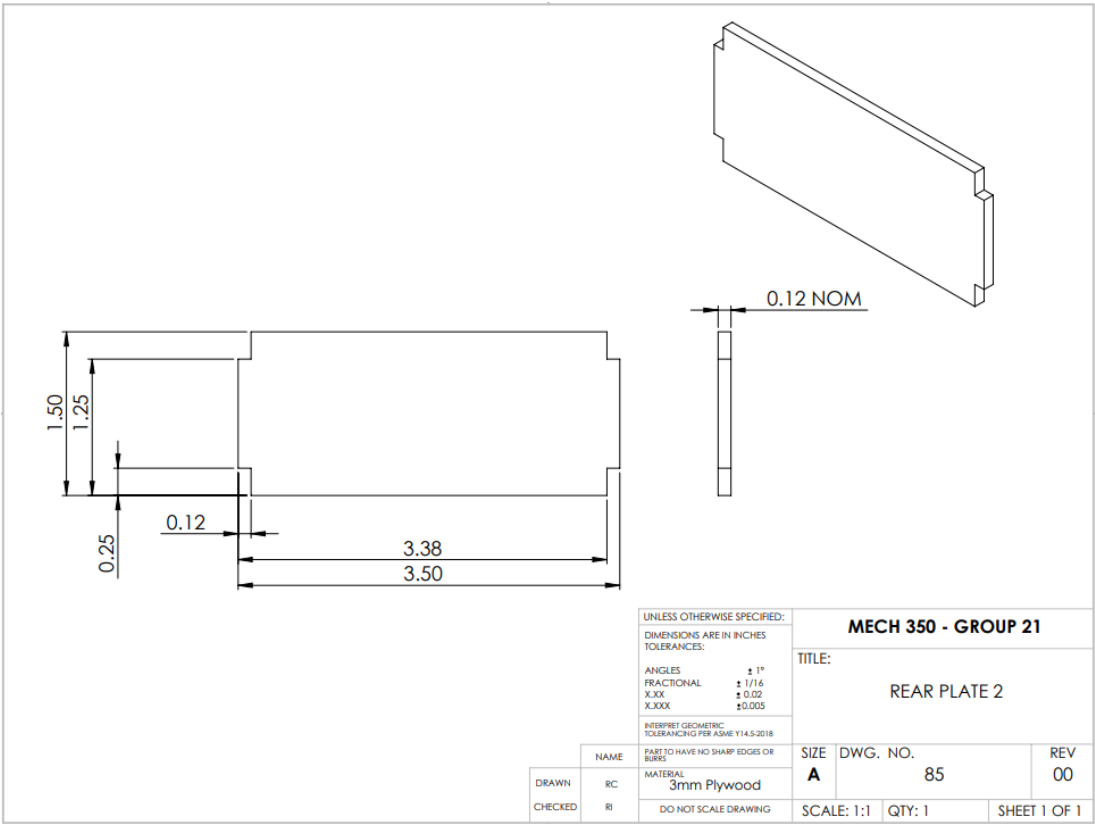


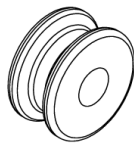
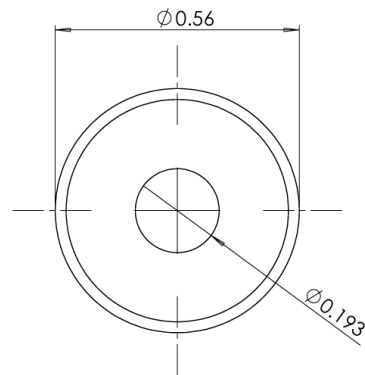
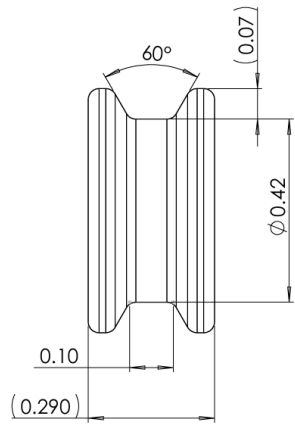
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				TITLE:		
				BOLT SUPPORT LH		
	NAME	PART TO HAVE NO SHARP EDGES OR BURRS		SIZE	DWG. NO.	
	MATERIAL	PLA		A	78	
DRAWN	RI				REV	
CHECKED	SL	DO NOT SCALE DRAWING		SCALE: 3:2	QTY: 1	
				SHEET 1 OF 1		



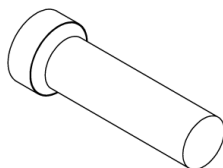
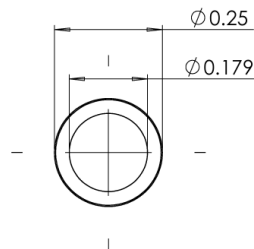
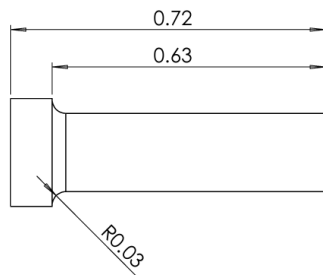
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		DIMENSIONS ARE IN INCHES			
		TOLERANCES:			
		ANGLES FRACTIONAL X.XX X.XXX			
		± 1° ± 1/16 ± 0.02 ± 0.005		TITLE: 	



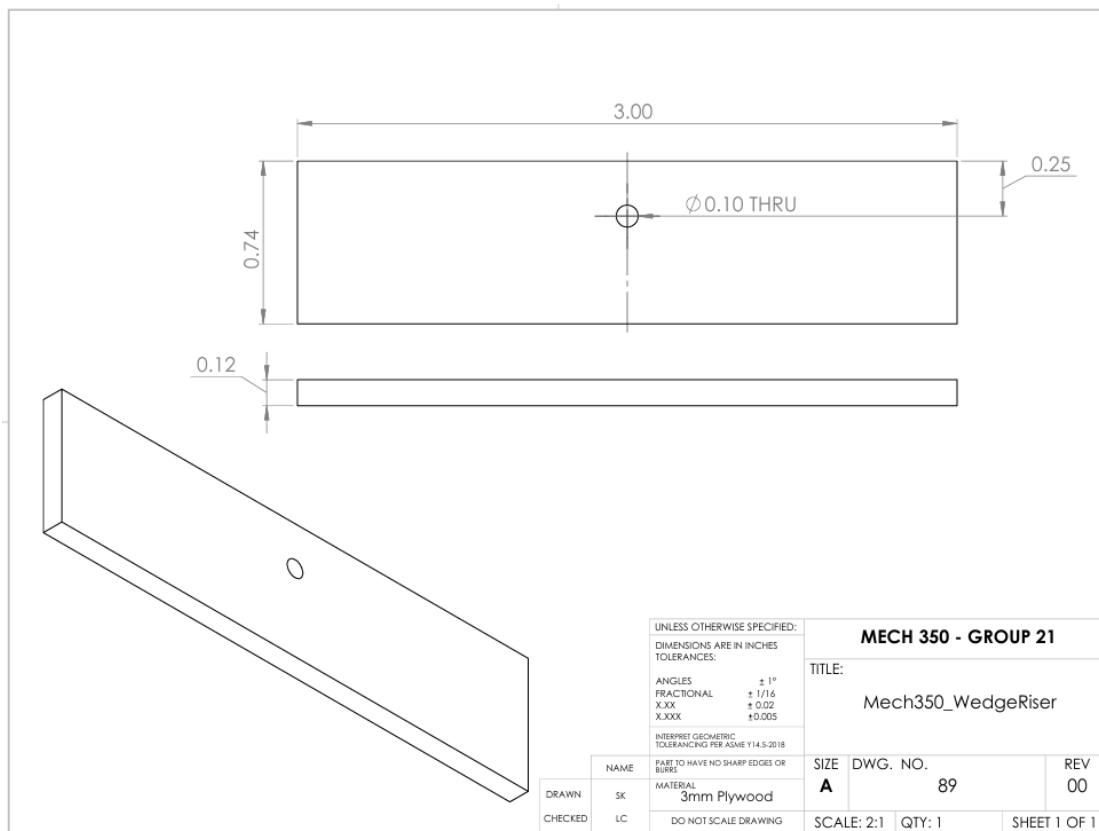




UNLESS OTHERWISE SPECIFIED:				MECH 350 - GROUP 21		
DIMENSIONS ARE IN INCHES				TITLE:		
TOLERANCES:				SECONDARY PULLEY		
ANGLES				SIZE	DWG. NO.	REV
FRACTIONAL				A	87	00
X.XX				SCALE: 4:1 QTY: 1 SHEET 1 OF 1		
X.XXX						
INTERPRET GEOMETRIC TOLERANCING PER ASME Y14.5-2018						
PART TO HAVE NO SHARP EDGES OR BURRS						
NAME						
MATERIAL						
PLA						
DO NOT SCALE DRAWING						



UNLESS OTHERWISE SPECIFIED:				MECH 350 - GROUP 21		
DIMENSIONS ARE IN INCHES				TITLE:		
TOLERANCES:				SECONDARY PULLEY PIN		
ANGLES				SIZE	DWG. NO.	REV
FRACTIONAL				A	88	00
X.XX				SCALE: 4:1 QTY: 1 SHEET 1 OF 1		
X.XXX						
INTERPRET GEOMETRIC TOLERANCING PER ASME Y14.5-2018						
PART TO HAVE NO SHARP EDGES OR BURRS						
NAME						
MATERIAL						
PLA						
DO NOT SCALE DRAWING						



Appendix E – Engineering Analysis

This appendix contains the analysis used for design decisions on the final bridge design.

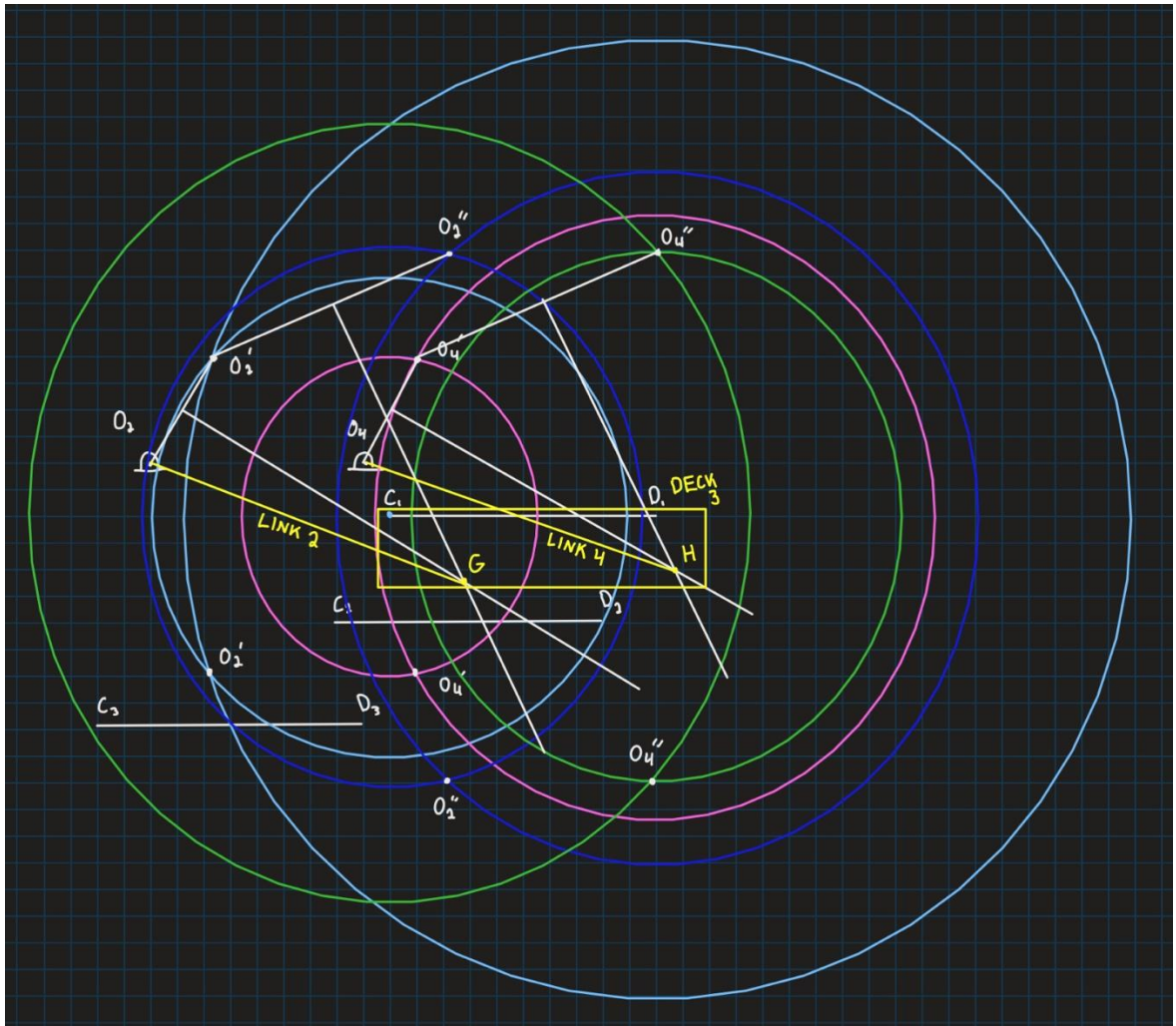
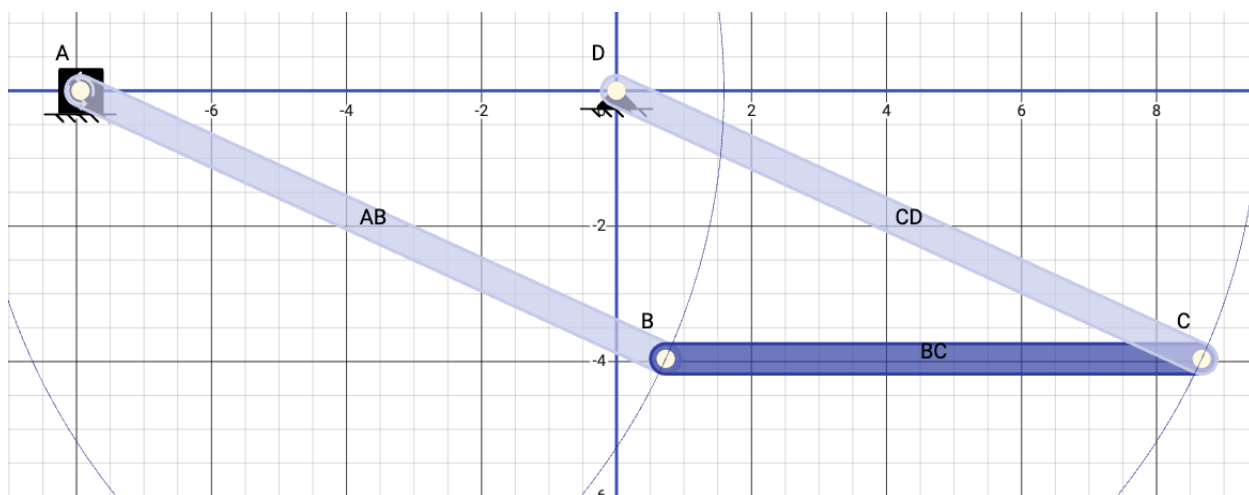


Figure 23: Graphical Synthesis of the Four-Bar Mechanism using Three-Position Motion Generation (Intersecting Circles Method).



Tool.

$$r_{B/A} = 0.123 \text{ m}, \quad r_{G/A} = 0.457 \text{ m}, \quad m = 1.12 \text{ kg}, \quad g = 9.81 \text{ m/s}^2$$

$$\Sigma F = -F_{N1} + F_{N2} - m_{steel}g$$

$$\Sigma M_A = (r_{B/A})(F_{N2}) + (r_{G/A})(-m_{steel}g)$$

$$F_{N2} = \left(\frac{r_{G/A}}{r_{B/A}} \right) (m_{steel}g)$$

$$F_{N2} = \left(\frac{0.457 \text{ m}}{0.123 \text{ m}} \right) ((1.12 \text{ kg}) (9.81 \text{ m/s}^2)) = 40.7 \text{ N}$$

$$F_{N1} = \left(\left(\frac{r_{G/A}}{r_{B/A}} \right) - 1 \right) m_{steel}g$$

$$F_{N1} = \left(\left(\frac{0.457 \text{ m}}{0.123 \text{ m}} \right) - 1 \right) ((1.12 \text{ kg}) (9.81 \text{ m/s}^2)) = 29.7 \text{ N}$$

...

$$\frac{1}{2}F_{N2}$$

$$\frac{1}{2}F_{N2}$$

$$\frac{1}{2}F_{N2}$$

$$\frac{1}{2}F_{N2}$$

$$r = 0.125 \text{ in} = 0.003175 \text{ m}$$

$$\tau_{max} = \frac{4V}{3A} = \frac{4 \left(\frac{1}{2} N_2 \right)}{3\pi r^2}$$

$$\tau_{max} = 856.6 \text{ KPa}$$

$$N = \frac{S_y}{\tau_{max}} = 250$$

≡

0.279 m

A

B

C

$$\frac{1}{4}F_{N2}$$

$$\frac{1}{2}F_{N2}$$

$$\frac{1}{4}F_{N2}$$

$$\frac{1}{4}F_{N2}$$

$$-\frac{1}{4}F_{N2}$$

$$-\frac{1}{4}F_{N2} \cdot r_{B/A}$$

$$h = 0.01538 \text{ m}$$

$$\sigma_b = \frac{Mc}{I} = \frac{6M}{bh^2}$$

$$\sigma_b = \frac{6(-\frac{1}{4}(40.7) \cdot 0.279)}{(0.003 \text{ m})(0.01538 \text{ m})} = -24 \text{ MPa}$$

$$N = \frac{\sigma_y}{\sigma_b} = \frac{45}{24} = 1.875 \Rightarrow \text{Low so add reinforcements}$$

$$\sigma_b = \frac{6\left(-\frac{1}{4}(40.7) \cdot 0.279\right)}{(0.012 \text{ m})(0.01538 \text{ m})} = -6 \text{ MPa}$$

$$N = \frac{\sigma_y}{\sigma_b} = \frac{45}{6} = 7.5$$